

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

SCHEM, TAUP0, M87

10/13/2007

REV	ZONE	ECN	DESCRIPTION OF CHANGE	CK APPD DATE	ENG APPD DATE
?		?	?	?	?

Page	(,csa)	Contents	Sync	Date
1	1	Table of Contents	N/A	N/A
2	2	System Block Diagram	(T9_MLB)	08/23/2006
3	3	Power Block Diagram	(T9_MLB)	08/23/2006
4	4	Power Block Diagram	N/A	N/A
5	5	BOM Configuration	N/A	N/A
6	6	Revision History	N/A	N/A
7	7	Functional / ICT Test	(MASTER)	(MASTER)
8	8	Power Aliases	(MASTER)	(MASTER)
9	9	Signal Aliases	(T9_MLB)	08/23/2006
10	10	CPU FSB	T9_NAME	03/16/2007
11	11	CPU Power & Ground	T9_NAME	03/16/2007
12	12	CPU Decoupling & VID	M76_MLB	03/19/2007
13	13	eXtended Debug Port (XDP)	T9_NAME	12/12/2006
14	14	NB CPU Interface	T9_NAME	03/16/2007
15	15	NB PEG / Video Interfaces	T9_NAME	03/16/2007
16	16	NB Misc Interfaces	T9_NAME	03/16/2007
17	17	NB DDR2 Interfaces	T9_NAME	03/16/2007
18	18	NB Power 1	T9_NAME	03/16/2007
19	19	NB Power 2	T9_NAME	03/16/2007
20	20	NB Grounds	T9_NAME	03/16/2007
21	21	NB Standard Decoupling	T9_NAME	01/17/2007
22	22	NB Graphics Decoupling	M76_MLB	03/12/2007
23	23	SB Enet, Disk, FSB, LPC	T9_NAME	03/16/2007
24	24	SB PCI, PCIE, DMI, USB	T9_NAME	03/16/2007
25	25	SB Pwr Mgt, GPIO, Clink	T9_NAME	03/16/2007
26	26	SB Power & Ground	T9_NAME	03/16/2007
27	27	SB Decoupling	T9_NAME	01/17/2007
28	28	SB Misc	(T9_MLB)	08/24/2006
29	29	Clock (CK505)	T9_NAME	03/16/2007
30	30	Clock Termination	(MASTER)	08/23/2006
31	31	DDR2 SO-DIMM Connector A	(M59_SYNC)	08/24/2006
32	32	DDR2 SO-DIMM Connector B	(M59_SYNC)	08/24/2006
33	33	Memory Active Termination	(T9_NAME)	11/14/2006
34	34	Left I/O Board Connector	(M59_SYNC)	08/24/2006
35	37	Ethernet (Yukon)	T9_NAME	03/16/2007
36	38	Yukon Power Control	T9_NAME	03/16/2007
37	39	Ethernet Connector	M76_MLB	03/19/2007
38	40	FireWire Link (TSB83AA22)	M76_MLB	03/19/2007
39	41	FireWire PHY (TSB83AA22)	M76_MLB	03/19/2007
40	42	FireWire Port Power	M76_MLB	03/19/2007
41	43	FireWire Ports	M76_MLB	03/19/2007
42	44	PATA Connector	(MASTER)	(MASTER)
43	46	External USB Connector	M88	08/02/2007
44	47	Left Clutch Barrel Interconnect	M87	07/16/2007
45	49	SMC	T9_NAME	01/17/2007

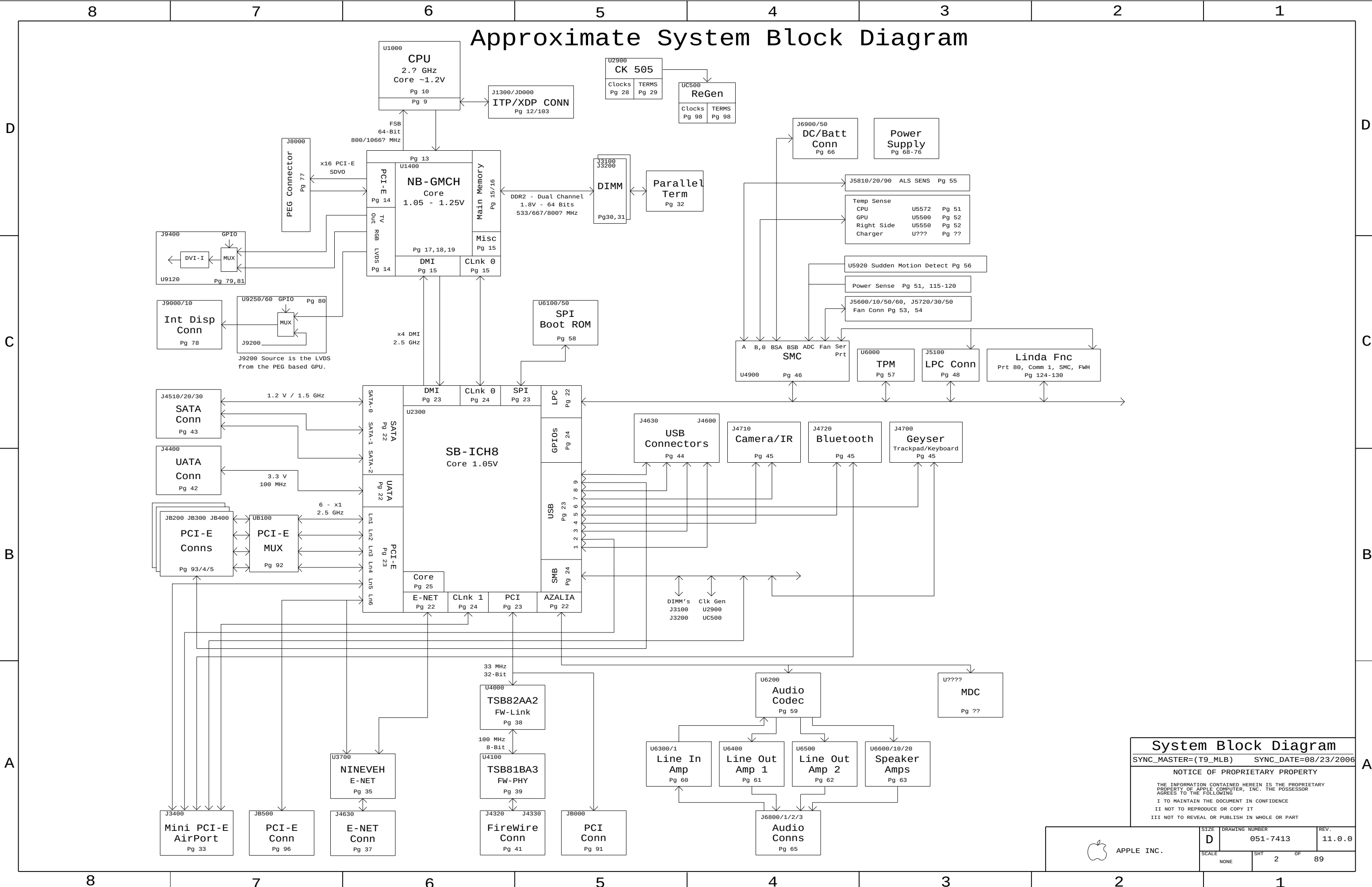
Page	Contents	Sync	Date
46	SMC Support	(MASTER)	
47	LPC+ Debug Connector	M76_MLB	03/19/2007
48	SMBus Connections	(MASTER)	(MASTER)
49	Current & Voltage Sensing	M87_MLB	05/22/2007
50	Current Sensing	M87_MLB	05/22/2007
51	Thermal Sensors	(MASTER)	(MASTER)
52	Fan Connectors	M76_MLB	03/19/2007
53	ALS Support	M76_MLB	03/19/2007
54	Sudden Motion Sensor (SMS)	M76_MLB	03/19/2007
55	SPI BootROM	T9_NAME	03/16/2007
56	PBus-In & Battery Connectors	(M59_SYNC)	09/09/2006
57	Power FETs	M76_MLB	03/19/2007
58	IMVP6 CPU VCore Regulator	M76_MLB	01/23/2007
59	5V / 3.3V Power Supply	M76_MLB	03/19/2007
60	1.25V / 1.05V Power Supply	M76_MLB	03/12/2007
61	1.8V DDR2 Supply	M76_MLB	03/19/2007
62	1.5V Power Supply	M76_MLB	03/12/2007
63	FW PHY Power Supplies	M76_MLB	03/19/2007
64	3.425V G3Hot Supply & Power Control	M88	08/02/2007
65	NV G84M PCI-E	(MASTER)	(MASTER)
66	NV G84M Core/FB Power	(MASTER)	(MASTER)
67	NV G84M Frame Buffer I/F	(MASTER)	(MASTER)
68	GDDR3 Frame Buffer A (Top)	(MASTER)	(MASTER)
69	GDDR3 Frame Buffer B (Top)	(MASTER)	(MASTER)
70	NV G84M GPIO/MIO/Misc	(MASTER)	(MASTER)
71	GPU Straps	M88	08/02/2007
72	NV G84M Video Interfaces	(MASTER)	(MASTER)
73	GPU (G84M) Core Supply	M88	08/02/2007
74	LVDS Display Connector	(MASTER)	(MASTER)
75	GDDR3 Frame Buffer A (Bot)	M88_MLB_VRAM_BOT	06/19/2007
76	GDDR3 Frame Buffer B (Bot)	M88_MLB_VRAM_BOT	06/19/2007
77	1.8V FB Power Supply	(MASTER)	(MASTER)
78	DVI Display Connector	(MASTER)	(MASTER)
79	Project Specific Connectors	(M59_SYNC)	08/24/2006
80	CPU/FSB Constraints	T9_NAME	01/17/2007
81	NB Constraints	T9_NAME	01/17/2007
82	Memory Constraints	T9_NAME	01/17/2007
83	SB Constraints (1 of 2)	T9_NAME	01/17/2007
84	SB Constraints (2 of 2)	T9_NAME	01/17/2007
85	Clock & SMC Constraints	T9_NAME	01/17/2007
86	FireWire Constraints	T9_NAME	01/17/2007
87	GPU (G84M) Constraints	(MASTER)	(MASTER)
88	Project Specific Constraints	(MASTER)	(MASTER)
89	PCB Rule Definitions	(MASTER)	(MASTER)

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-7413	1	SCHEM, TAUP0, M87	SCH	CRITICAL	
820-2249	1	PCBF, TAUP0, M87	PCB	CRITICAL	

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DRAWING
TITLE=MLB
ABBREV=DRAWING
LAST_MODIFIED=Fri Oct 12 18:10:11 2007
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	ENG APPD		MFG APPD			
	QA APPD		DESIGNER			
RELEASE		SCALE	NONE			
 THIRD ANGLE PROJECTION	MATERIAL/FINISH NOTED AS APPLICABLE		SIZE	DRAWING NUMBER		
			D	051-7413		
				REV. 11.0.0		
				SHT 1 OF 89		



System Block Diagram

SYNC_MASTER=(T9_MLB) SYNC_DATE=08/23/2006

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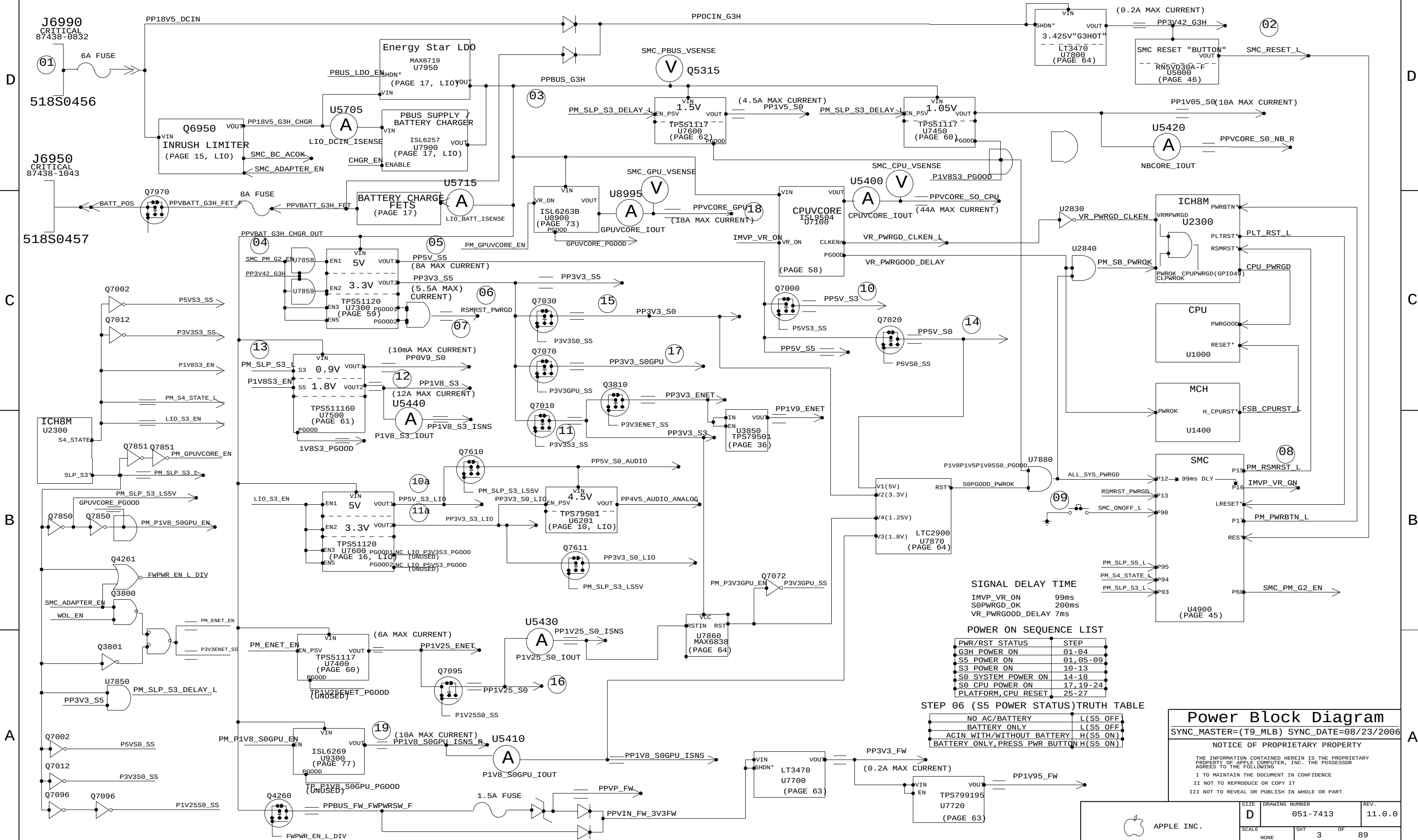
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SCALE		SHT	OF
NONE		2	89

M87 POWER SYSTEM ARCHITECTURE



Power Block Diagram

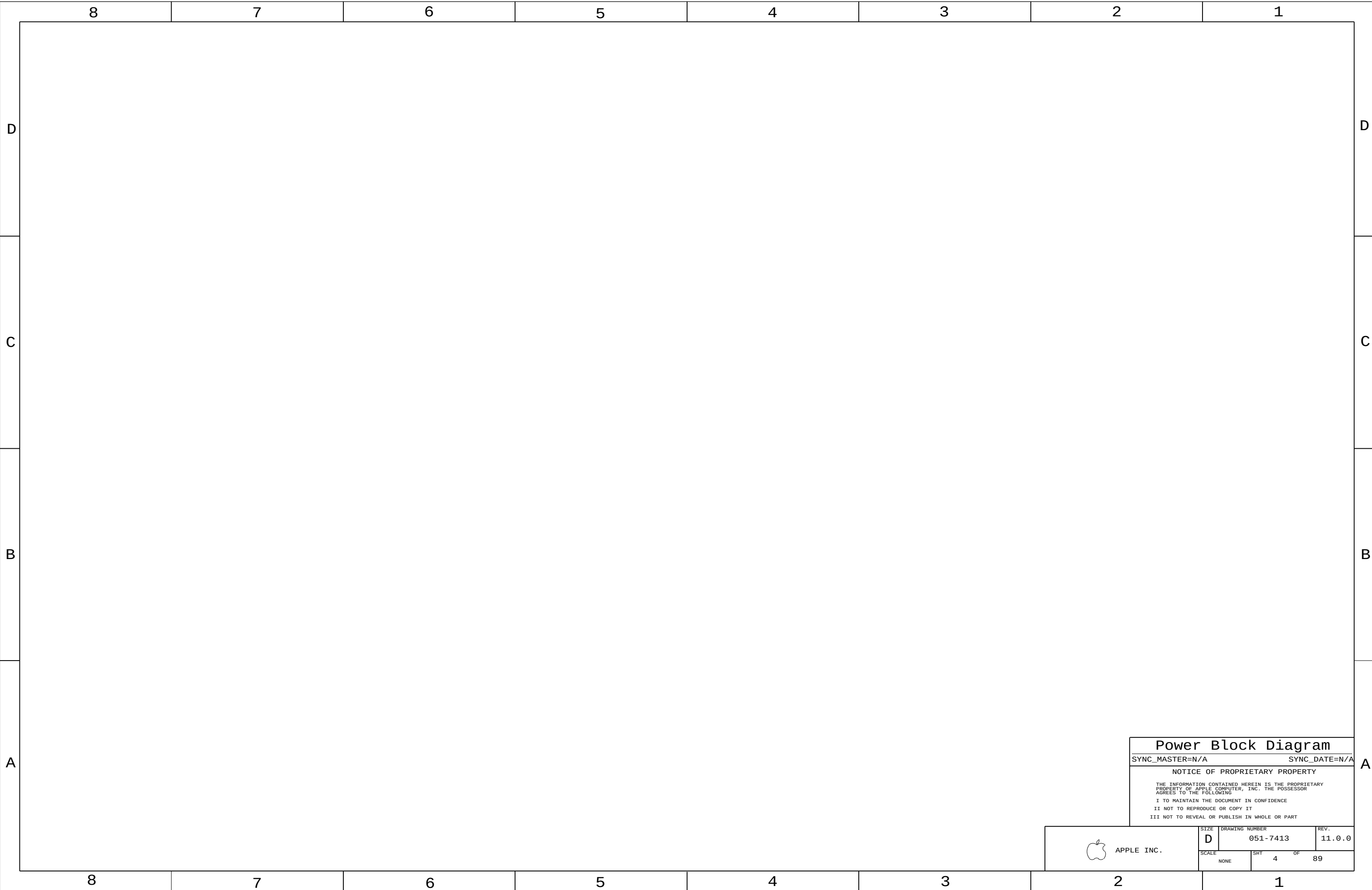
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SCALE	SHT	OF
NONE	3	89



Power Block Diagram

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SYNC_DATE=N/A

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 APPLE INC.	SIZE D	DRAWING NUMBER 051-7413	REV. 11.0.0
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	<table><tr><th>BOM NUMBER</th><th colspan="4">BOM NAME</th><th colspan="6">BOM OPTIONS</th></tr><tr><td>630-9089</td><td colspan="4">PCBA, 2.5GHZ, 512SAM_VRAM, M87</td><td colspan="6">M87_COMMON, EEE_Z3G, CPU_2_5GHZ, FB_512_SAMSUNG</td></tr><tr><td>630-9091</td><td colspan="4">PCBA, 2.6GHZ, 512SAM_VRAM, M87</td><td colspan="6">M87_COMMON, EEE_Z3J, CPU_2_6GHZ, FB_512_SAMSUNG</td></tr><tr><td>630-9088</td><td colspan="4">PCBA, 2.5GHZ, 512HY_VRAM, M87</td><td colspan="6">M87_COMMON, EEE_Z3F, CPU_2_5GHZ, FB_512_HYNIX</td></tr><tr><td>630-9090</td><td colspan="4">PCBA, 2.6GHZ, 512HY_VRAM, M87</td><td colspan="6">M87_COMMON, EEE_Z3H, CPU_2_6GHZ, FB_512_HYNIX</td></tr><tr><td>630-9213</td><td colspan="4">PCBA, 2.4GHZ, 256SAM_VRAM, M87</td><td colspan="6">M87_COMMON, EEE_ZUP, CPU_2_4GHZ, FB_256_SAMSUNG</td></tr><tr><td>630-9238</td><td colspan="4">PCBA, 2.4GHZ, 256HY_VRAM, M87</td><td colspan="6">M87_COMMON, EEE_043, CPU_2_4GHZ, FB_256_HYNIX</td></tr></table>																BOM NUMBER	BOM NAME				BOM OPTIONS						630-9089	PCBA, 2.5GHZ, 512SAM_VRAM, M87				M87_COMMON, EEE_Z3G, CPU_2_5GHZ, FB_512_SAMSUNG						630-9091	PCBA, 2.6GHZ, 512SAM_VRAM, M87				M87_COMMON, EEE_Z3J, CPU_2_6GHZ, FB_512_SAMSUNG						630-9088	PCBA, 2.5GHZ, 512HY_VRAM, M87				M87_COMMON, EEE_Z3F, CPU_2_5GHZ, FB_512_HYNIX						630-9090	PCBA, 2.6GHZ, 512HY_VRAM, M87				M87_COMMON, EEE_Z3H, CPU_2_6GHZ, FB_512_HYNIX						630-9213	PCBA, 2.4GHZ, 256SAM_VRAM, M87				M87_COMMON, EEE_ZUP, CPU_2_4GHZ, FB_256_SAMSUNG						630-9238	PCBA, 2.4GHZ, 256HY_VRAM, M87				M87_COMMON, EEE_043, CPU_2_4GHZ, FB_256_HYNIX																
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M75 BOM Groups																																																																																																								
<table><tr><th>BOM GROUP</th><th colspan="14">BOM OPTIONS</th></tr><tr><td>M87_COMMON</td><td colspan="14">ALTERNATE, COMMON, M87_COMMON1, M87_COMMON2, M87_DEBUG, M87_PROGPARTS</td></tr><tr><td>M87_COMMON1</td><td colspan="14">ISL9504B, ONEWIRE_PU</td></tr><tr><td>M87_COMMON2</td><td colspan="14">GPUVID_1P13V, P1V8S3_1V8, SMS_M0T_DIS, YUKON_ULTRA, VGA_TERM_CONN, GPU_TMP401</td></tr><tr><td>M87_DEBUG</td><td colspan="14">SMC_DEBUG_YES, XDP, XDP_CONN, LPCPLUS</td></tr><tr><td>M87_PROGPARTS</td><td colspan="14">BOOTROM_PROG, SMC_PROG</td></tr></table>															BOM GROUP	BOM OPTIONS														M87_COMMON	ALTERNATE, COMMON, M87_COMMON1, M87_COMMON2, M87_DEBUG, M87_PROGPARTS														M87_COMMON1	ISL9504B, ONEWIRE_PU														M87_COMMON2	GPUVID_1P13V, P1V8S3_1V8, SMS_M0T_DIS, YUKON_ULTRA, VGA_TERM_CONN, GPU_TMP401														M87_DEBUG	SMC_DEBUG_YES, XDP, XDP_CONN, LPCPLUS														M87_PROGPARTS	BOOTROM_PROG, SMC_PROG													
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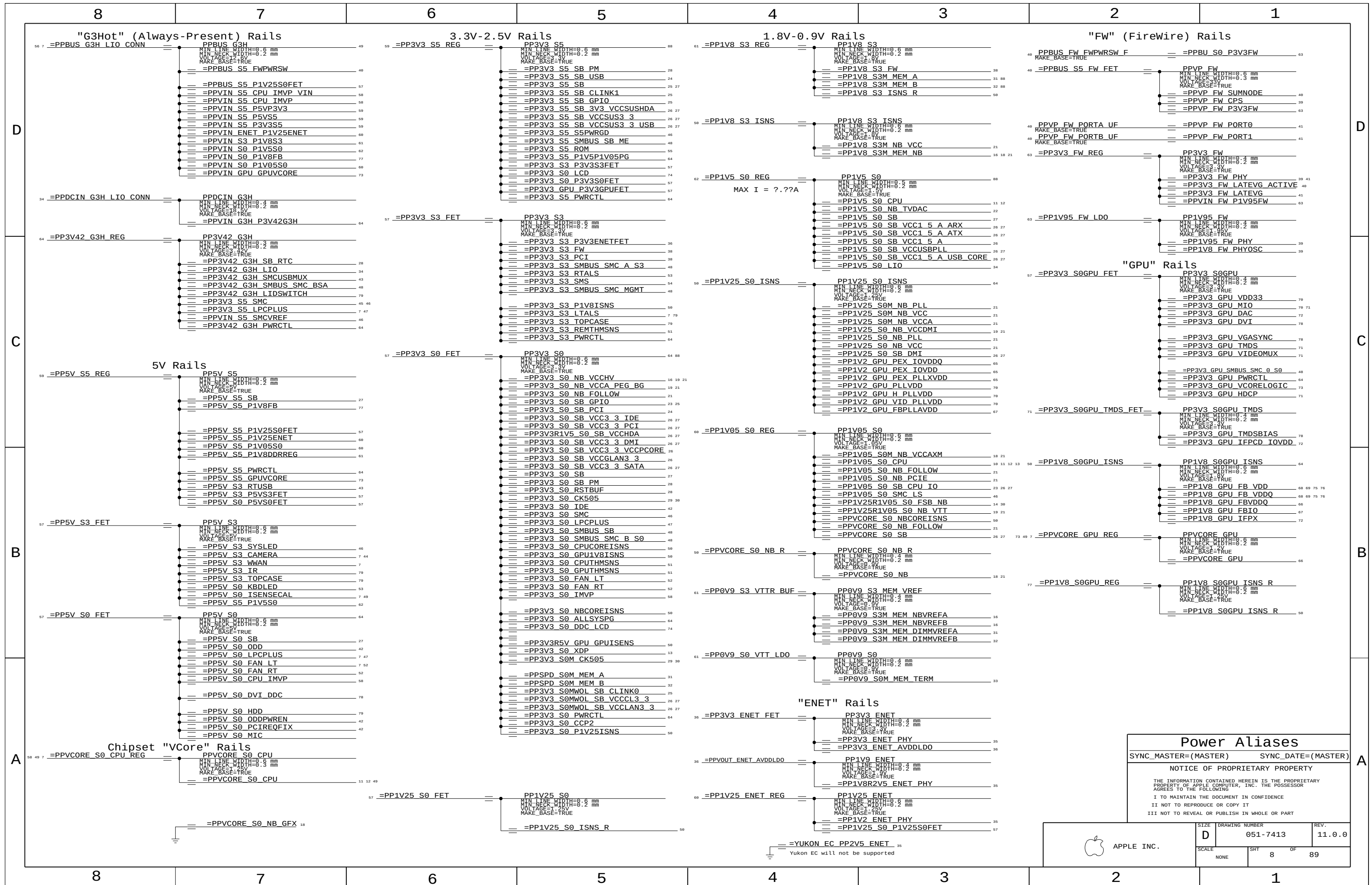
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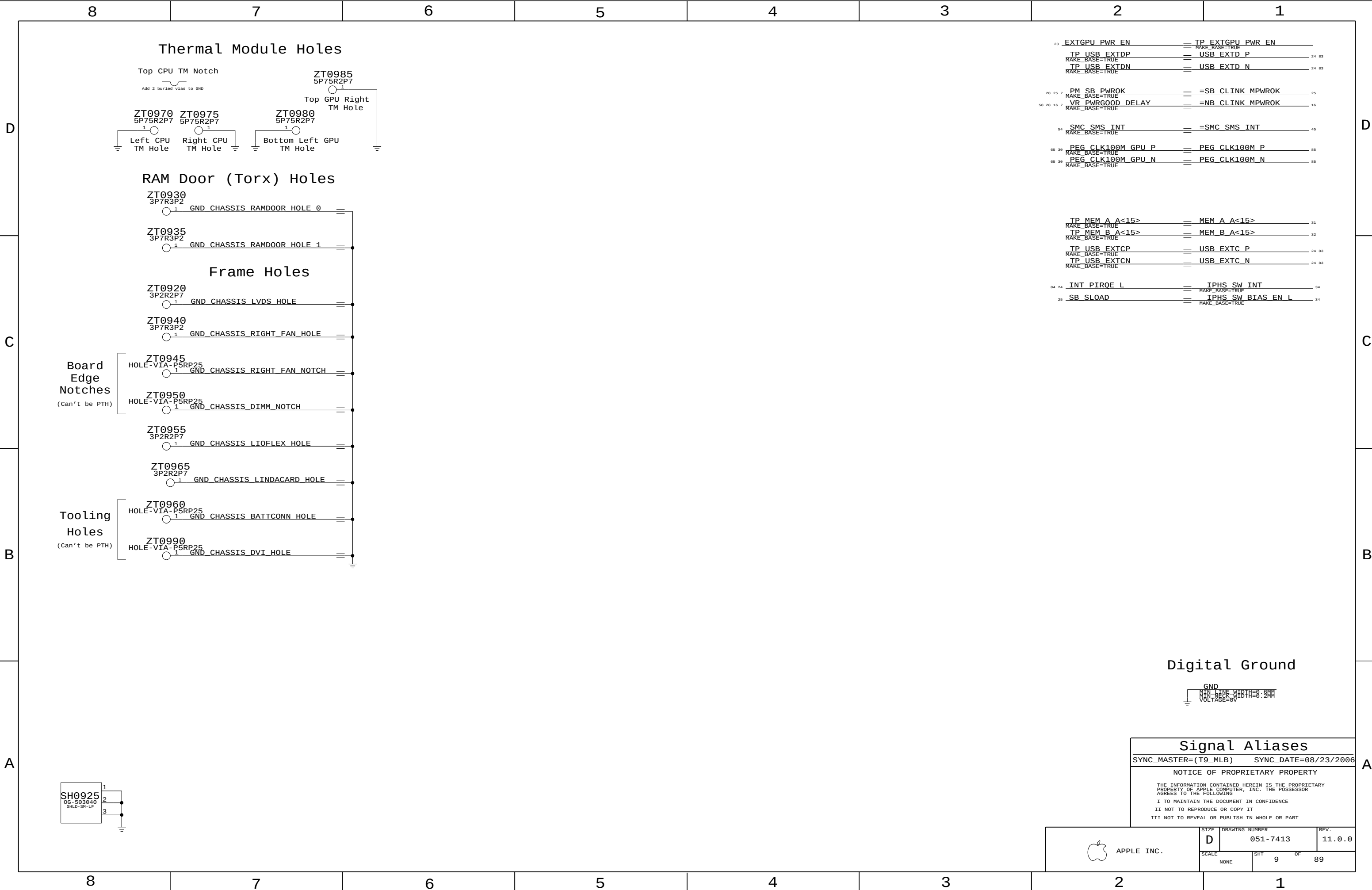
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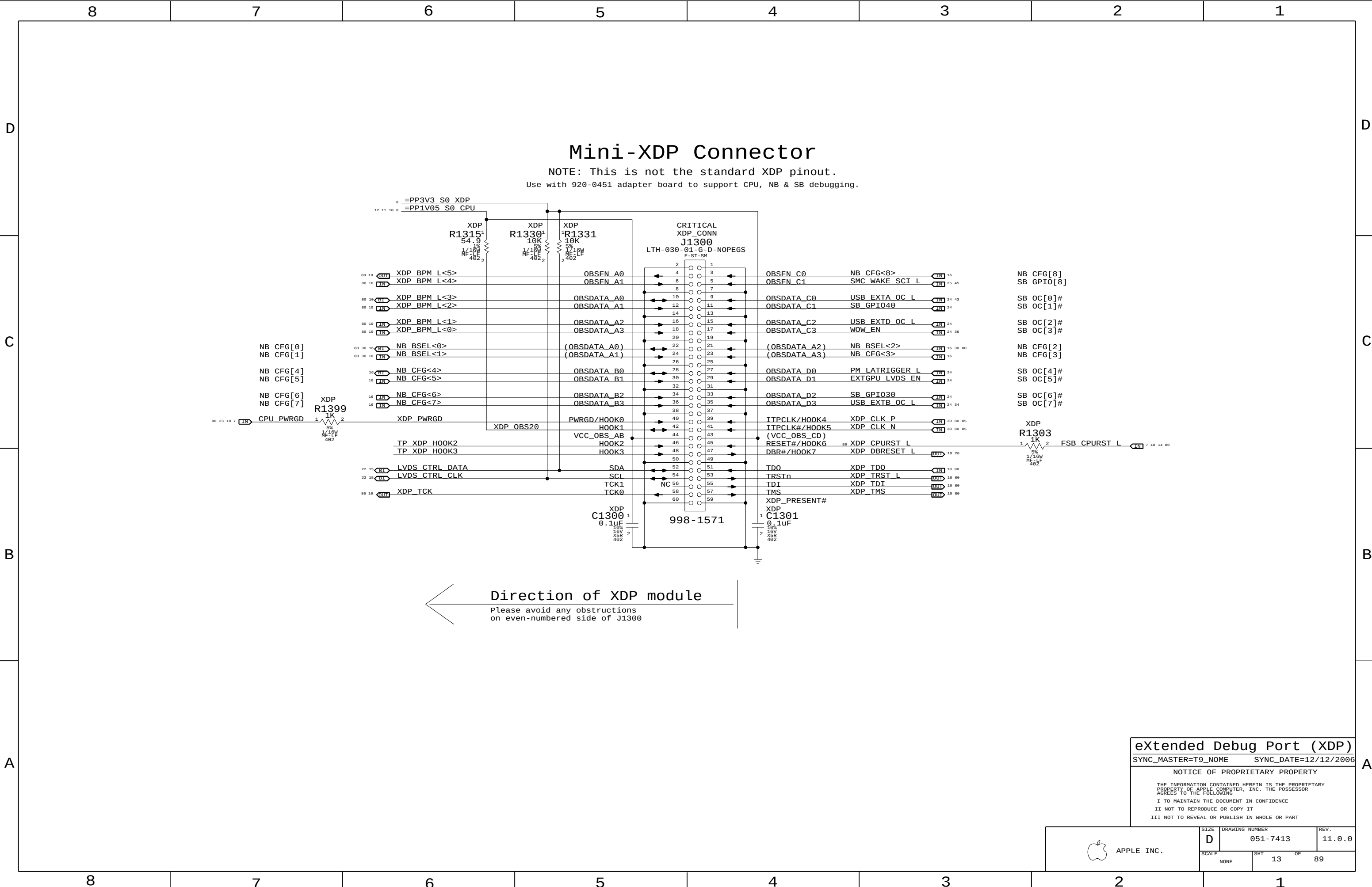
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SIZE D	DRAWING NUMBER 051-7413	REV. 11.0.0
SCALE NONE	SHT 5 OF 89	

[illegible]







Mini-XDP Connector

NOTE: This is not the standard XDP pinout.
Use with 920-0451 adapter board to support CPU, NB & SB debugging.

Direction of XDP module

Please avoid any obstructions
on even-numbered side of J1300

eXtended Debug Port (XDP)

SYNC_MASTER=T9_NOME SYNC_DATE=12/12/2006

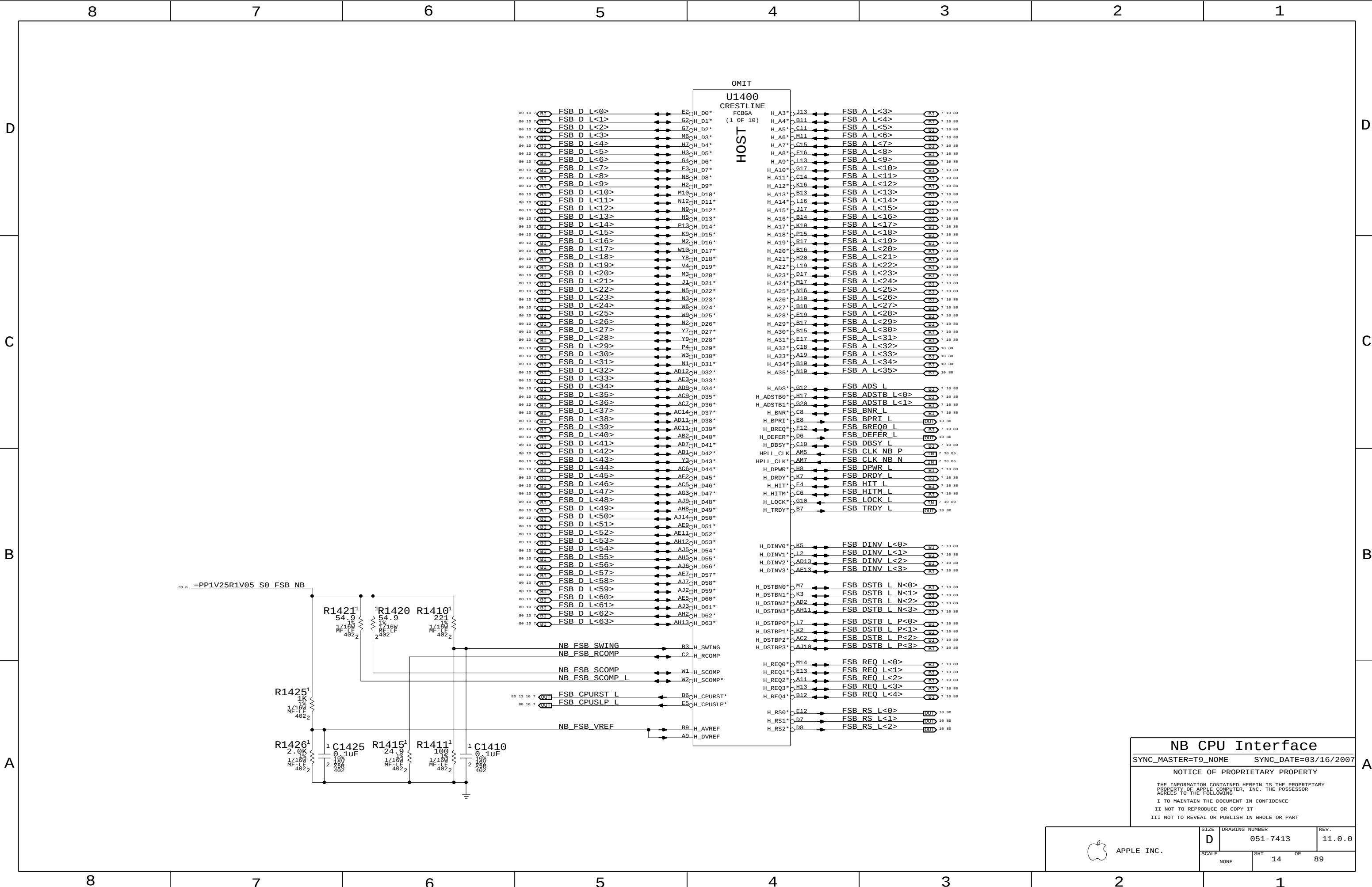
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NONE	13	89



NB CPU Interface

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SYNC_DATE=03/16/2007

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D

C

B

A

LVDS Disable

Can leave all signals NC if LVDS is not implemented.
Tie VCC_TX_LVDS and VCCA_LVDS to GND.

If SDVO is used, VCCD_LVDS must remain powered with proper decoupling. Otherwise, tie VCCD_LVDS to GND also.

Note: SR DG says to tie LVDS_VREFH/L to GND. This causes a glitch during wake-up on LVDS DATA/CLK pairs. New recommendation is to float both signals, see Radar #5067636.

TV-Out Signal Usage:

Composite: DACA only
S-Video: DACB & DACC only
Component: DACA, DACB & DACC

Unused DAC outputs must remain powered, but can omit filtering components. Unused DAC outputs should connect to GND through 75-ohm resistors.

TV-Out Disable / CRT Enable

Tie TVx_DAC and TVx_RTN to GND. Must power all TVDAC rails. VCCA_TVx_DAC and VCCA_DAC_BG can share filtering with VCCA_CRT_DAC.

CRT Disable / TV-Out Enable

Tie R/R#/G/G#/B/B#, HSYNC and VSYNC to GND. All CRT/TVDAC rails must be powered. All rails must be filtered except for VCCA_CRT.

CRT & TV-Out Disable

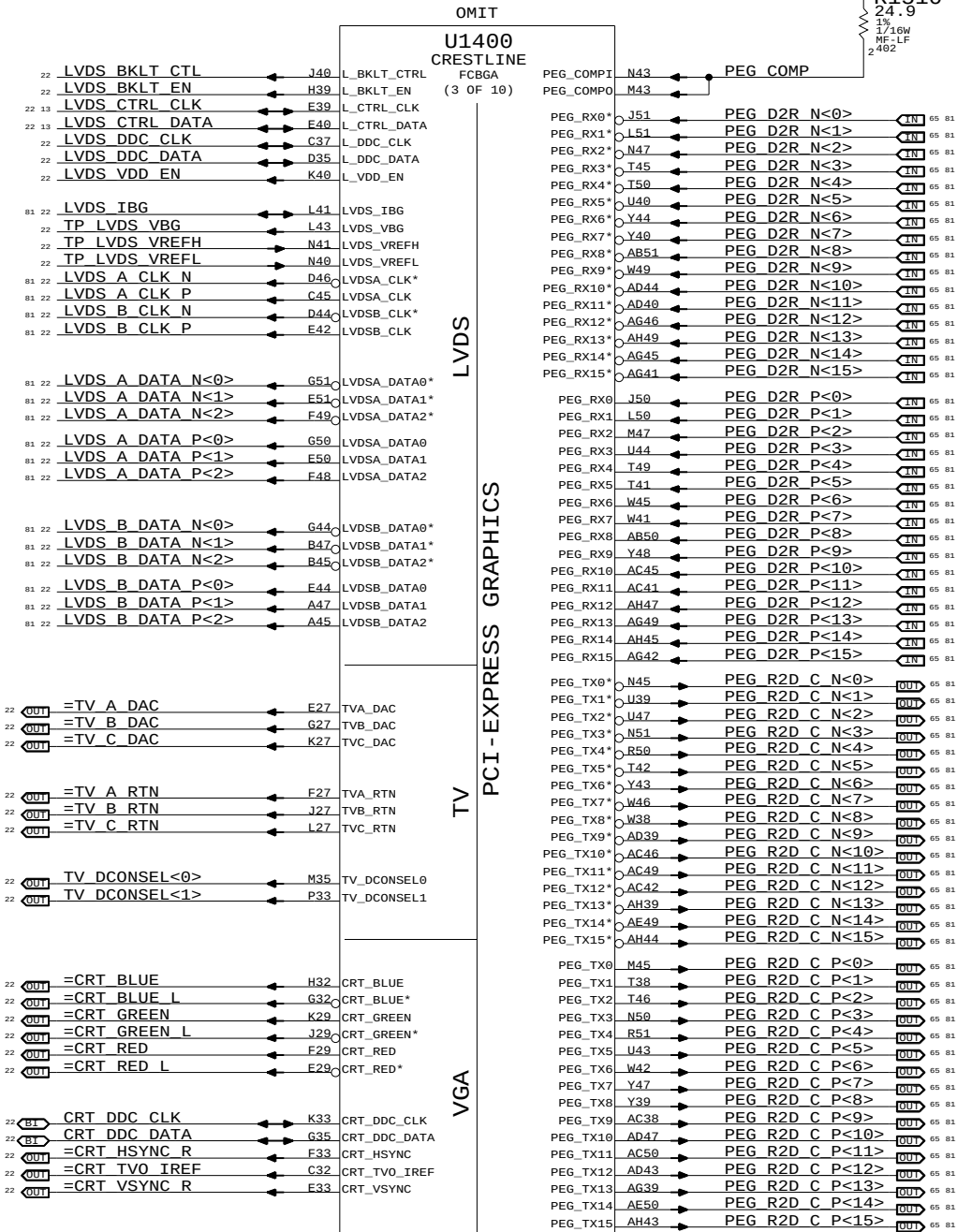
Tie TVx_DAC, TVx_RTN, R/R#/G/G#/B/B#, HSYNC, VSYNC and CRT_TV0_IREF to GND.
Can tie the following rails to GND:
VCCA_CRT_DAC, VCCA_DAC_BG, VCCA_TVx_DAC, VCCD_CRT, VCCD_QDAC and VCC_SYNC.

NOTE: Must keep VDDC_TVDAC powered and filtered at all times!

Internal Graphics Disable

Follow instructions for LVDS and CRT & TV-Out Disable above. Can also tie CRT_DDC_*, L_CTRL_*, L_DDC_*, SDVO_CTRL_* and TV_DCONSELx to GND.

Tie DPLL_REF_CLK and DPLL_REF_SSCLK to GND.
Tie DPLL_REF_CLK* and DPLL_REF_SSCLK* to VCC (VCore).
Tie VCCA_DPLLA and VCCA_DPLLB to VCC (VCore).
Tie VCC_AXG and VCC_AXG_NCTF to GND.
Leave GFX_VID<3..0> and GFX_VR_EN as NC.



SDVO Alternate Function

SDVO_TVCLKIN#
SDVO_INT#
SDVO_FLDSTALL#

SDVO_TVCLKIN
SDVO_INT
SDVO_FLDSTALL

SDVOB_RED#
SDVOB_GREEN#
SDVOB_BLUE#
SDVOB_CLKN
SDVOC_RED#
SDVOC_GREEN#
SDVOC_BLUE#
SDVOC_CLKN

SDVOB_RED
SDVOB_GREEN
SDVOB_BLUE
SDVOB_CLKP
SDVOC_RED
SDVOC_GREEN
SDVOC_BLUE
SDVOC_CLKP

NB PEG / Video Interfaces

SYNC_MASTER=T9_NAME SYNC_DATE=03/16/2007

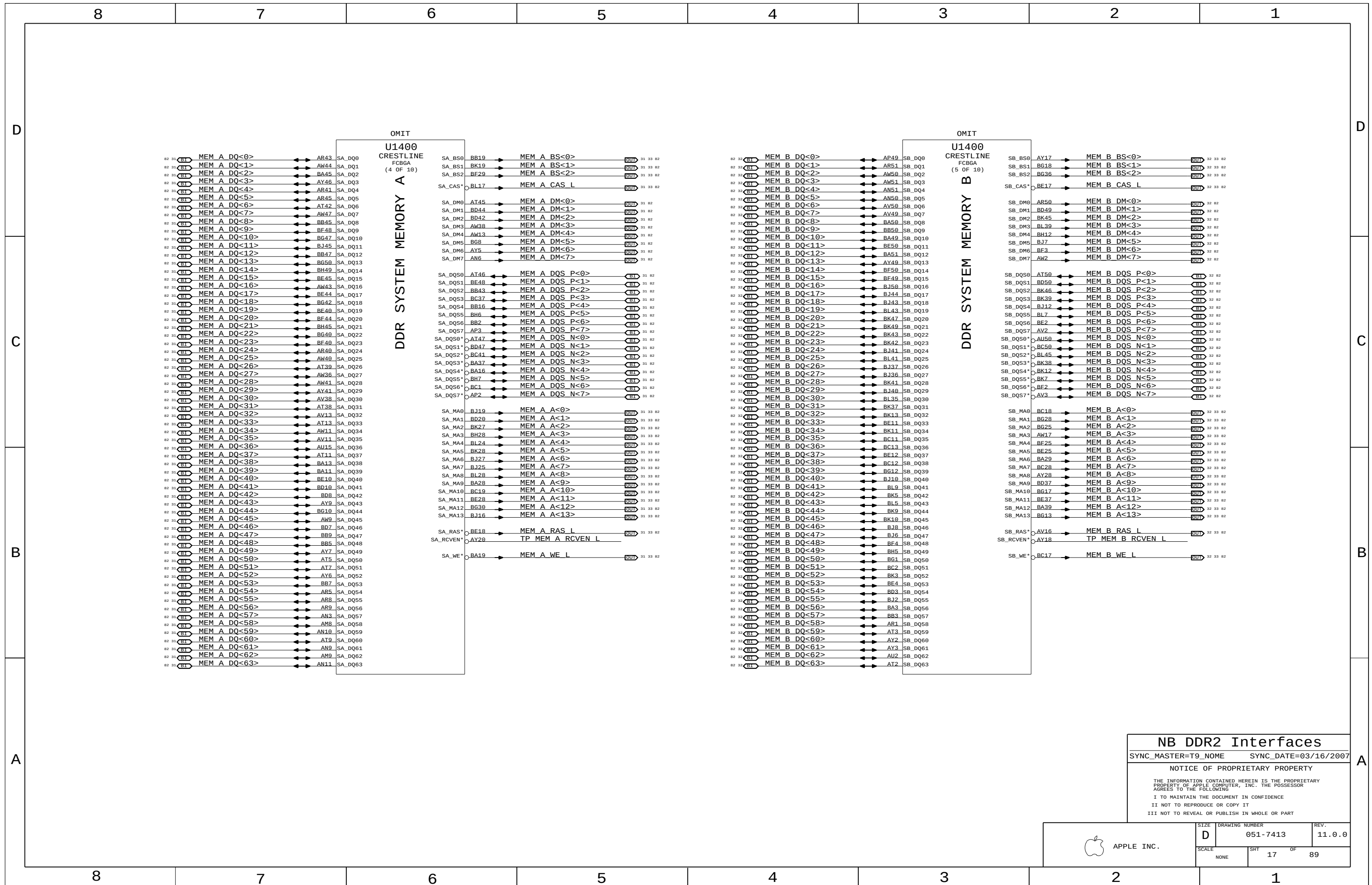
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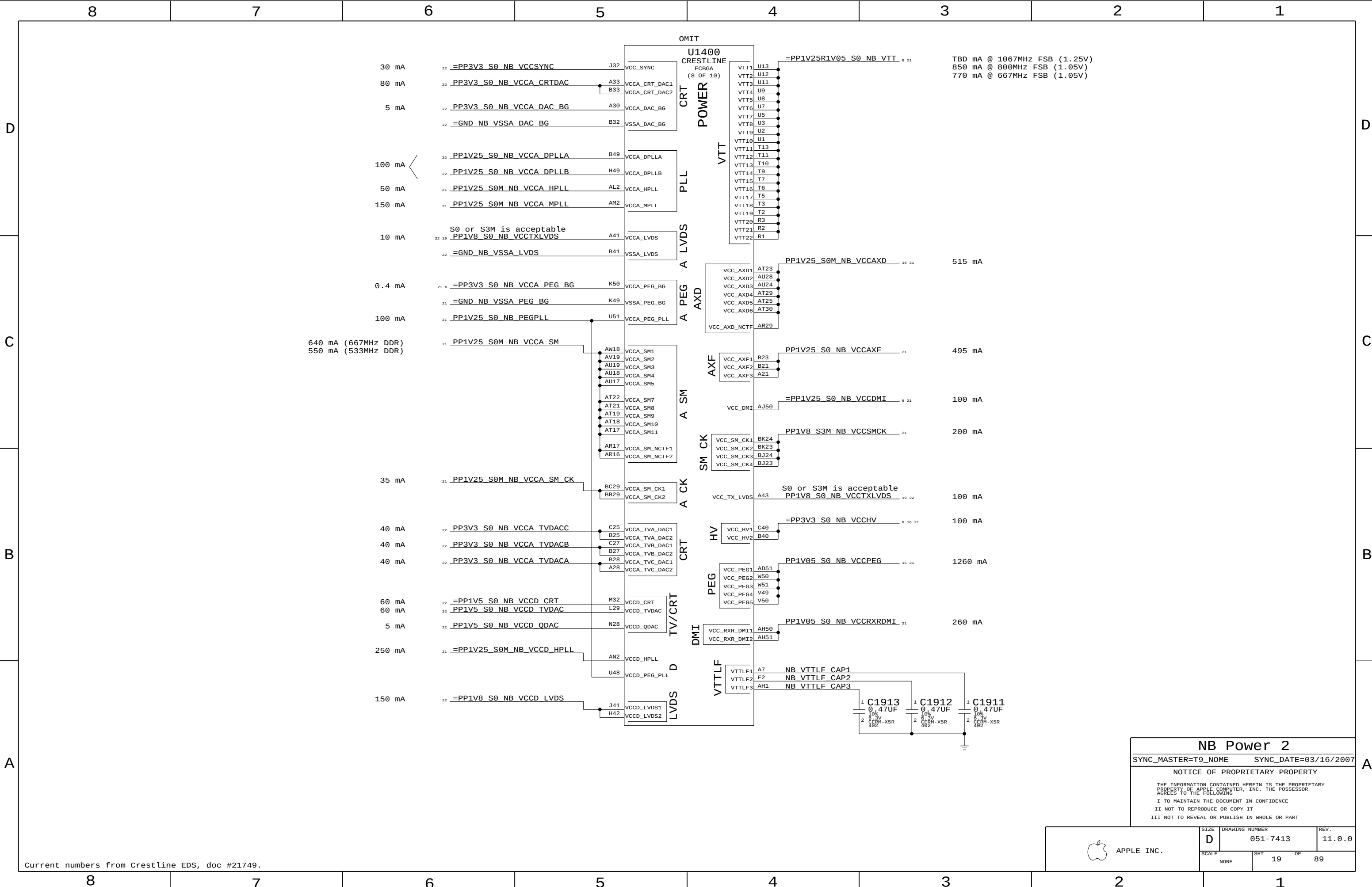
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NONE	15	89





NB Power 2

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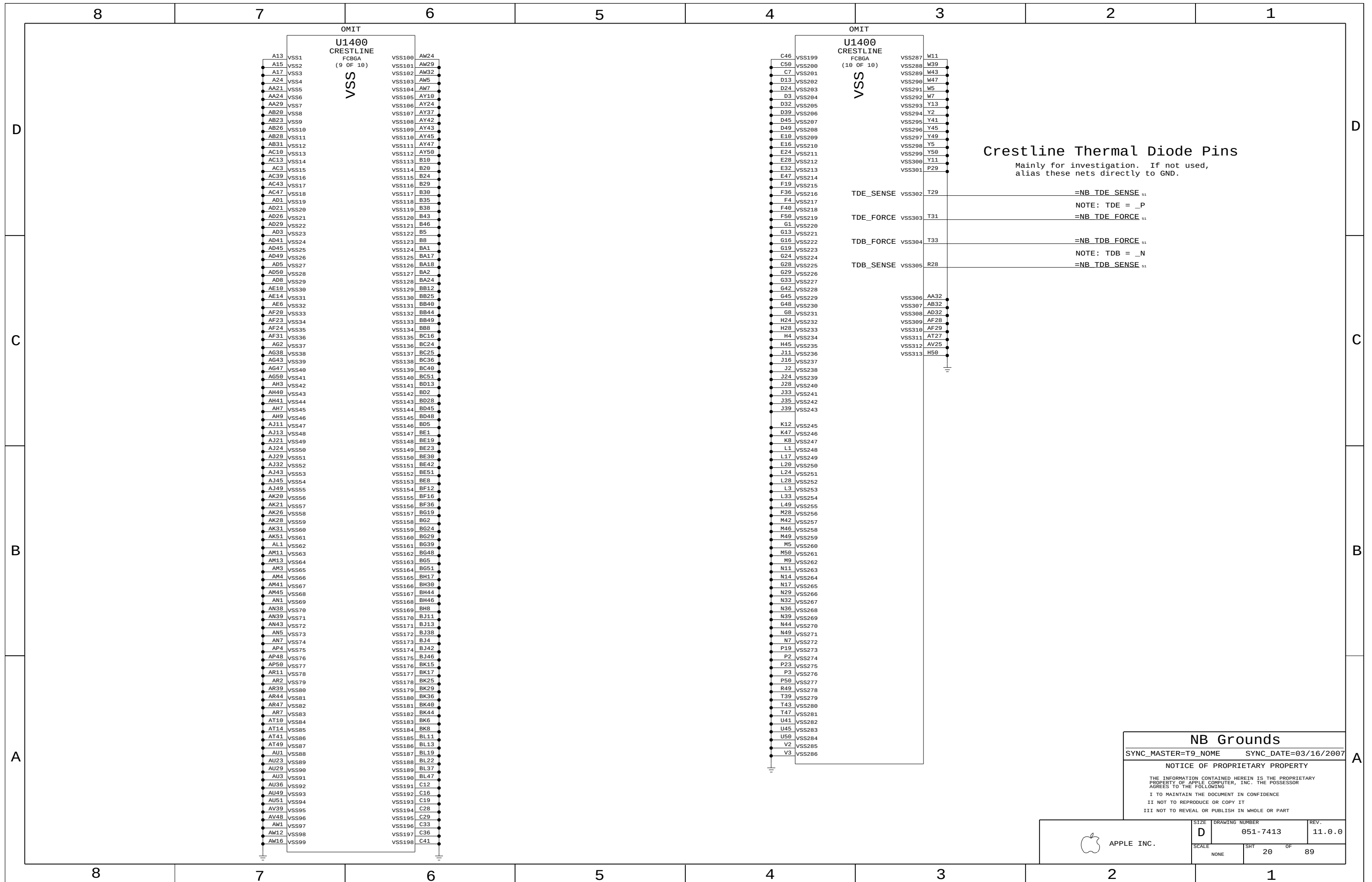
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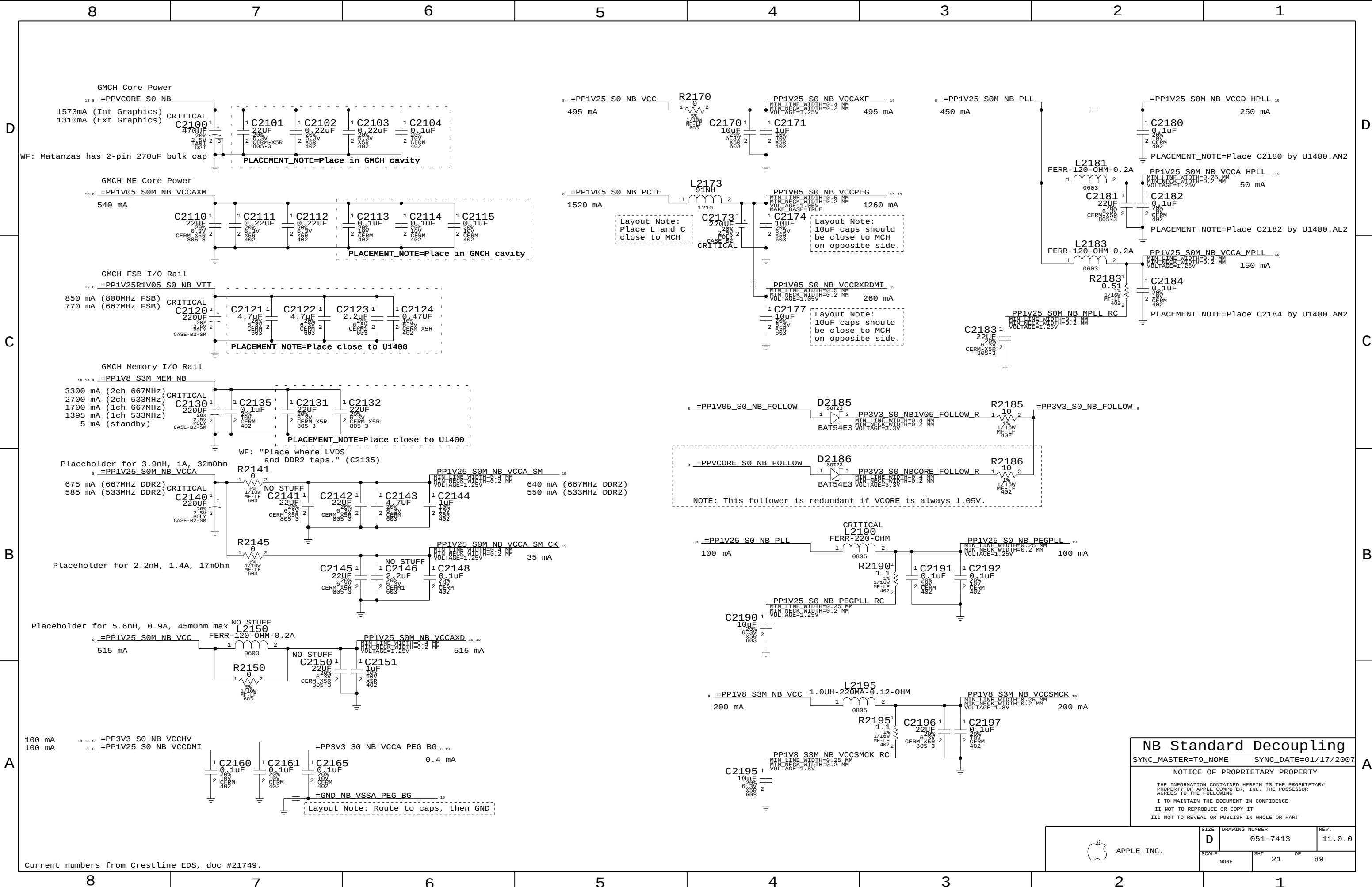
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SCALE		SHT	OF
NONE		19	89





NB Standard Decoupling

SYNC_MASTER=T9_NOME SYNC_DATE=01/17/2007

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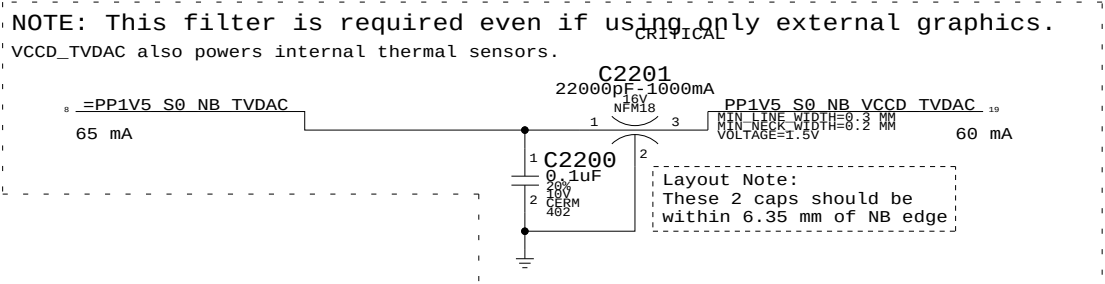


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SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	21	89

Current numbers from Crestline EDS, doc #21749.

Crestline LVDS Strapping



Layout Note:
These 2 caps should be
within 6.35 mm of NB edge

15	LVDS_DDC_CLK	
15	LVDS_DDC_DATA	
15 13	LVDS_CTRL_CLK	
15 13	LVDS_CTRL_DATA	
15	=CRT_RED	
15	=CRT_RED_L	
15	=CRT_GREEN	
15	=CRT_GREEN_L	
15	=CRT_BLUE	
15	=CRT_BLUE_L	
15	=CRT_HSYNC_R	
15	=CRT_VSYNC_R	
15	=CRT_TV0_IREF	
15		
15	=TV_A_DAC	
15	=TV_A_RTN	
15	=TV_B_DAC	
15	=TV_B_RTN	
15	=TV_C_DAC	
15	=TV_C_RTN	
15		
15	CRT_DDC_CLK	
15	CRT_DDC_DATA	
15	SDVO_CTRLCLK	
15	SDVO_CTRLDATA	
15	TV_DCONSEL<0>	
15	TV_DCONSEL<1>	
15		
16	=NB_CLK96M_DOT_N	
16	=NB_CLK96M_DOT_P	
16	=NB_CLK100M_DPLLSS_P	
16	=NB_CLK100M_DPLLSS_N	

```

=GNB_NB_VSSA_LVDS 19
PP1V8_S0_NB_VCCTXLVDS 19
PP1V25_S0_NB_VCCA_DPLL 19
PP1V25_S0_NB_VCCA_DPLLA 19
=PP1V8_S0_NB_VCCD_LVDS 19
=PP1V5_S0_NB_VCCD_CRT 19
PP3V3_S0_NB_VCCA_CRTDAC 19
=PP3V3_S0_NB_VCCSYNC 19
PP1V5_S0_NB_VCCD_QDAC 19
PP3V3_S0_NB_VCCA_DAC_BG 19
=GNB_NB_VSSA_DAC_BG 19
PP3V3_S0_NB_VCCA_TVADACA 19
PP3V3_S0_NB_VCCA_TVADACB 19
PP3V3_S0_NB_VCCA_TVADACC 19

```

15	<u>LVDS BKLT CTL</u>	==	NC LVDS BKLT CTL	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS BKLT EN</u>	==	NC LVDS BKLT EN	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS VDD EN</u>	==	NC LVDS VDD EN	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS IBG</u>	==	NC LVDS IBG	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>TP LVDS VBG</u>	==	NC LVDS VBG	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A CLK N</u>	==	NC LVDS A CLKN	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A CLK P</u>	==	NC LVDS A CLKP	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B CLK N</u>	==	NC LVDS B CLKN	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B CLK P</u>	==	NC LVDS B CLKP	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A DATA N<0></u>	==	NC LVDS A DATAN<0>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A DATA N<1></u>	==	NC LVDS A DATAN<1>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A DATA N<2></u>	==	NC LVDS A DATAN<2>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A DATA P<0></u>	==	NC LVDS A DATAP<0>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A DATA P<1></u>	==	NC LVDS A DATAP<1>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS A DATA P<2></u>	==	NC LVDS A DATAP<2>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B DATA N<0></u>	==	NC LVDS B DATAN<0>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B DATA N<1></u>	==	NC LVDS B DATAN<1>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B DATA N<2></u>	==	NC LVDS B DATAN<2>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B DATA P<0></u>	==	NC LVDS B DATAP<0>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B DATA P<1></u>	==	NC LVDS B DATAP<1>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>LVDS B DATA P<2></u>	==	NC LVDS B DATAP<2>	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>TP LVDS VREFH</u>	==	NC LVDS VREFH	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>TP LVDS VREFL</u>	==	NC LVDS VREFL	==	MAKE_BASE=TRUE	NO_TEST=TRUE
15	<u>GFX VID<1></u>	==	TP GFX VID<1>	==	MAKE_BASE=TRUE	
15	<u>GFX VID<2></u>	==	TP GFX VID<2>	==	MAKE_BASE=TRUE	
15	<u>GFX VID<3></u>	==	TP GFX VID<3>	==	MAKE_BASE=TRUE	
15	<u>GFX VID<4></u>	==	TP GFX VID<4>	==	MAKE_BASE=TRUE	
15	<u>=GFX VR EN</u>	==	TP GFX VR EN	==	MAKE_BASE=TRUE	

NB Graphics Decoupling

SYNC_MASTER=M76_MLB	SYNC_DATE=03/12/2007
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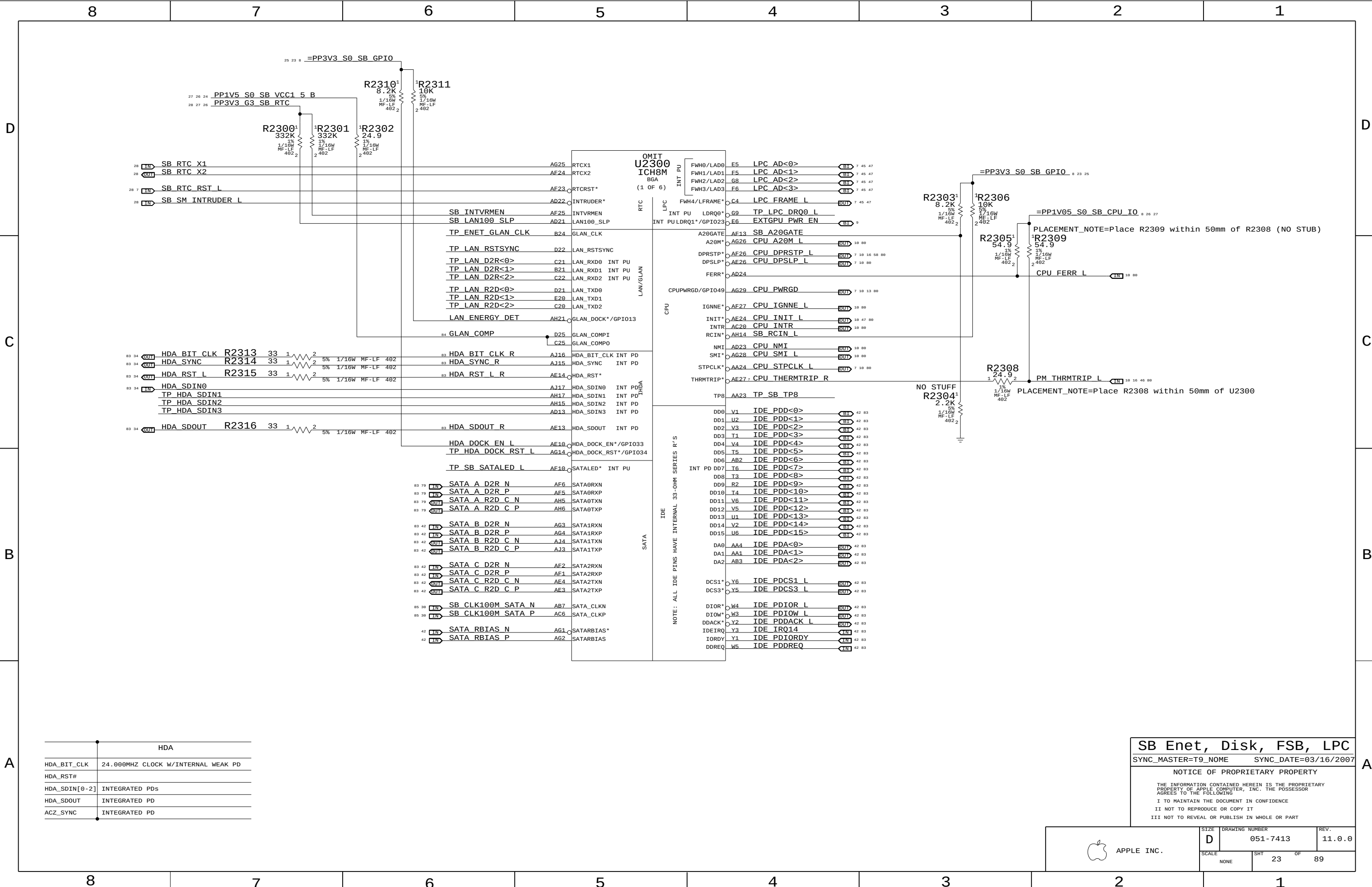
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Current numbers from Crestline EDS Addendum, doc #20127.



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D	051-7413	11.0.0	
SCALE		SHT	OF
NONE		22	89



HDA	
HDA_BIT_CLK	24.000MHZ CLOCK W/INTERNAL WEAK PD
HDA_RST#	
HDA_SDIN[0-2]	INTEGRATED PDS
HDA_SDOUT	INTEGRATED PD
ACZ_SYNC	INTEGRATED PD

SB Enet, Disk, FSB, LPC

SYNC_MASTER=T9_NAME SYNC_DATE=03/16/2007

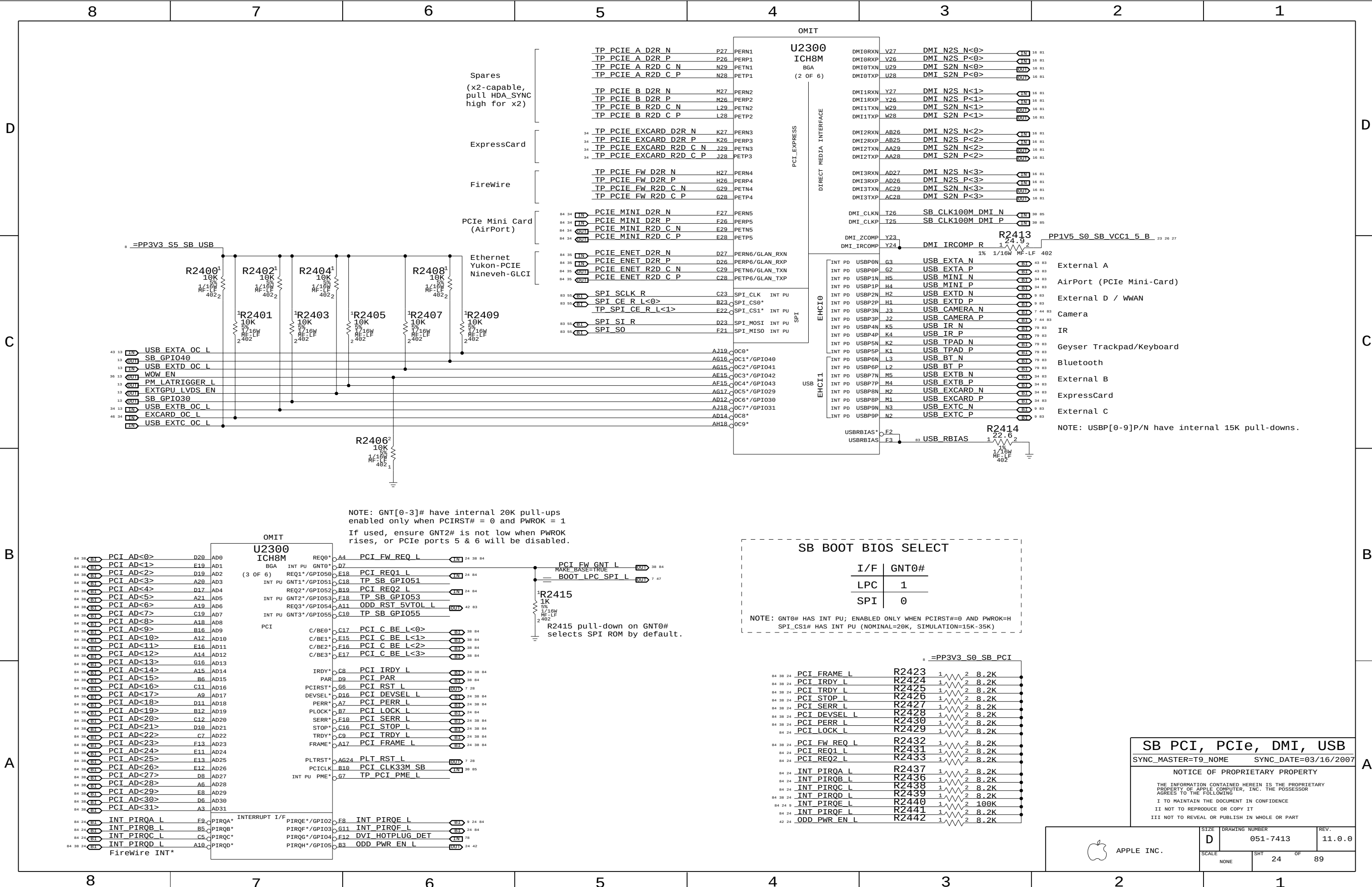
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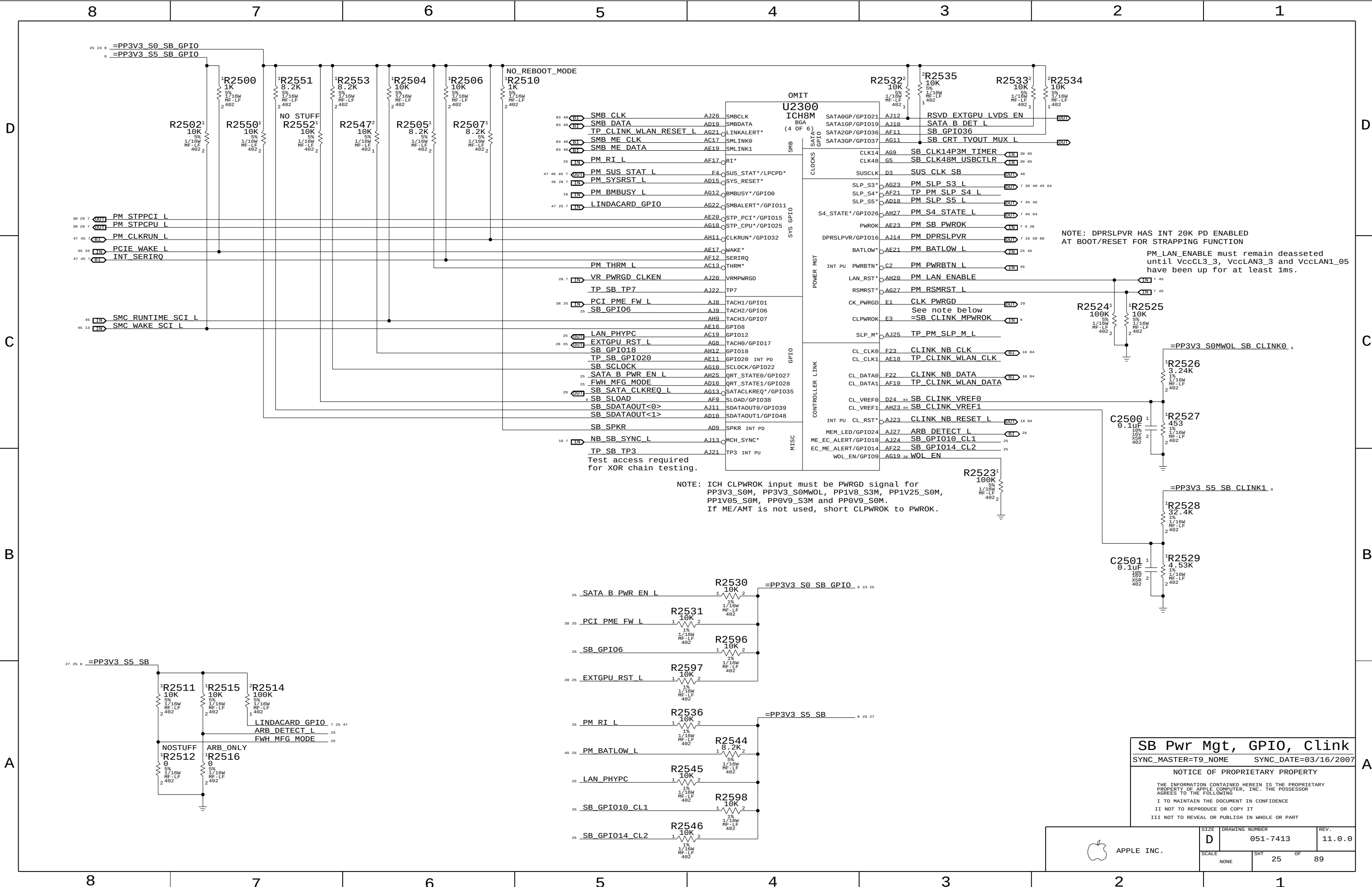


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NONE	23	89



SB PCI, PCIe, DMI, USB		
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APPLE INC.	SIZE D	DRAWING NUMBER 051-7413
	SCALE NONE	SHT 24 OF 89



NOTE: DPRSLPVR HAS INT 20K PD ENABLED AT BOOT/RESET FOR STRAPPING FUNCTION

PM_LAN_ENABLE must remain deasserted until VccCL3_3, VccLAN3_3 and VccLAN1_05 have been up for at least 1ms.

NOTE: ICH CLPWROK input must be PWRGD signal for PP3V3_S0M, PP3V3_S0MWOL, PP1V8_S3M, PP1V25_S0M, PP1V05_S0M, PP0V9_S3M and PP0V9_S0M. If ME/AMT is not used, short CLPWROK to PWROK.

SB Pwr Mgt, GPIO, Clink

SYNC_MASTER=T9_NOME SYNC_DATE=03/16/2007

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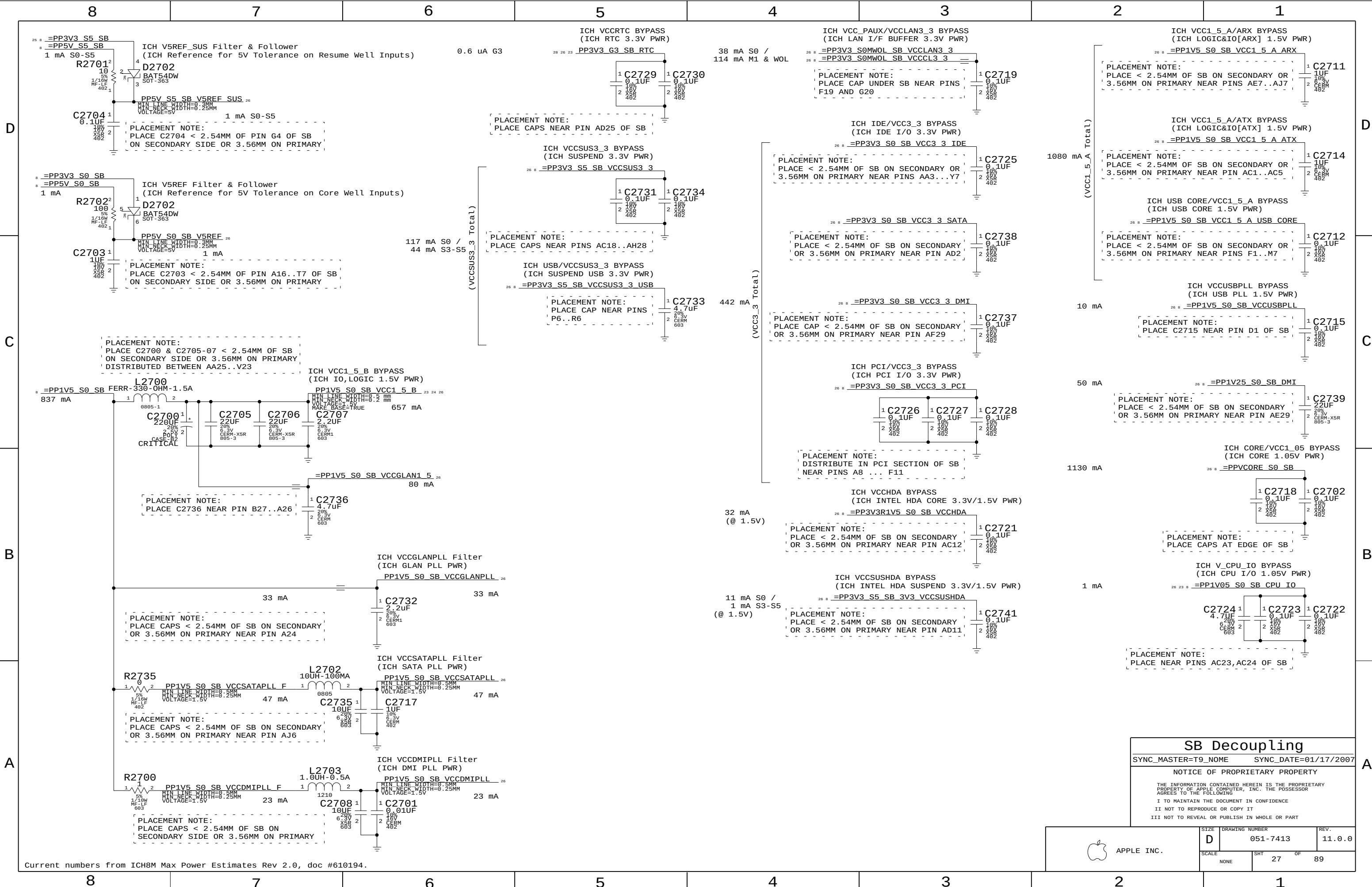
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SCALE	NONE	SHT	25	OF	89



SB Decoupling

SYNC_MASTER=T9_NAME

SYNC_DATE=01/17/2007

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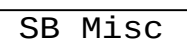
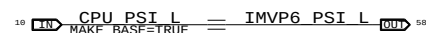
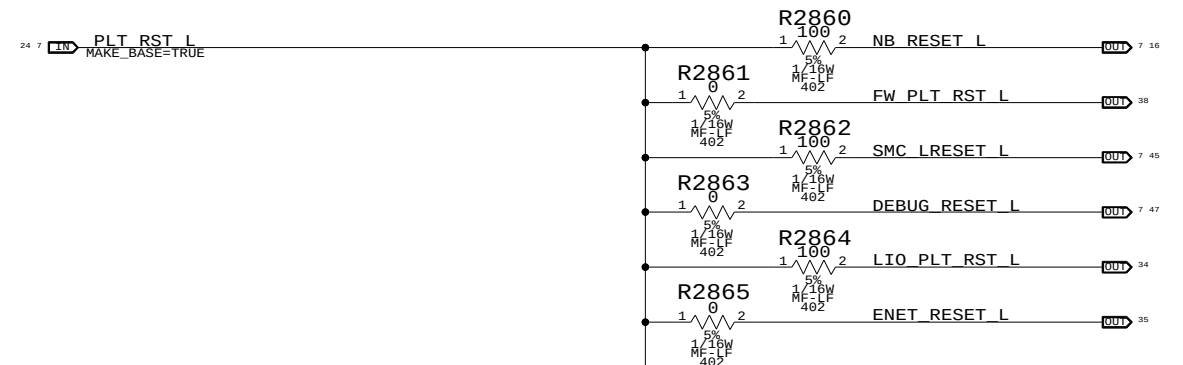
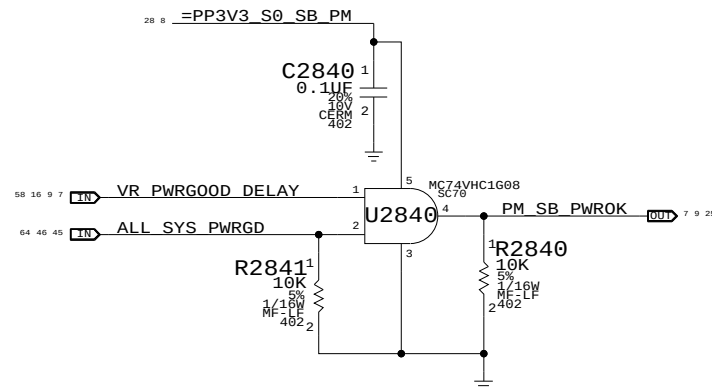
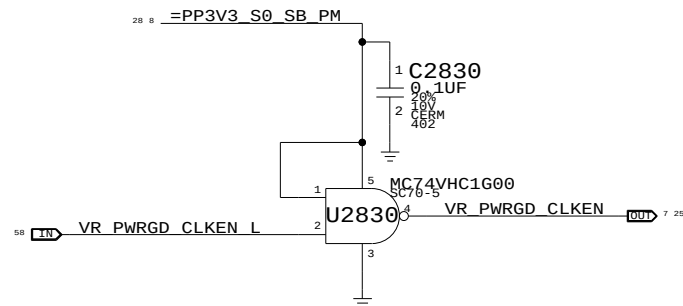
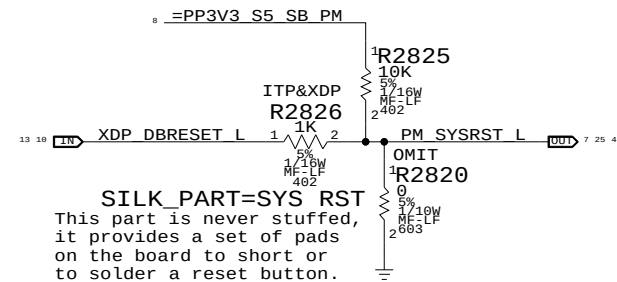
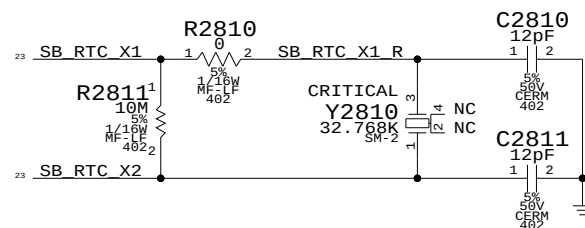
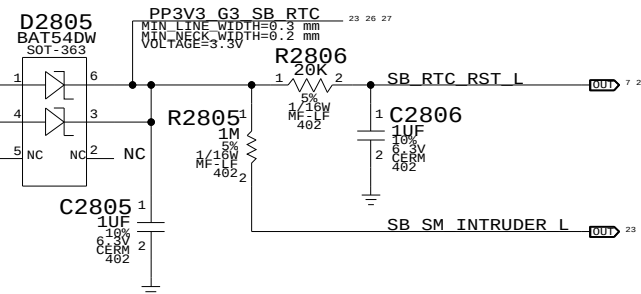
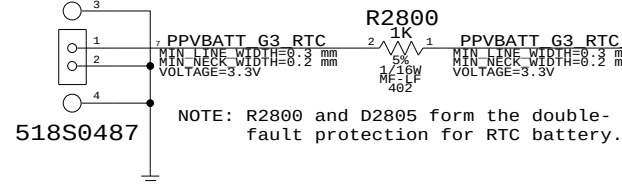
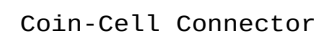
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	D	051-7413	11.0.0
SCALE		SHT	OF
NONE		27	89



SYNC_MASTER=(T9_MLB)	SYNC_DATE=08/24/2006
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APPLE INC.

SIZE
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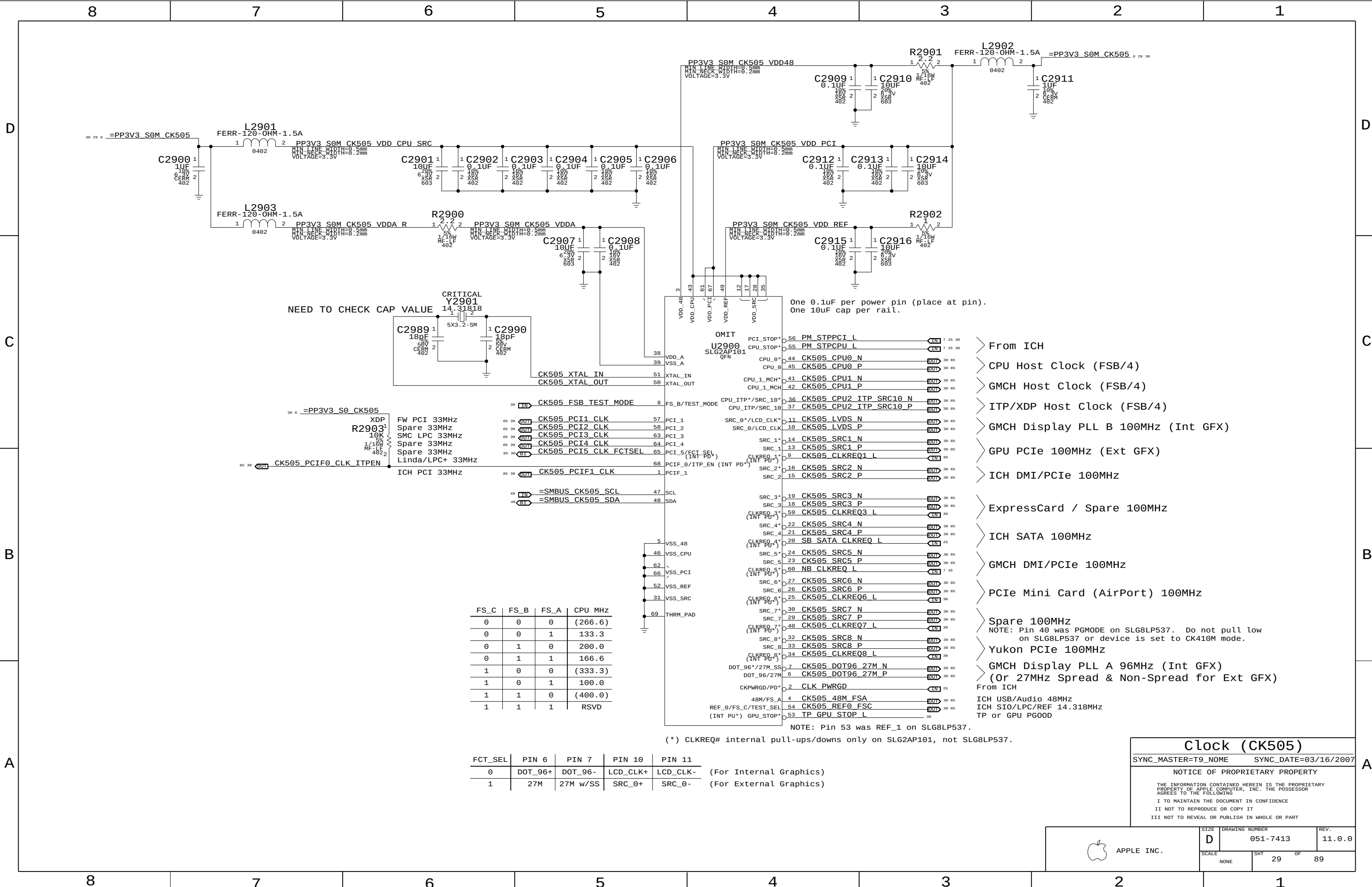
SIZE	DRAWING NUMBER
D	051-741

REV.	
11.0.0	

SCALE	
	NON

SHT	28
-----	----

9



FS_C	FS_B	FS_A	CPU MHz
0	0	0	(266.6)
0	0	1	133.3
0	1	0	200.0
0	1	1	166.6
1	0	0	(333.3)
1	0	1	100.0
1	1	0	(400.0)
1	1	1	RSVD

FCT_SEL	PIN 6	PIN 7	PIN 10	PIN 11
0	DOT_96+	DOT_96-	LCD_CLK+	LCD_CLK-
1	27M	27M w/SS	SRC_0+	SRC_0-

(For Internal Graphics)
(For External Graphics)

Clock (CK505)

SYNC_MASTER=T9_NAME SYNC_DATE=03/16/2007

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SIZE D

DRAWING NUMBER 051-7413

REV. 11.0.0

SCALE NONE

SHT 29 OF 89

D

C

B

A

(Only 100-200MHz supported by
SLG8LP536 and CY28545-5)

(Note: HOST/SRC/GFX clock termination removed. Silago SL8GLP536 or equiv. support only)





Power aliases required by this page:
- =PP1V8_S3M_MEM_A
- =PP0V9_S3M_MEM_DIMMVREFA
- =PPSPD_S0M_MEM_A (2.5V - 3.3V)

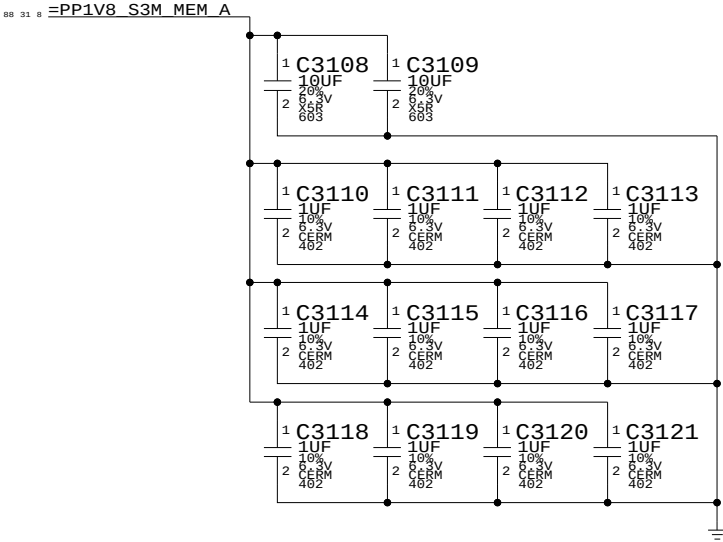
Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

"Factory" (thru-hole) slot

DDR2 Bypass Caps

(For return current)



DDR2 SO-DIMM Connector A
SYNC_MASTER=(M59_SYNC) SYNC_DATE=08/24/2006

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APPLE INC.

SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	31	89



Page Notes

Power aliases required by this page:
- =PP1V8_S3M_MEM_B
- =PP0V9_S3M_MEM_DIMMVREFB
- =PPSPD_S0M_MEM_B (2.5V - 3.3V)

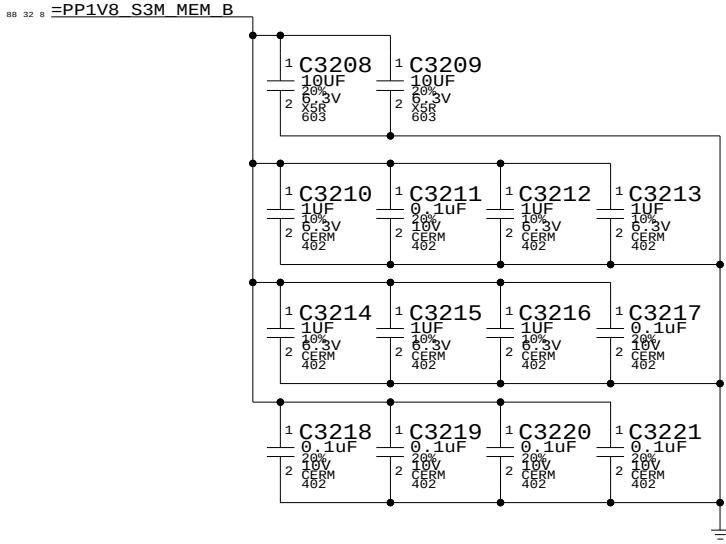
Signal aliases required by this page:
- =I2C_SODIMMB_SCL
- =I2C_SODIMMB_SDA

BOM options provided by this page:
(NONE)

"Expansion" (surface-mount) slot

DDR2 Bypass Caps

(For return current)



DDR2 SO-DIMM Connector B

SYNC_MASTER=(M59_SYNC) SYNC_DATE=08/24/2006

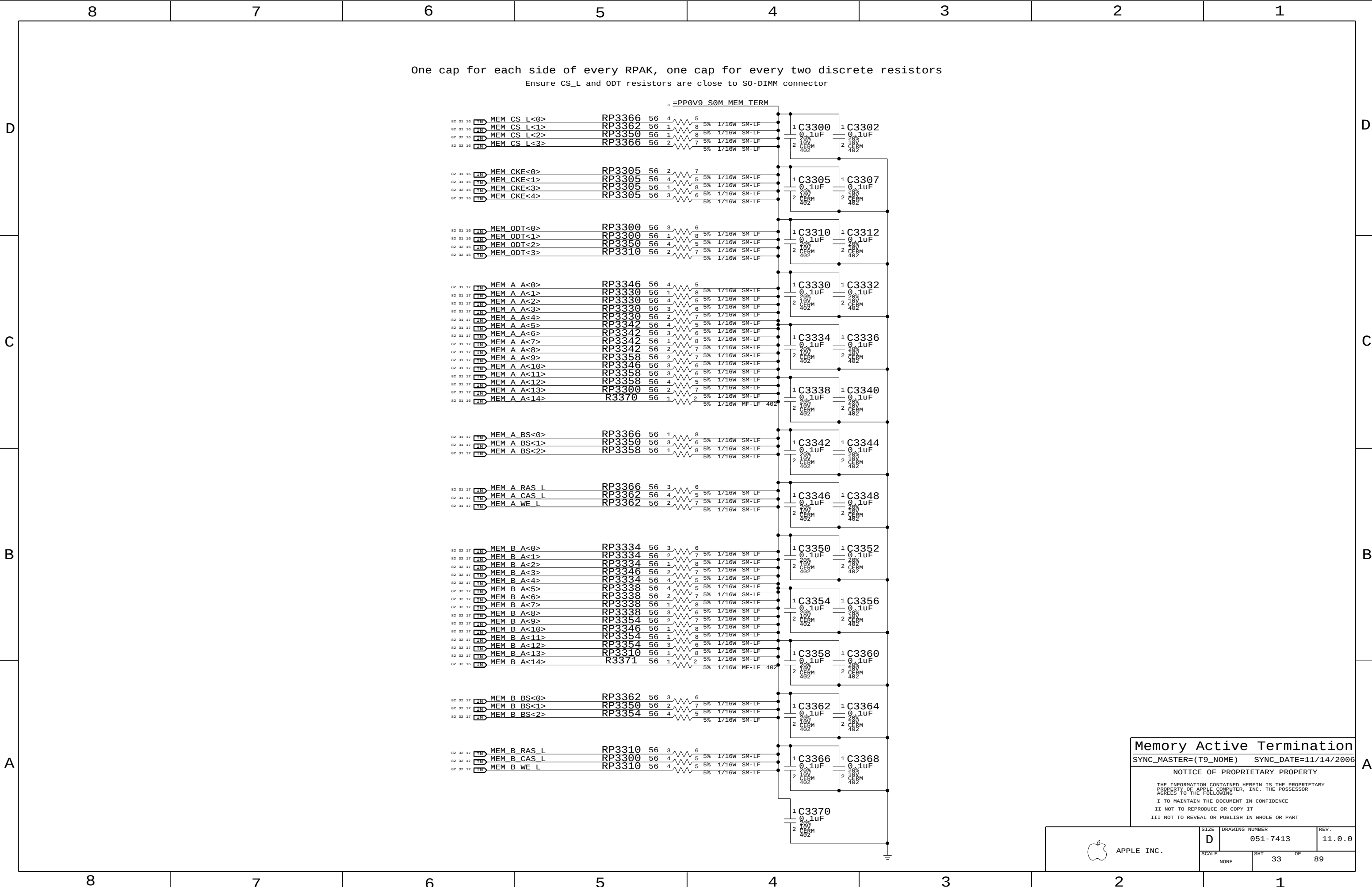
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SCALE	SHT	OF
NONE	32	89



Memory Active Termination

SYNC_MASTER=(T9_NOME) SYNC_DATE=11/14/2006

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SIZE

D

DRAWING NUMBER

051-7413

REV.

11.0.0

SCALE

NONE

SHT

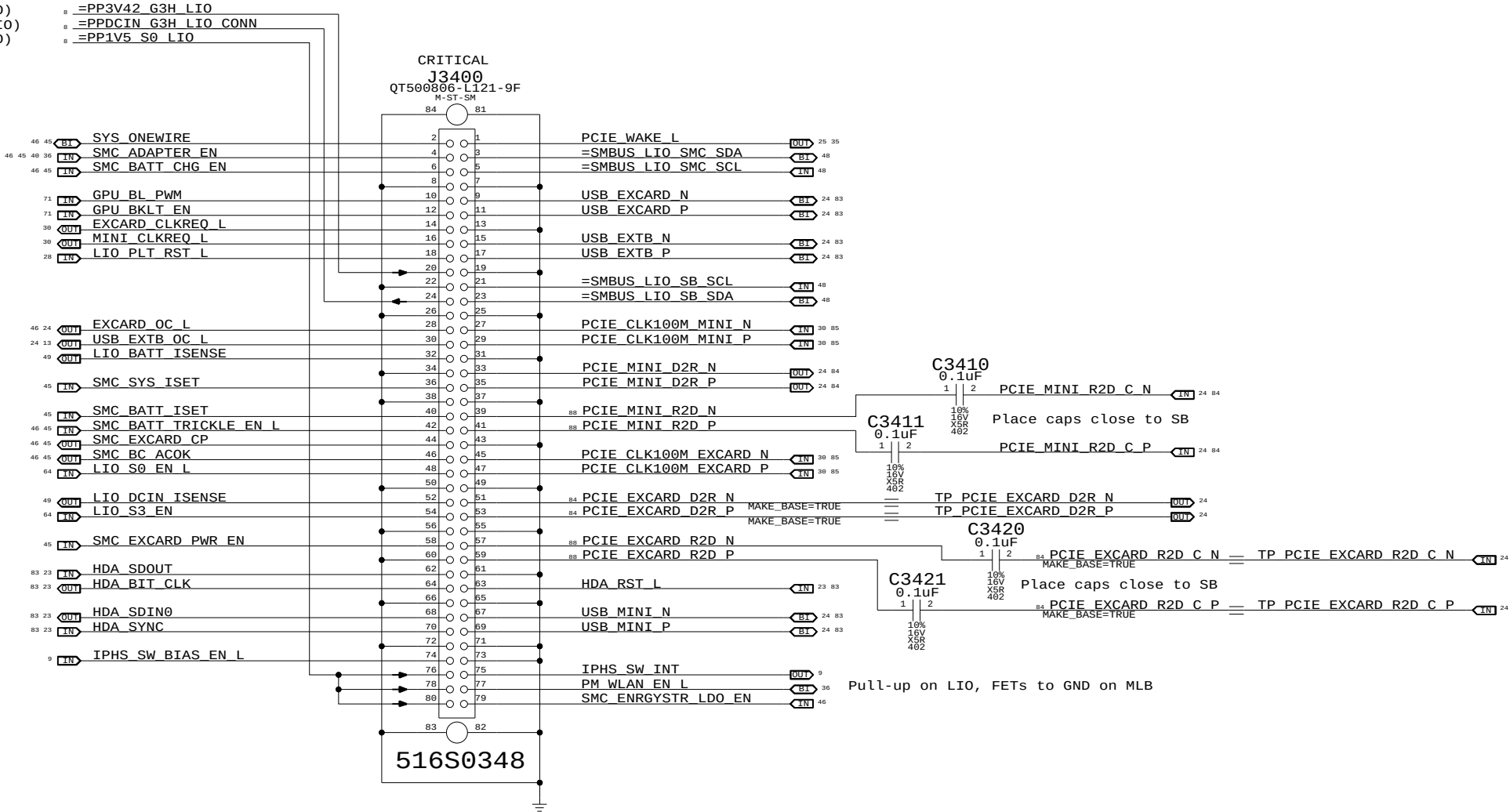
33

OF

89

Left I/O Board Connector

(Output to LIO)
(Input from LIO)
(Output to LIO)



Left I/O Board Connector

SYNC_MASTER=(M59_SYNC) SYNC_DATE=08/24/2006

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D	051-7413	11.0.0
SCALE	SHT	OF
NONE	34	89

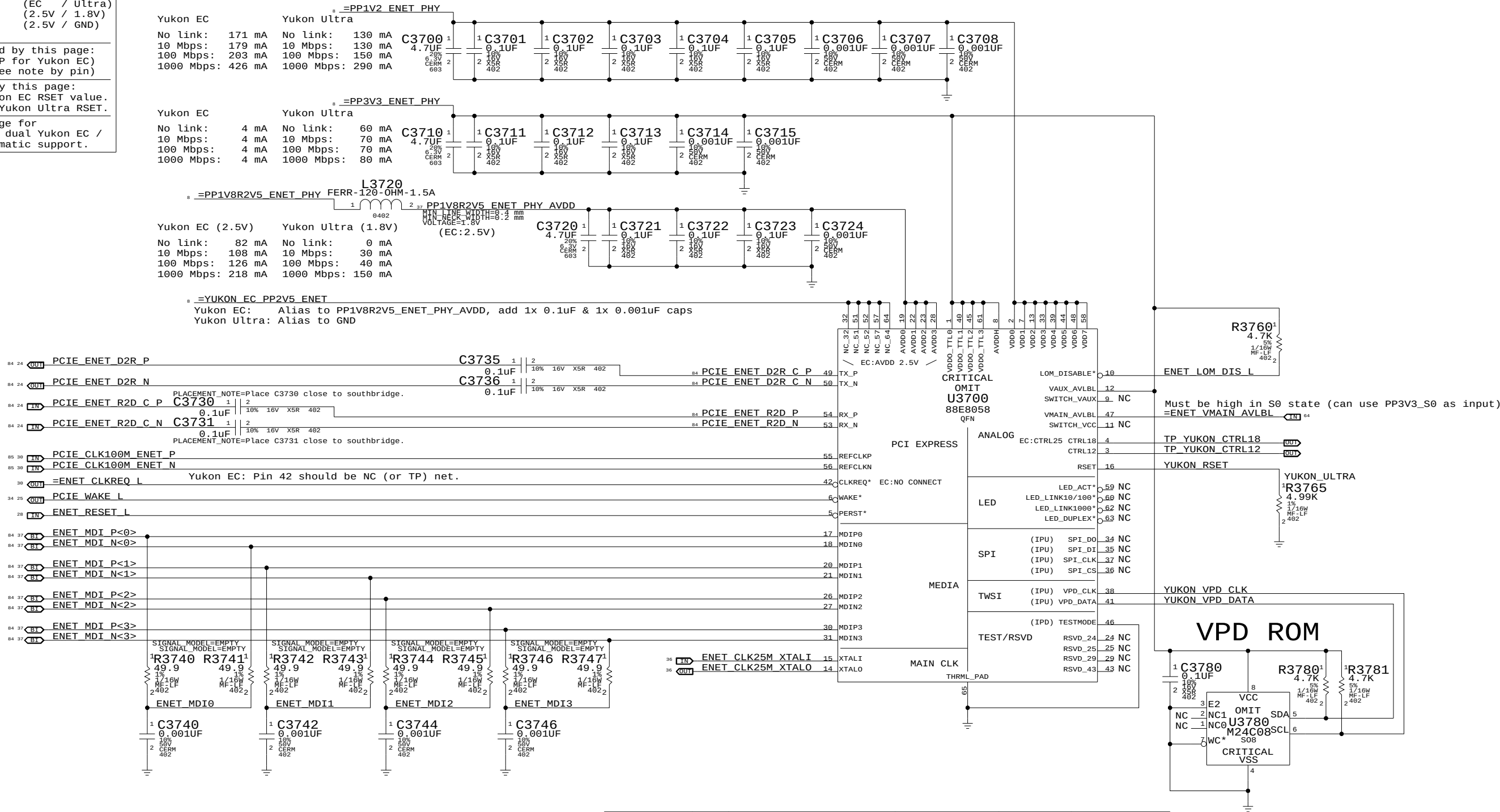
Page Notes

Power aliases required by this page:
- =PP3V3_ENET_PHY (EC / Ultra)
- =PP1V8R2V5_ENET_PHY (2.5V / 1.8V)
- =YUKON_EC_PP2V5_ENET (2.5V / GND)
- =PP1V2_ENET_PHY

Signal aliases required by this page:
- =ENET_CLKREQ_L (NC/TP for Yukon EC)
- =ENET_VMAIN_AVLBL (See note by pin)

BOM options provided by this page:
YUKON_EC - Selects Yukon EC RSET value.
YUKON_ULTRA - Selects Yukon Ultra RSET.

NOTE: See bottom of page for instructions for dual yukon EC / Yukon Ultra schematic support.



To support Yukon EC and Ultra on the same board:

- Alias =YUKON_EC_PP2V5_ENET to PP1V8R2V5_ENET_PHY_AVDD, add 1x 0.1uF and 1x 0.001uF caps
- Use 0-ohm resistors or variable supply to provide 1.8V or 2.5V to =PP1V8R2V5_ENET_PHY and magnetics. Can also use BCP69T1 connected to CTRL18 pin 4 for internal VR.
- Connect =ENET_CLKREQ_L to clock generator via 0-ohm resistor (BOMOPTION: YUKON_ULTRA)
- Use YUKON_EC and YUKON_ULTRA BOMOPTIONS to select stuffed part

Ethernet (Yukon)

SYNC_MASTER=T9_NAME SYNC_DATE=03/16/2007

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D	051-7413	11.0.0
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NONE	35	89

D

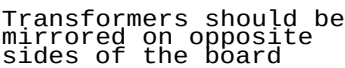
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B



SCALE	SHI	OF
NONE	36	89

Power aliases required by this page: - =GND_CHASSIS_ENET
Signal aliases required by this page: (NONE)
BOM options provided by this page: (NONE)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
157S0053	2	XFMR, ISO, HALF-PORT, 1000T, 16P, SMD, 2MM	T3900, T3901	CRITICAL	

SYNC_MASTER=M76_MLB SYNC_DATE=03/19/2007

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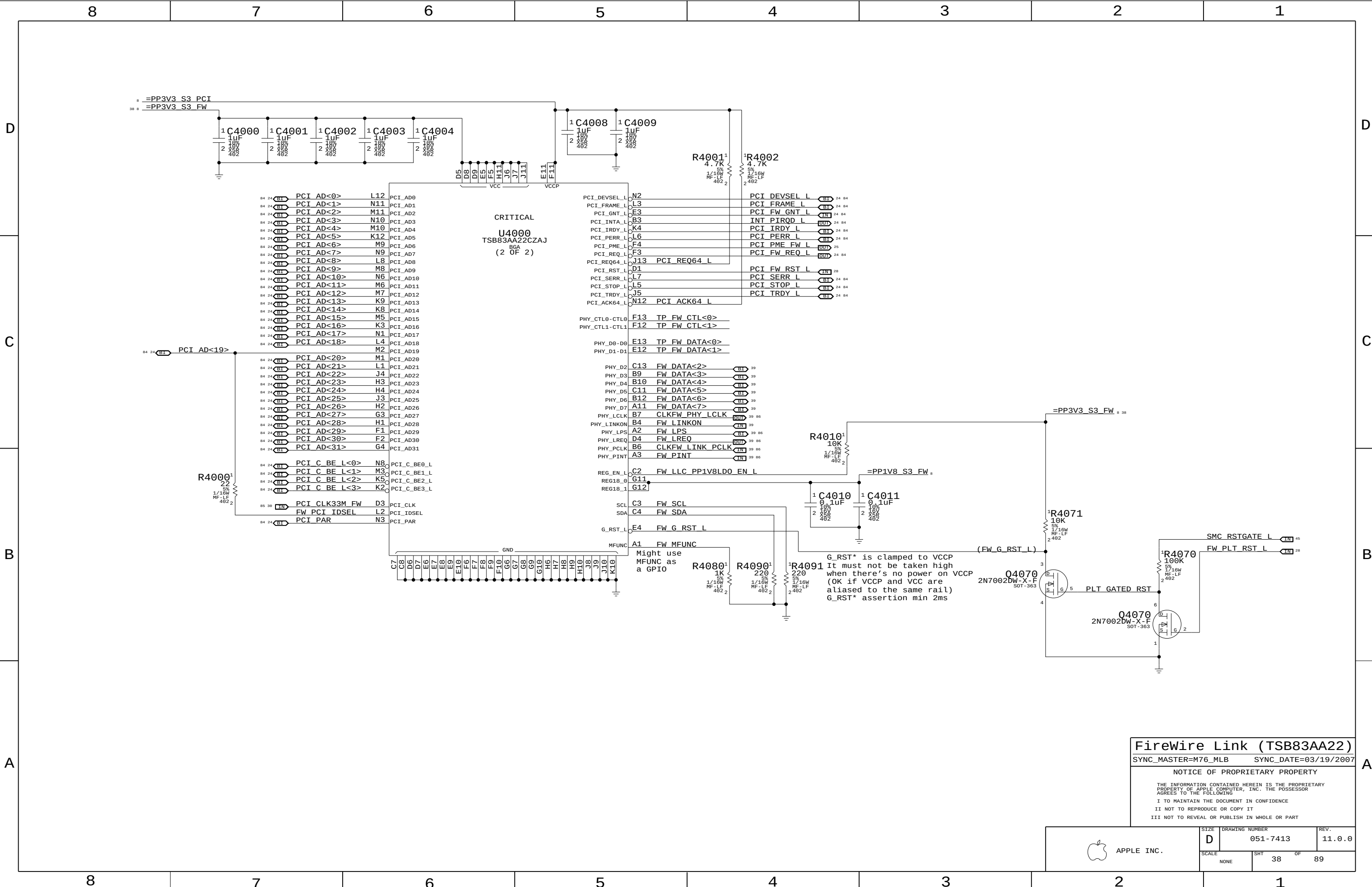
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SIZE D	DRAWING NUMBER 051-7413	REV. 11.0.0
SCALE NONE	SHT 37	OF 89



FireWire Link (TSB83AA22)

SYNC_MASTER=M76_MLB SYNC_DATE=03/19/2007

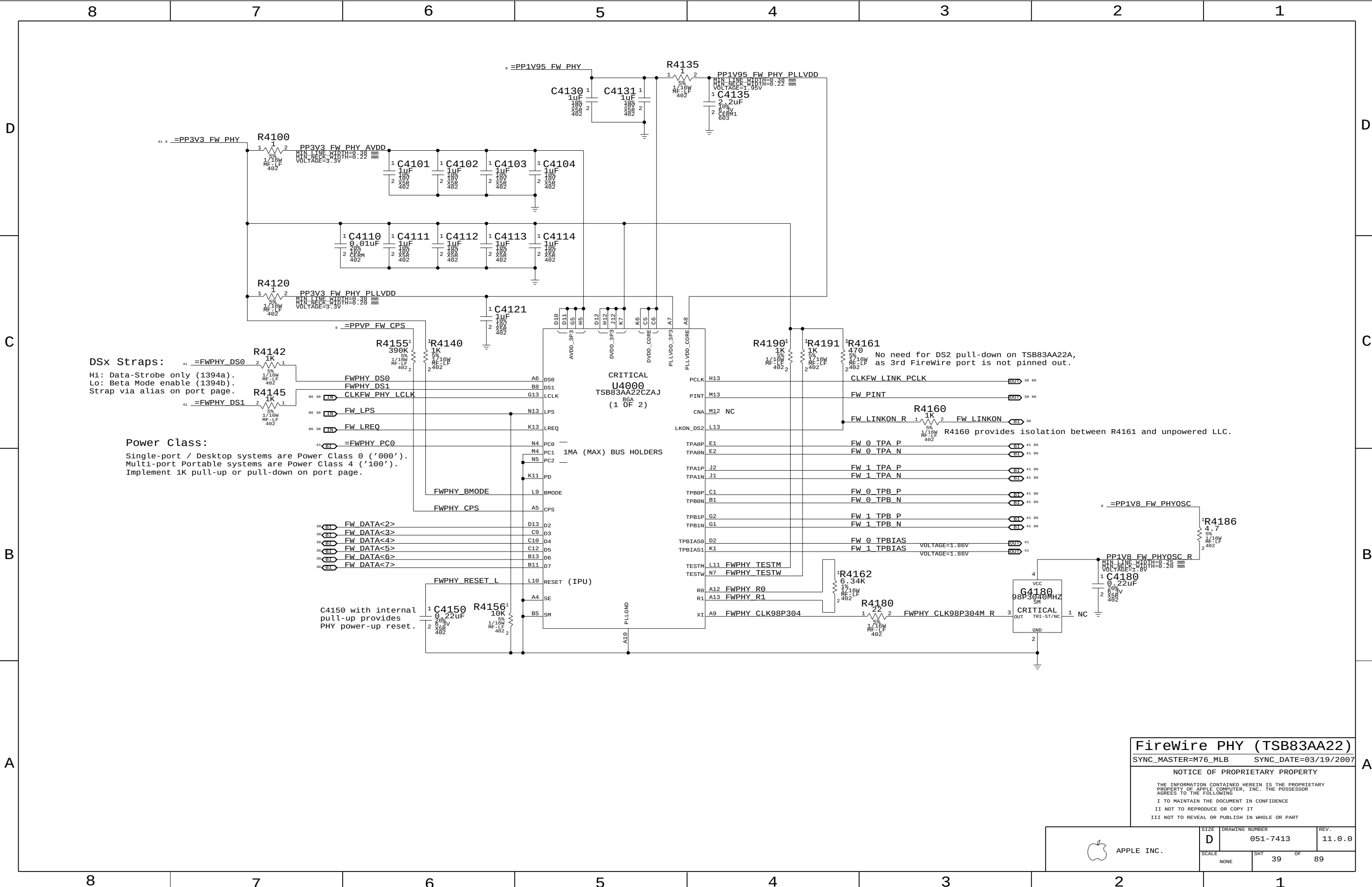
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SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	38	89



8 7 6 5 4 3 2 1

D

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8 7 6 5 4 3 2 1

FireWire PHY (TSB83AA22)

SYNC_MASTER=M76_MLB SYNC_DATE=03/19/2007

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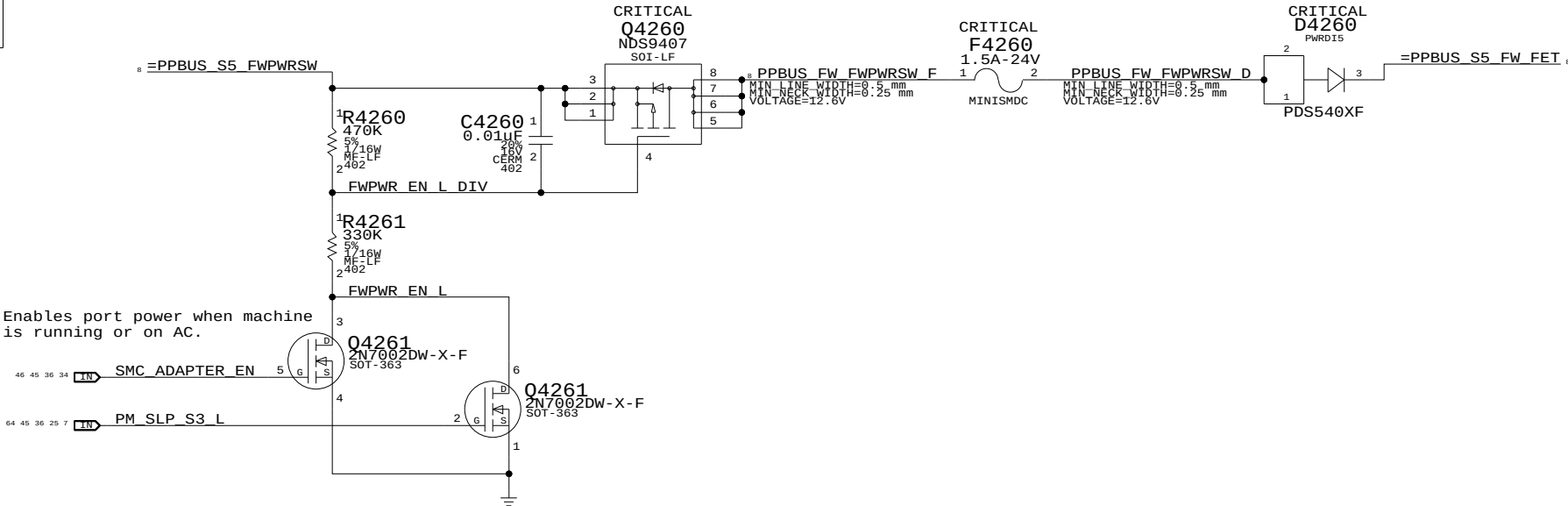
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SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	39	89

Page Notes

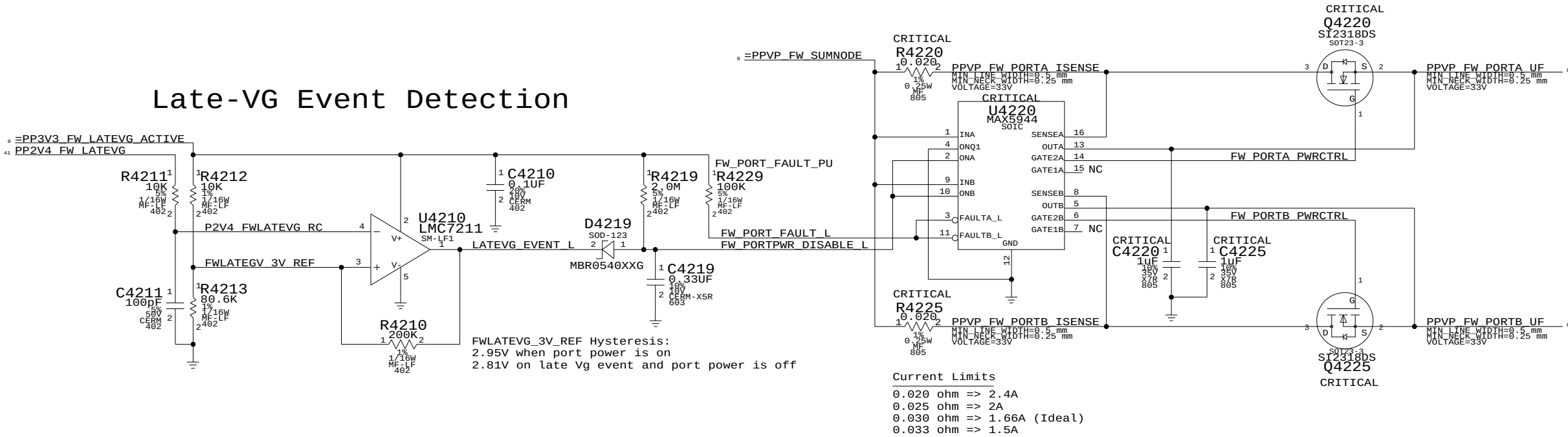
Power aliases required by this page:
- =PPBUS_S5_FWPWSW (system supply for bus power)
- =PP3V3_FW_LATEVG_ACTIVE
- =PPVP_FW_SUMNODE (power passthru summation node)
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
- FW_PORT_FAULT_PU

FireWire Port Power Switch



Current Limit/Active Late-VG Protection

Late-VG Event Detection



Current Limits
0.020 ohm => 2.4A
0.025 ohm => 2A
0.030 ohm => 1.66A (Ideal)
0.033 ohm => 1.5A

MAX5944 current limiter trips if integrator (counter) reaches 16. A new sample (taken every 125 us) is weighted as +1 if over the limit (at any point during the period) and -1/128 if under the limit. As a result, the device tends to trip easily on devices that produce periodic current spikes. Current limit has been set higher to compensate.

FireWire Port Power

SYNC_MASTER=M76_MLB SYNC_DATE=03/19/2007

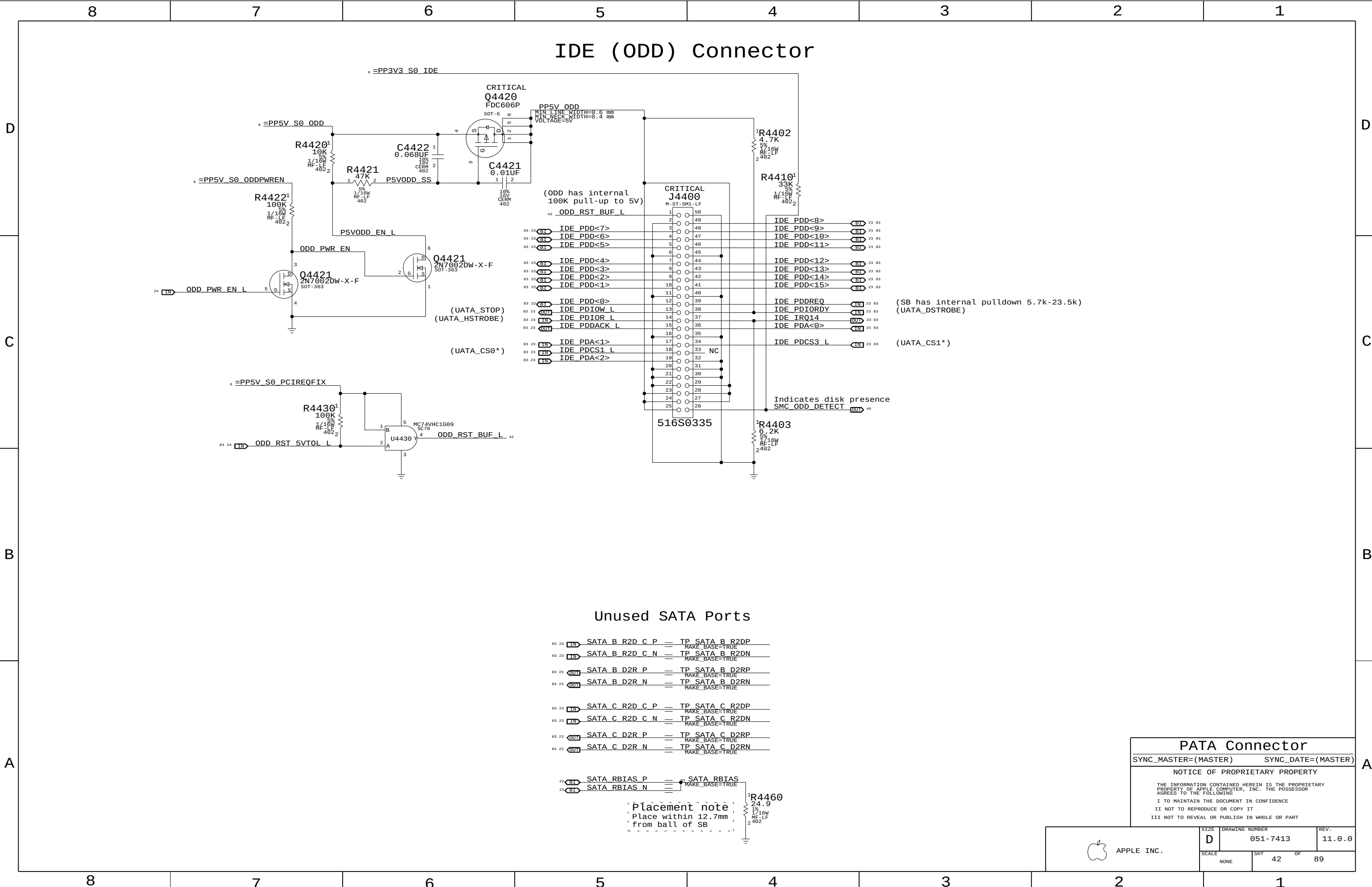
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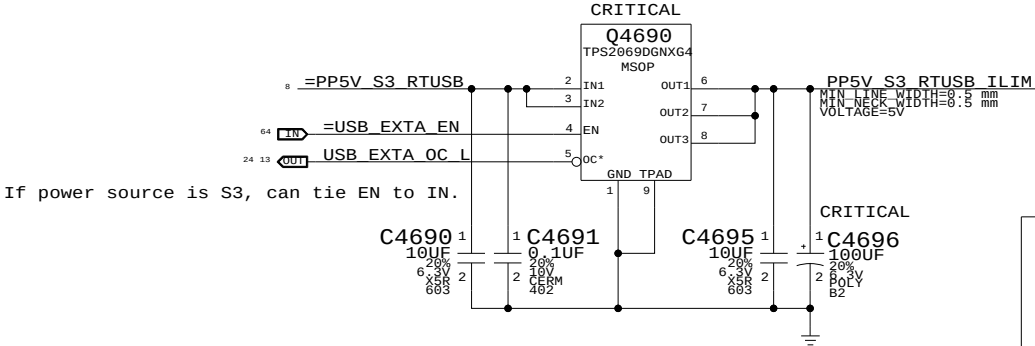


APPLE INC.

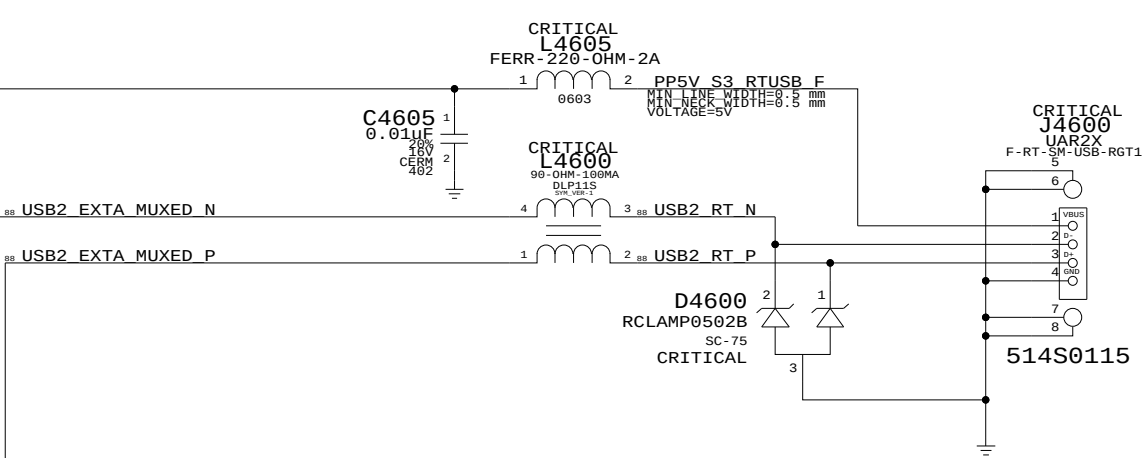
SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	40	89



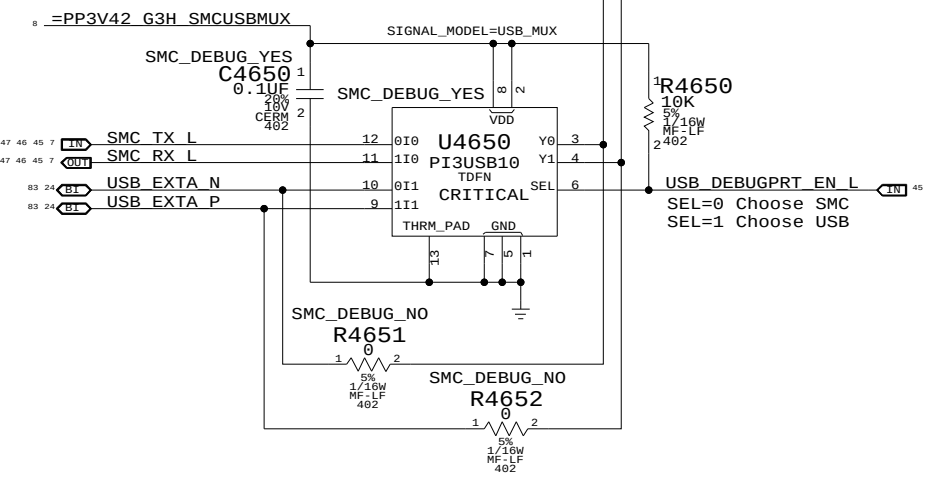
Port Power Switch



Right USB Port



USB/SMC Debug Mux



External USB Connector

SYNC_MASTER=M88 SYNC_DATE=08/02/2007

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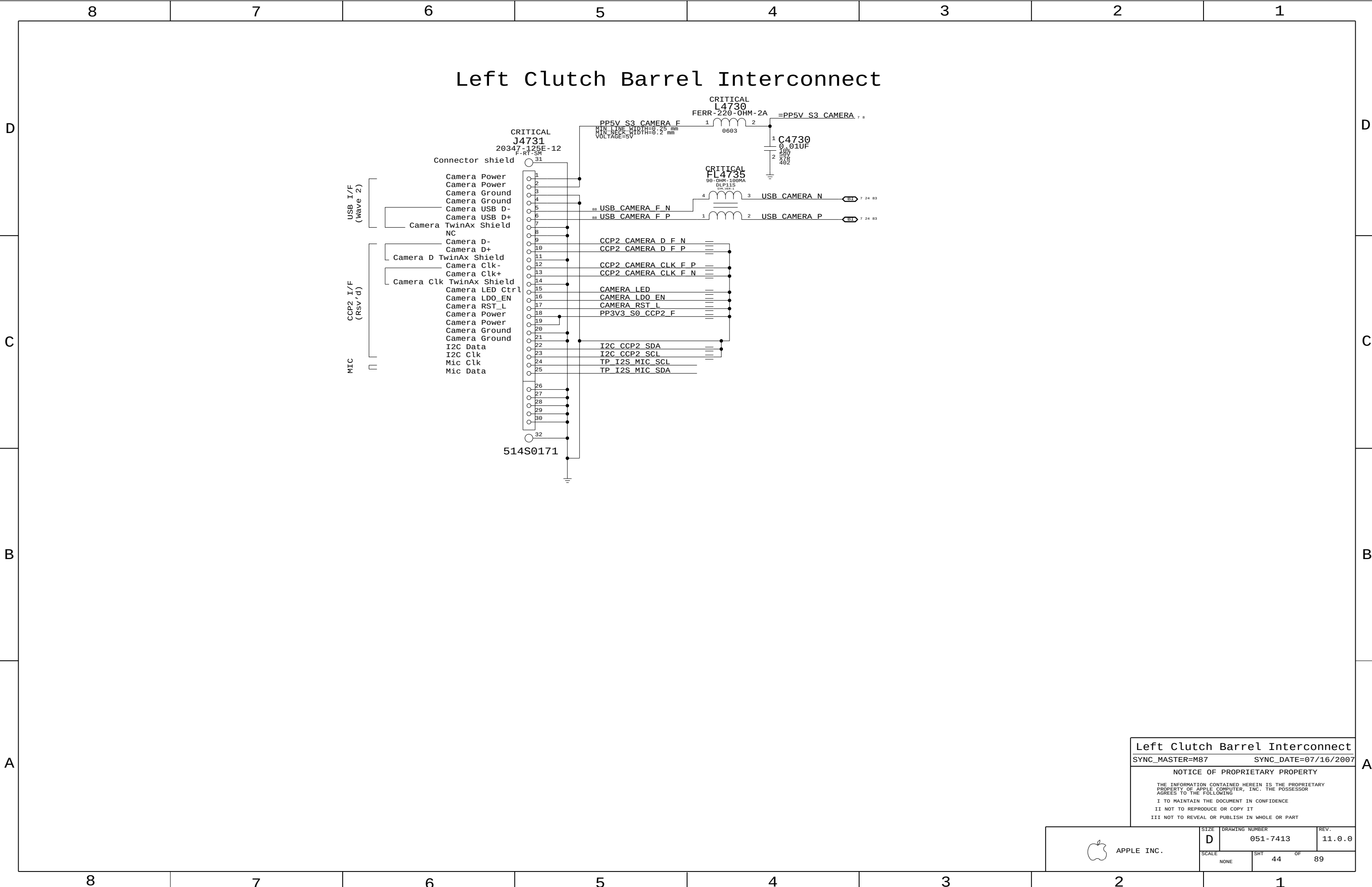
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SIZE	DRAWING NUMBER	REV.
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SCALE	SHT	OF
NONE	43	89



Left Clutch Barrel Interconnect

SYNC_MASTER=M87 SYNC_DATE=07/16/2007

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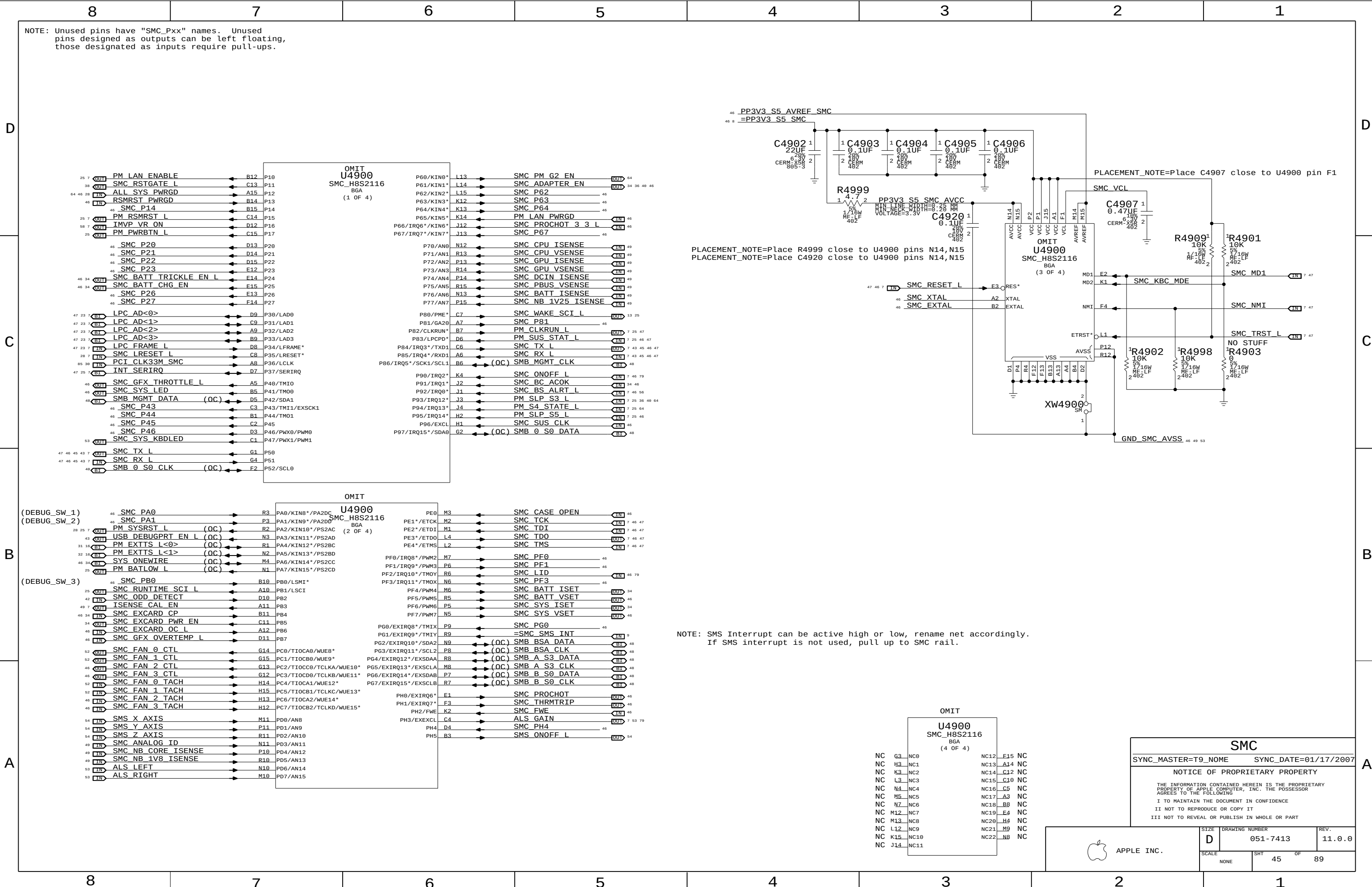
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II NOT TO REPRODUCE OR COPY IT

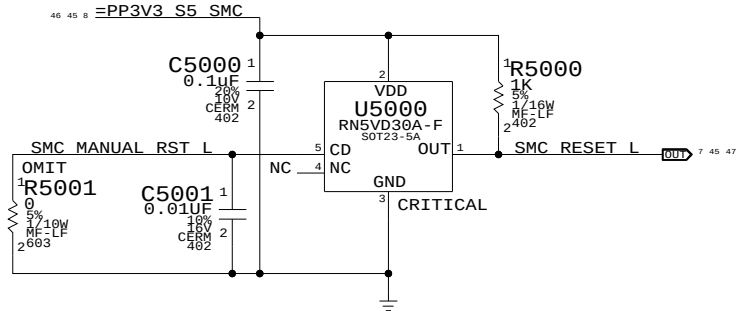
III NOT TO REVEAL OR PUBLISH IN WHOLE OR PART



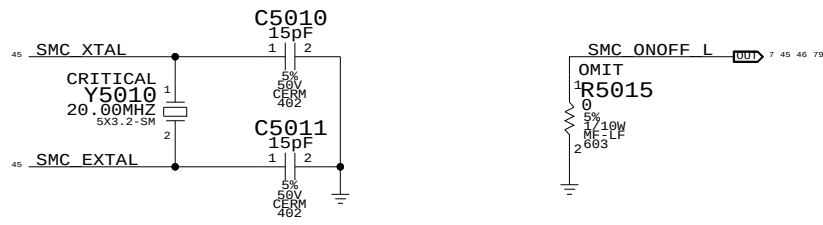
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SCALE	SHT	OF
NONE	44	89



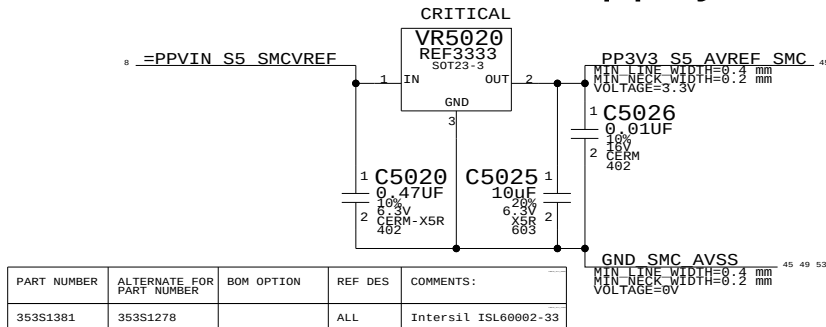
SMC Reset "Button" / Brownout Detect



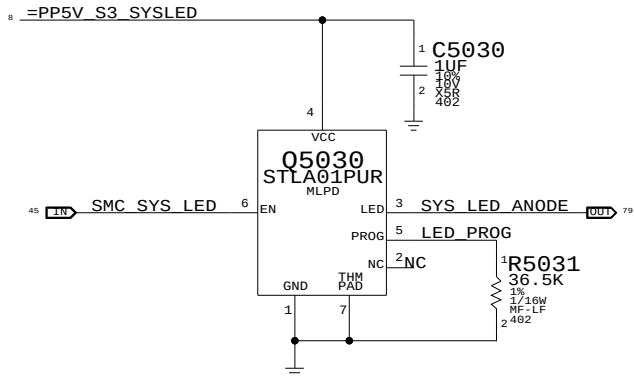
SMC Crystal Circuit Debug Power "Button"



SMC AVREF Supply



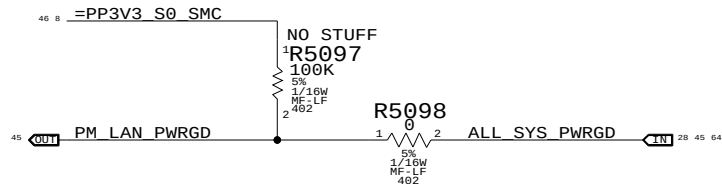
System (Sleep) LED Circuit



SMC_FAN_2_CTL	=	TP_SMC_FAN_2_CTL
SMC_FAN_2_TACH	=	TP_SMC_FAN_2_TACH
SMC_FAN_3_CTL	=	TP_SMC_FAN_3_CTL
SMC_FAN_3_TACH	=	TP_SMC_FAN_3_TACH
SMC_GFX_OVERTEMP_L	=	TP_SMC_GFX_OVERTEMP_L
SMC_GFX_THROTTLE_L	=	TP_SMC_GFX_THROTTLE_L
SMC_BATT_VSET	=	TP_SMC_BATT_VSET
SMC_SYS_VSET	=	TP_SMC_SYS_VSET
SMC_P14	=	TP_SMC_P14
SMC_P20	=	TP_SMC_P20
SMC_P21	=	TP_SMC_P21
SMC_P22	=	TP_SMC_P22
SMC_P23	=	TP_SMC_P23
SMC_P26	=	TP_SMC_P26
SMC_P27	=	TP_SMC_P27
SMC_P43	=	TP_SMC_P43
SMC_P44	=	TP_SMC_P44
SMC_P46	=	TP_SMC_P46
SMC_P62	=	TP_SMC_P62
SMC_P63	=	TP_SMC_P63
SMC_P64	=	TP_SMC_P64
SMC_P81	=	TP_SMC_P81
SMC_PF0	=	TP_SMC_PF0
SMC_PF1	=	TP_SMC_PF1

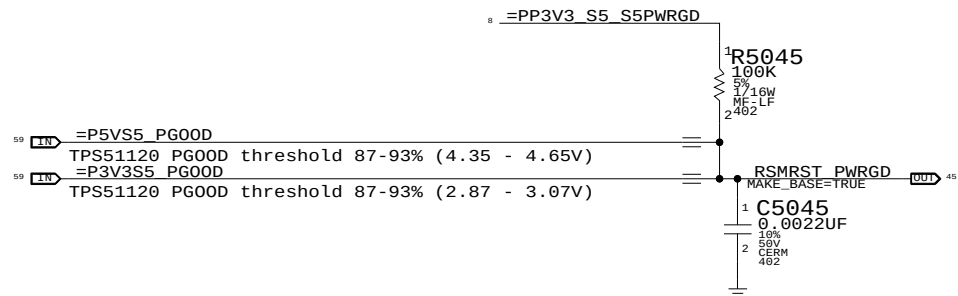
SMC_EXCARD_OC_L	=	EXCARD_OC_L
SMC_SUS_CLK	=	SUS_CLK_SB
SMC_P45	=	SMC_ENRGYSTRLDO_EN

LAN PWRGD Circuit

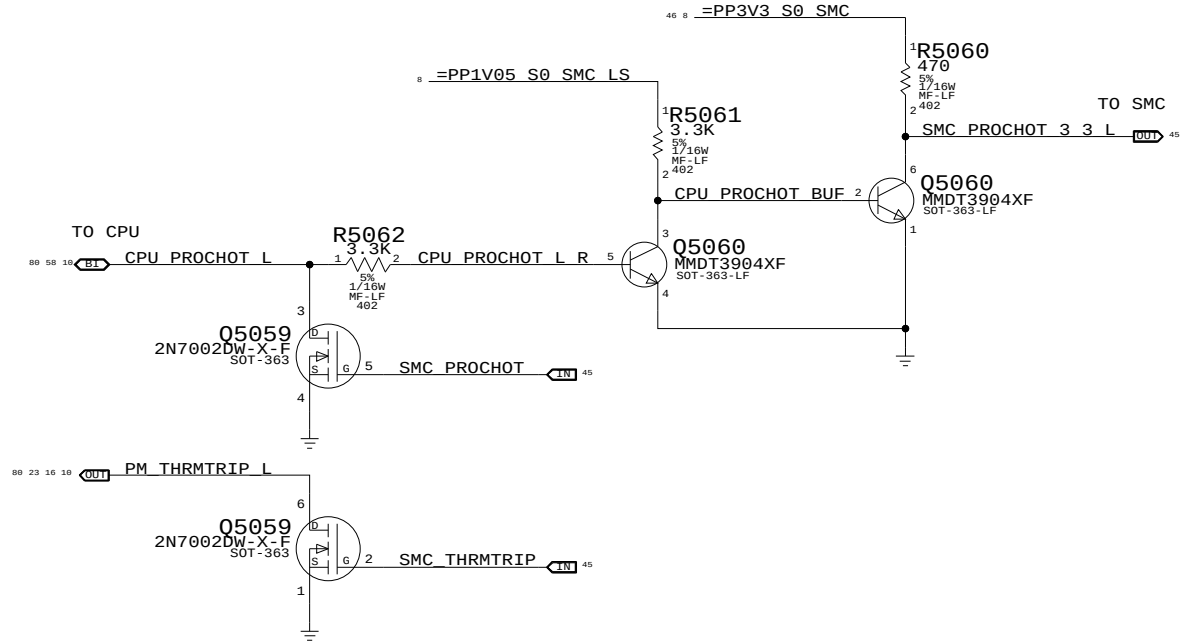


S5 Rail PWRGD Circuit

Reports when 5V S5 and 3.3V S5 are in regulation



SMC FSB to 3.3V Level Shifting



SMC_PA0	R5091	100K	1	2	5%	1/16W	MF-LF	482
SMC_PA1	R5092	100K	1	2	5%	1/16W	MF-LF	482
SMC_PB0	R5093	100K	1	2	5%	1/16W	MF-LF	482
SMC_ONOFF_L	R5070	10K	1	2	5%	1/16W	MF-LF	482
SMC_LID	R5071	100K	1	2	5%	1/16W	MF-LF	482
SMC_FWE	R5072	10K	1	2	5%	1/16W	MF-LF	482
SMC_TX_L	R5073	10K	1	2	5%	1/16W	MF-LF	482
SMC_RX_L	R5074	100K	1	2	5%	1/16W	MF-LF	482
SYS_ONEWIRE	R5075	2.0K	1	2	5%	1/16W	MF-LF	482
SMC_BS_ALRT_L	R5076	100K	1	2	5%	1/16W	MF-LF	482
SMC_TMS	R5077	10K	1	2	5%	1/16W	MF-LF	482
SMC_TDO	R5078	10K	1	2	5%	1/16W	MF-LF	482
SMC_TDI	R5079	10K	1	2	5%	1/16W	MF-LF	482
SMC_TCK	R5080	10K	1	2	5%	1/16W	MF-LF	482
SMC_P67	R5094	10K	1	2	5%	1/16W	MF-LF	482
SMC_PF3	R5081	10K	1	2	5%	1/16W	MF-LF	482
SMC_PG0	R5096	10K	1	2	5%	1/16W	MF-LF	482
SMC_PH4	R5082	10K	1	2	5%	1/16W	MF-LF	482
SMC_BATT_TRICKLE_EN_L	R5083	10K	1	2	5%	1/16W	MF-LF	482
SMC_BATT_CHG_EN	R5084	10K	1	2	5%	1/16W	MF-LF	482
SMC_ADAPTER_EN	R5085	10K	1	2	5%	1/16W	MF-LF	482
SMC_CASE_OPEN	R5086	10K	1	2	5%	1/16W	MF-LF	482
SMC_BC_ACOK	R5087	470K	1	2	5%	1/16W	MF-LF	482
SMC_EXCARD_CP	R5088	10K	1	2	5%	1/16W	MF-LF	482
PM_SUS_STAT_L	R5089	100K	1	2	5%	1/16W	MF-LF	482
PM_SLP_S5_L	R5090	100K	1	2	5%	1/16W	MF-LF	482

SMC Support

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

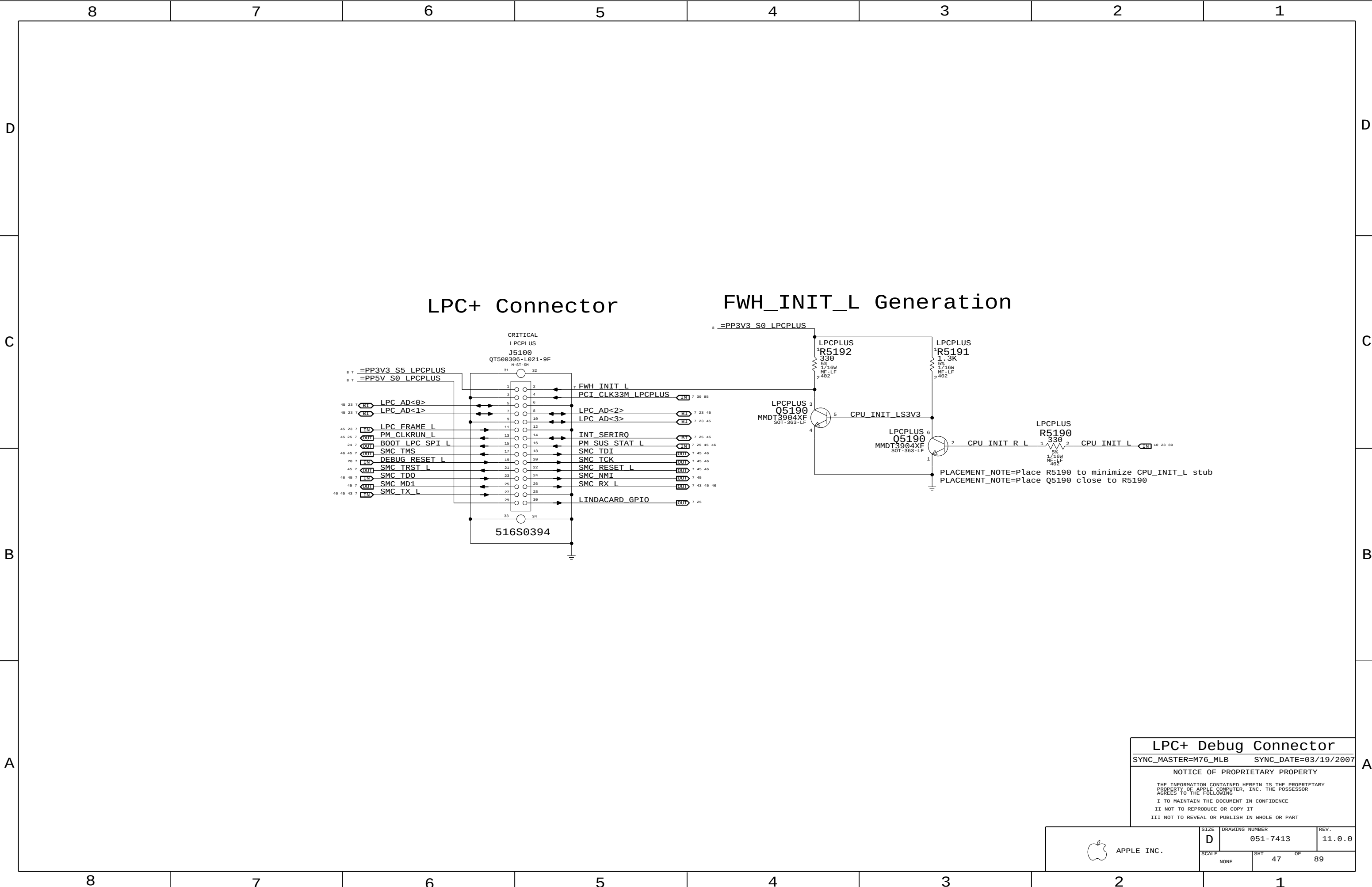
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SIZE	D	DRAWING NUMBER	051-7413	REV.	11.0.0
SCALE	NONE	SHT	46	OF	89



LPC+ Debug Connector

SYNC_MASTER=M76_MLB

SYNC_DATE=03/19/2007

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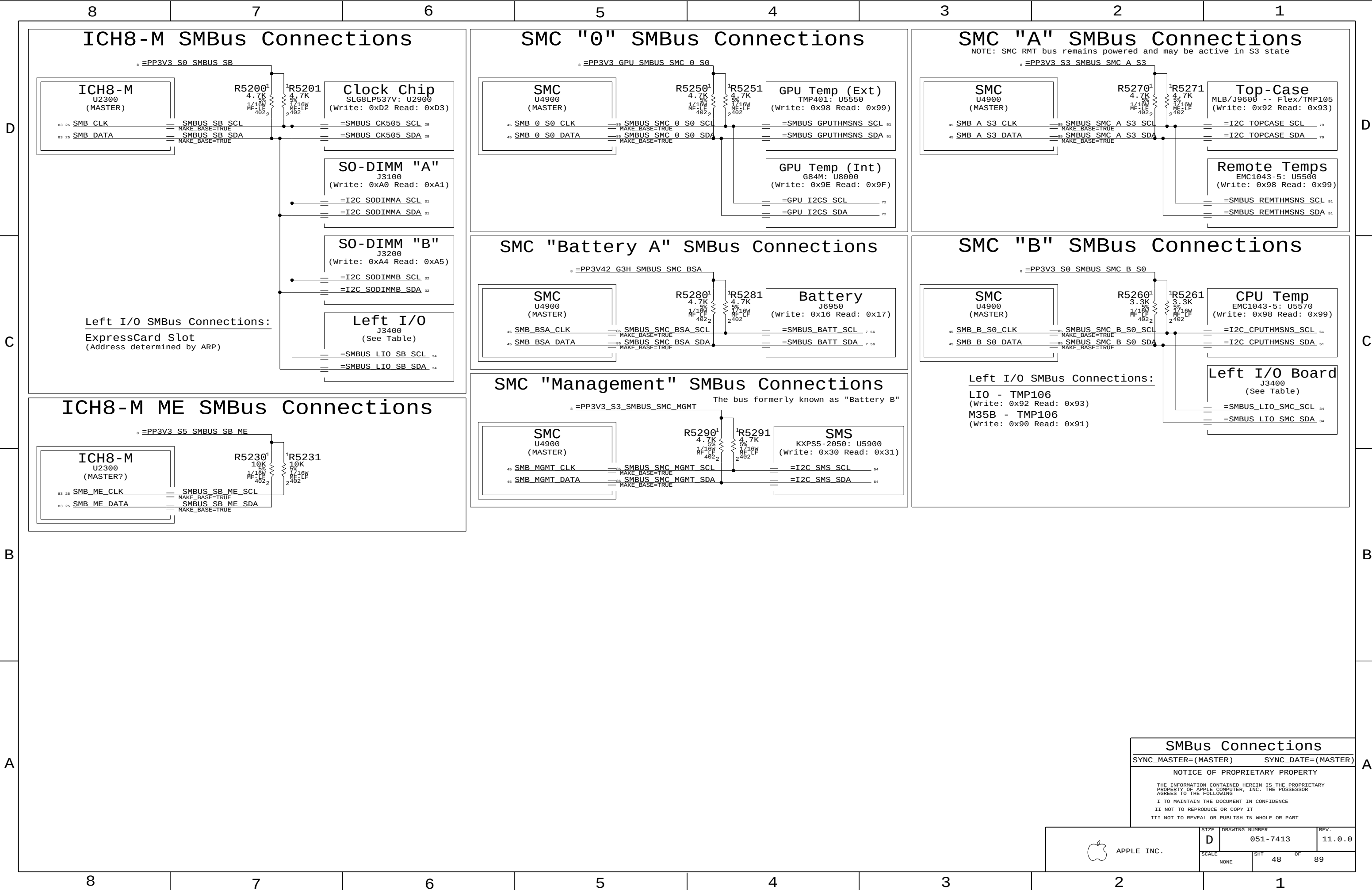
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	SCALE NONE	SHT 47	OF 89



SMBus Connections

SYNC_MASTER=(MASTER)

SYNC_DATE=(MASTER)

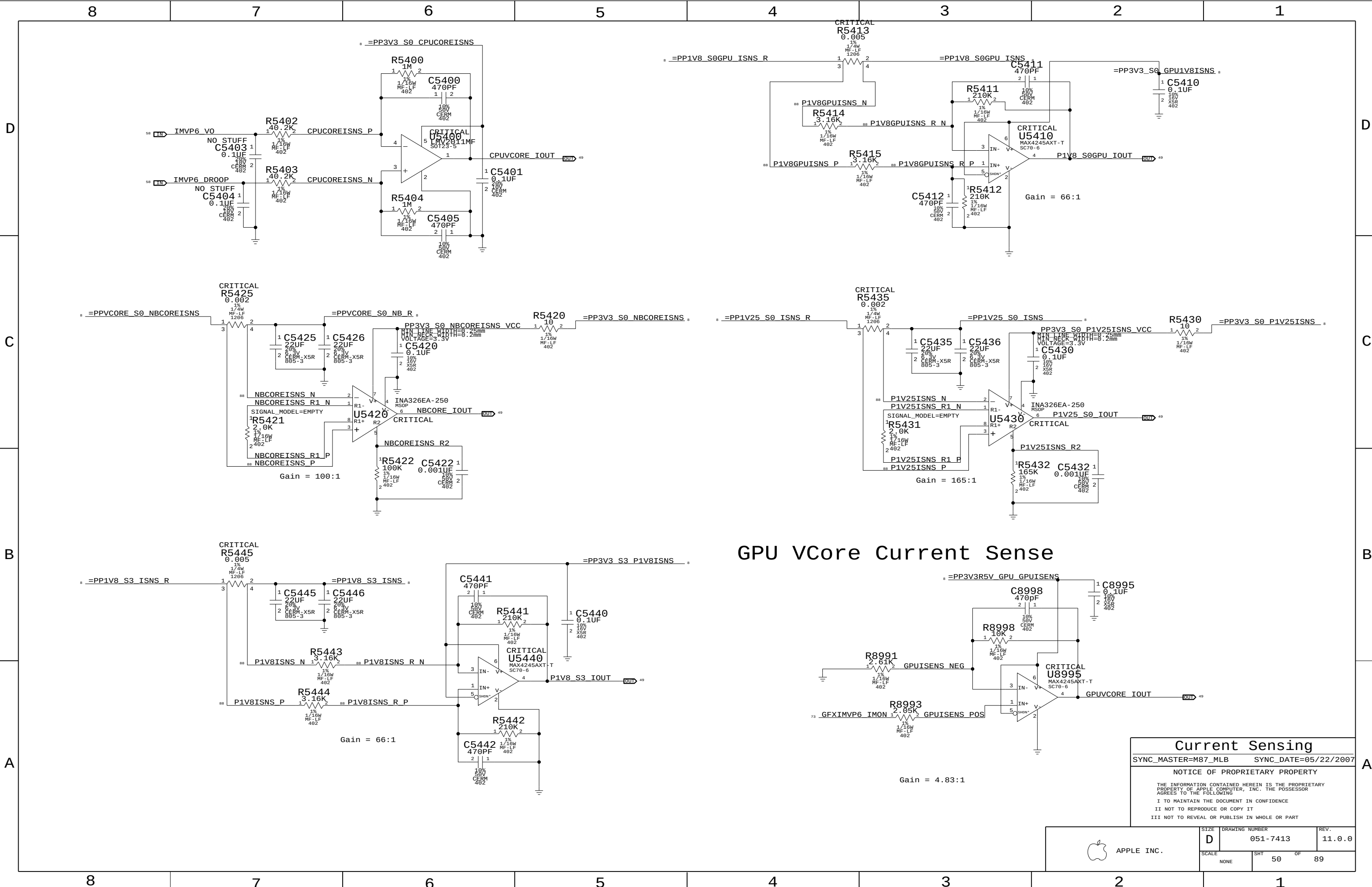
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GPU VCore Current Sense

Current Sensing

SYNC_MASTER=M87_MLB

SYNC_DATE=05/22/2007

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	SCALE NONE	SHT 50	OF 89



APPLE INC.

[illegible]

26	BI	=NB	TDB	FORCE	---
26	BI	=NB	TDB	SENSE	---
26	BI	=NB	TDE	SENSE	---
26	BI	=NB	TDE	FORCE	---

(Th1H)

CRITICAL
BM02B-ACHKS-GAN-TF-LF-SN-M
M-RT-SM
Place on left side of fan cutout

518S0487

NO STUFF
C5510
18PF
50V
10%
CERM
402

SIGNAL_MODEL=EMPTY
C5511
0.0022uF
10%
50V
CERM
402

RSFSTHMSNS_D_P
RSFSTHMSNS_D_N
HSTHMSNS_D_P
HSTHMSNS_D_N

(Th0H)

PP3V3 S3 REMTHMSNS R
MIN LINE WIDTH=0.38 mm
MIN LINE SPACING=0.25 mm
VOLTAGE=3.3V

R5500
47
1/16W
MF
402

C5500
0.1uF
10V
10%
CERM
402

R5501
10K
1/16W
MF
402

R5502
10K
1/16W
MF
402

U5500
EMC1403-1-AIZL
TSSOP
CRITICAL

VDD
THERM*
DP1
DN1
DP2
DN2
GND

7 REMTHMSNS_THM_L
8 REMTHMSNS_ALERT_L
9 =SMBUS_REMTHMSNS_SDA
10 =SMBUS_REMTHMSNS_SCL

(TG0H)

CRITICAL
BM02B-ACHKS-GAN-TF-LF-SN-M
M-RT-SM
Place near GPU

518S0487

NO STUFF
C5520
18PF
50V
10%
CERM
402

SIGNAL_MODEL=EMPTY
C5521
0.0022uF
10%
50V
CERM
402

HSTHMSNS_D_P
HSTHMSNS_D_N

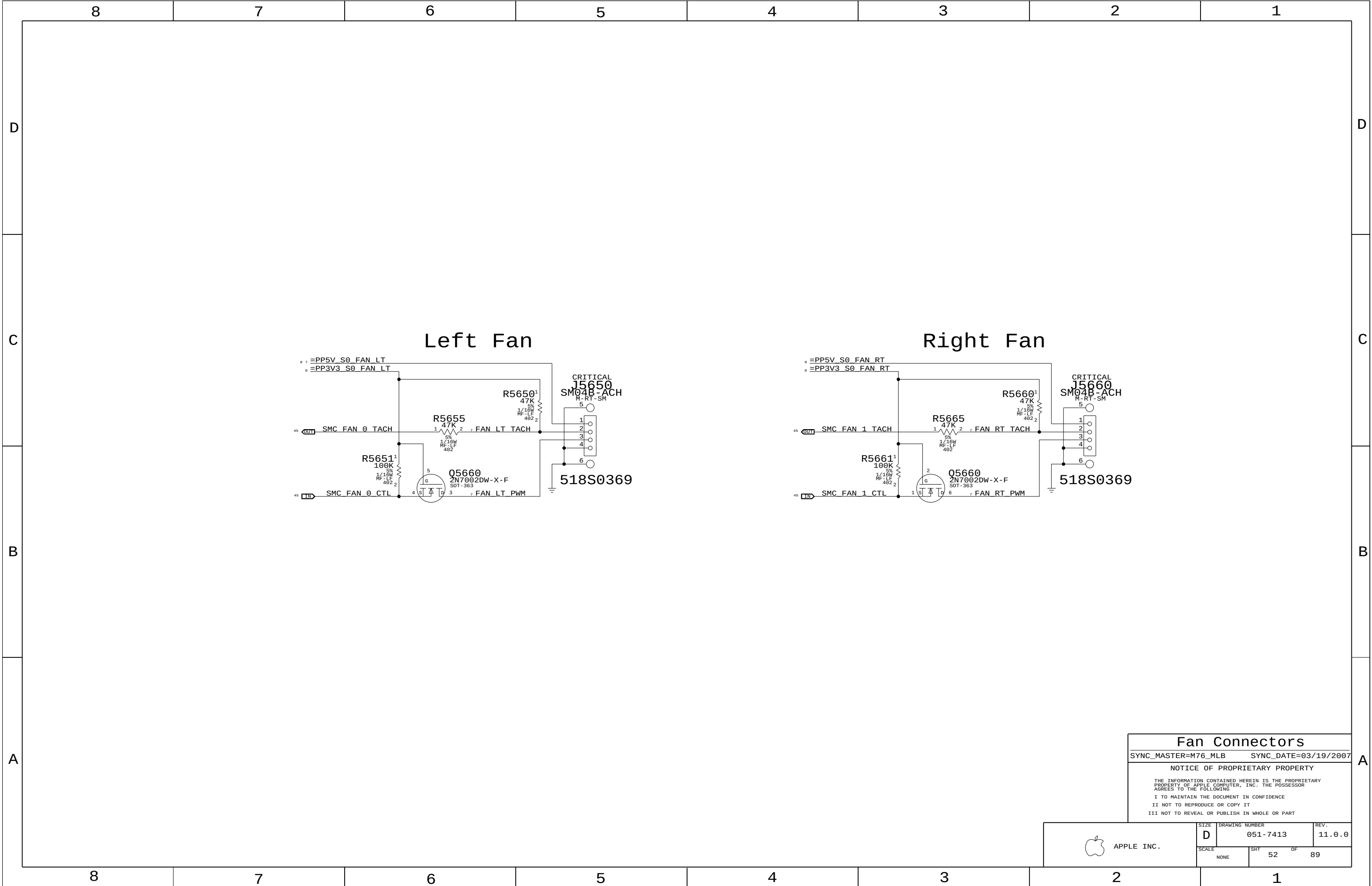
Placement note:
Keep 2 caps as close to
connectors as possible

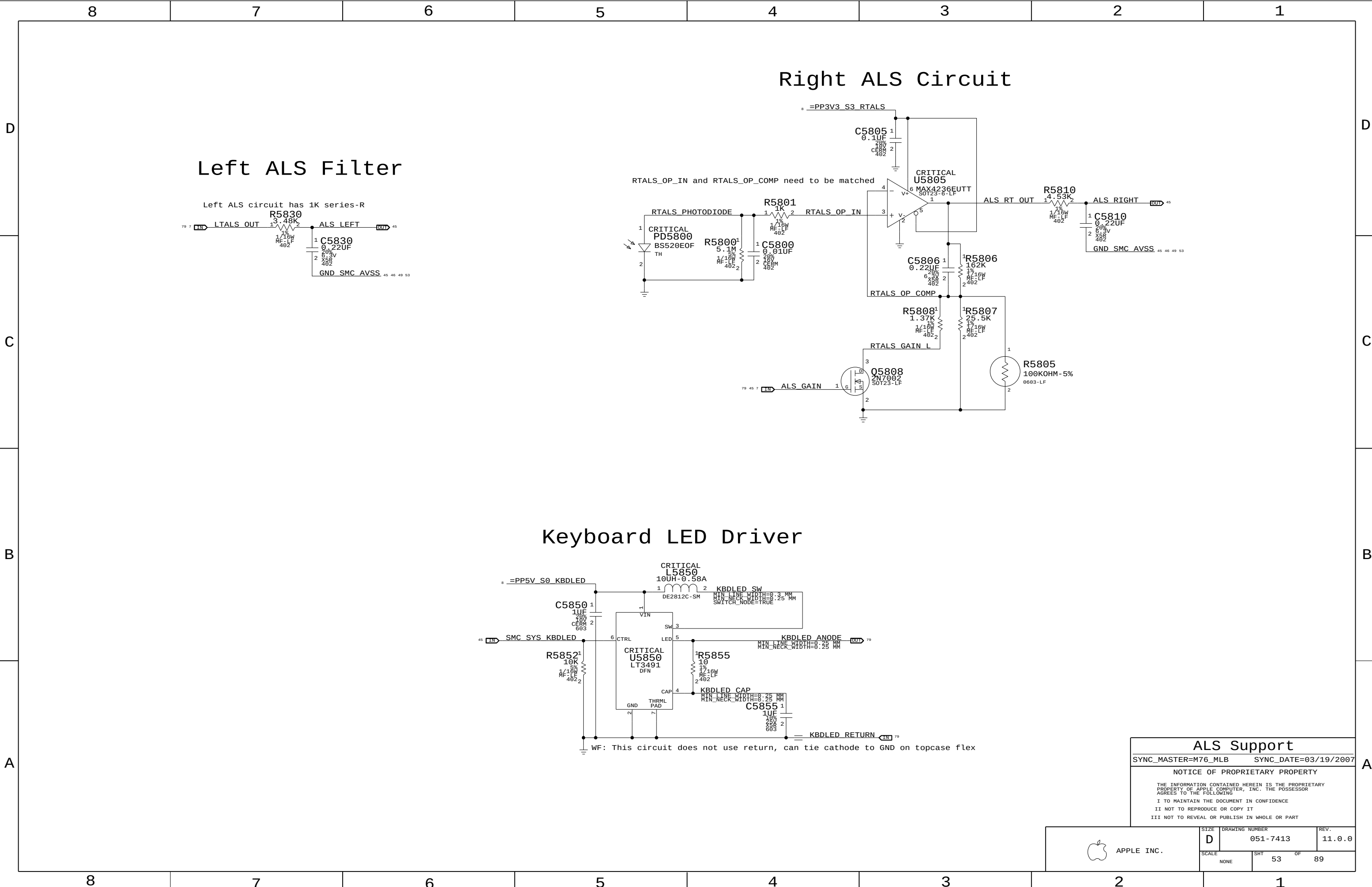
Placement note:
Keep 2 caps as close
to IC pins as possible

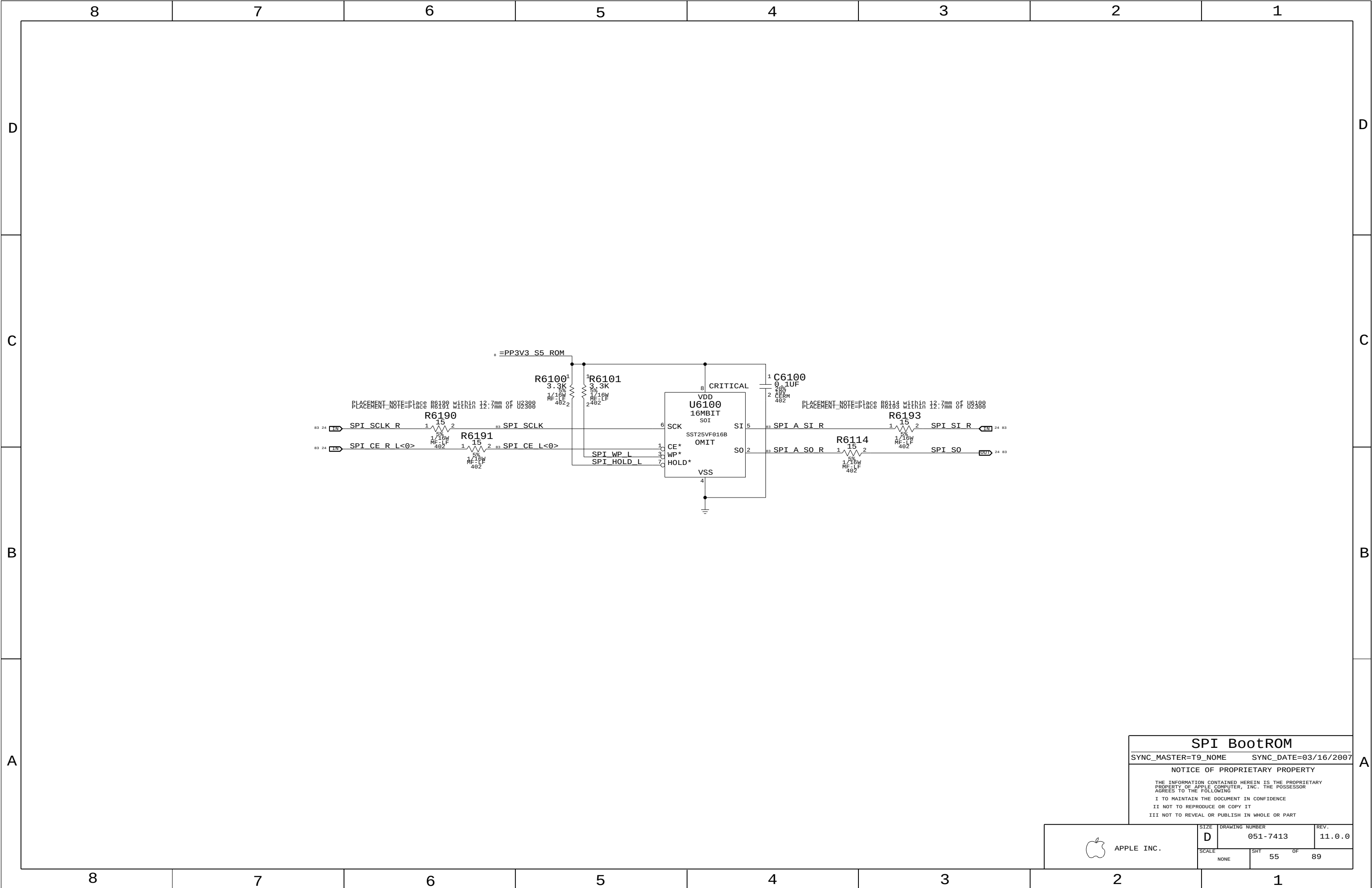
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 APPLE INC.	SIZE	DRAWING NUMBER		REV.
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	SCALE	SHT	OF	
	NONE	51	89	



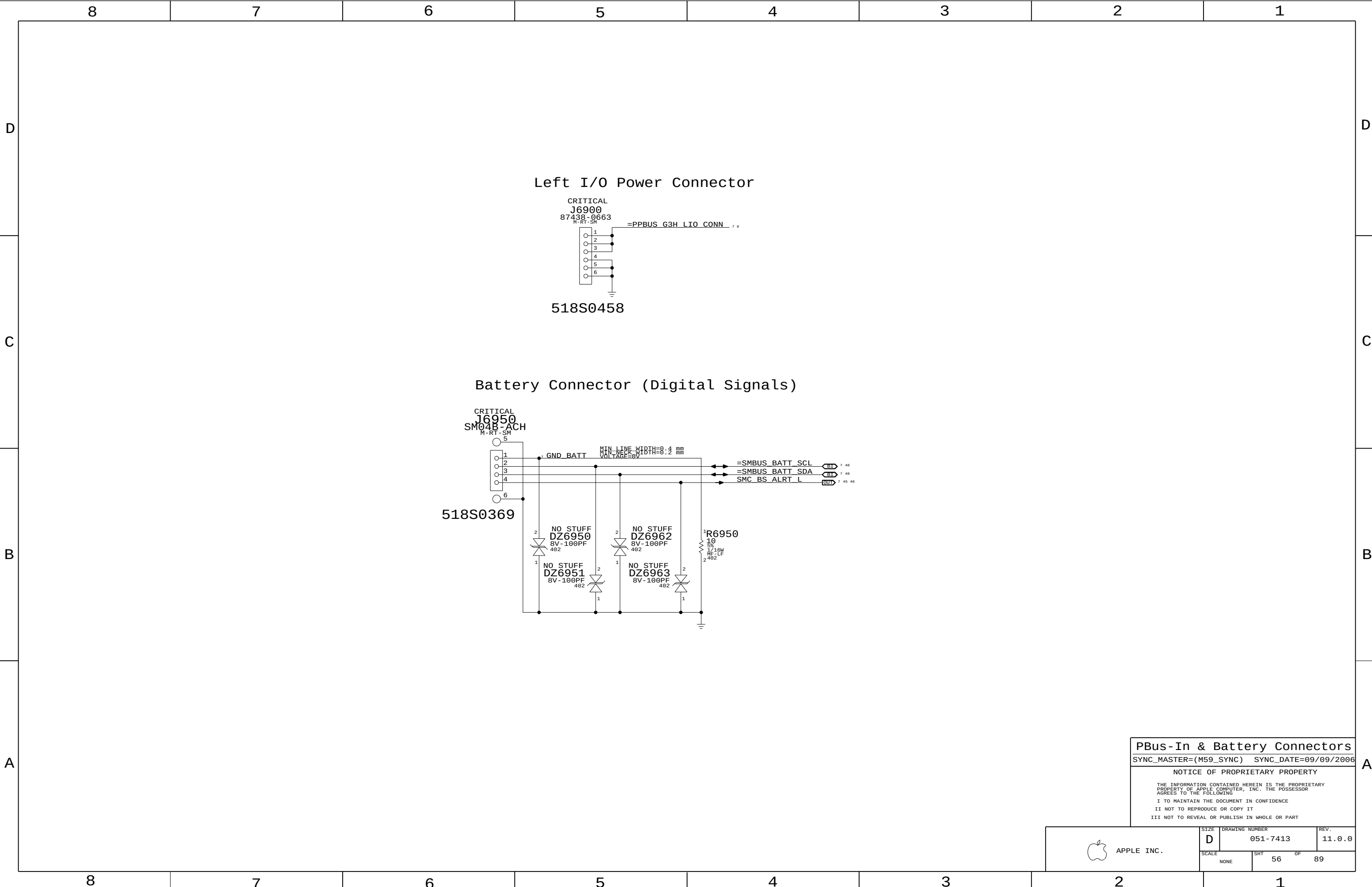






SPI BootROM		
SYNC_MASTER=T9_NOME		SYNC_DATE=03/16/2007
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SIZE D	DRAWING NUMBER 051-7413	REV. 11.0.0
SCALE NONE	SHT 55	OF 89





PBus-In & Battery Connectors

SYNC_MASTER=(M59_SYNC) SYNC_DATE=09/09/2006

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SIZE

D

DRAWING NUMBER

051-7413

REV.

11.0.0

SCALE

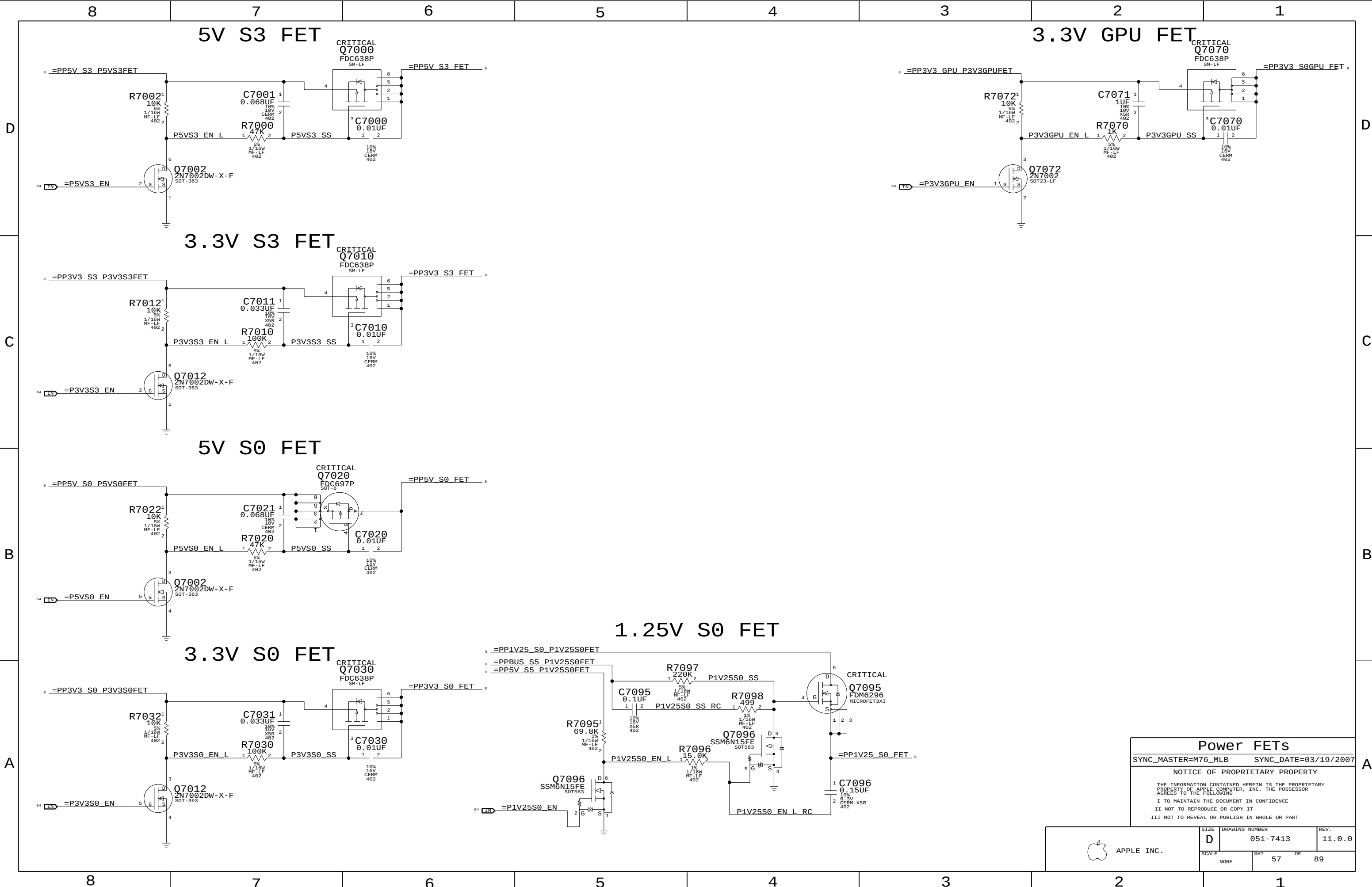
NONE

SHT

56

OF

89



Power FETs

SYNC_MASTER=M76_MLB

SYNC_DATE=03/19/2007

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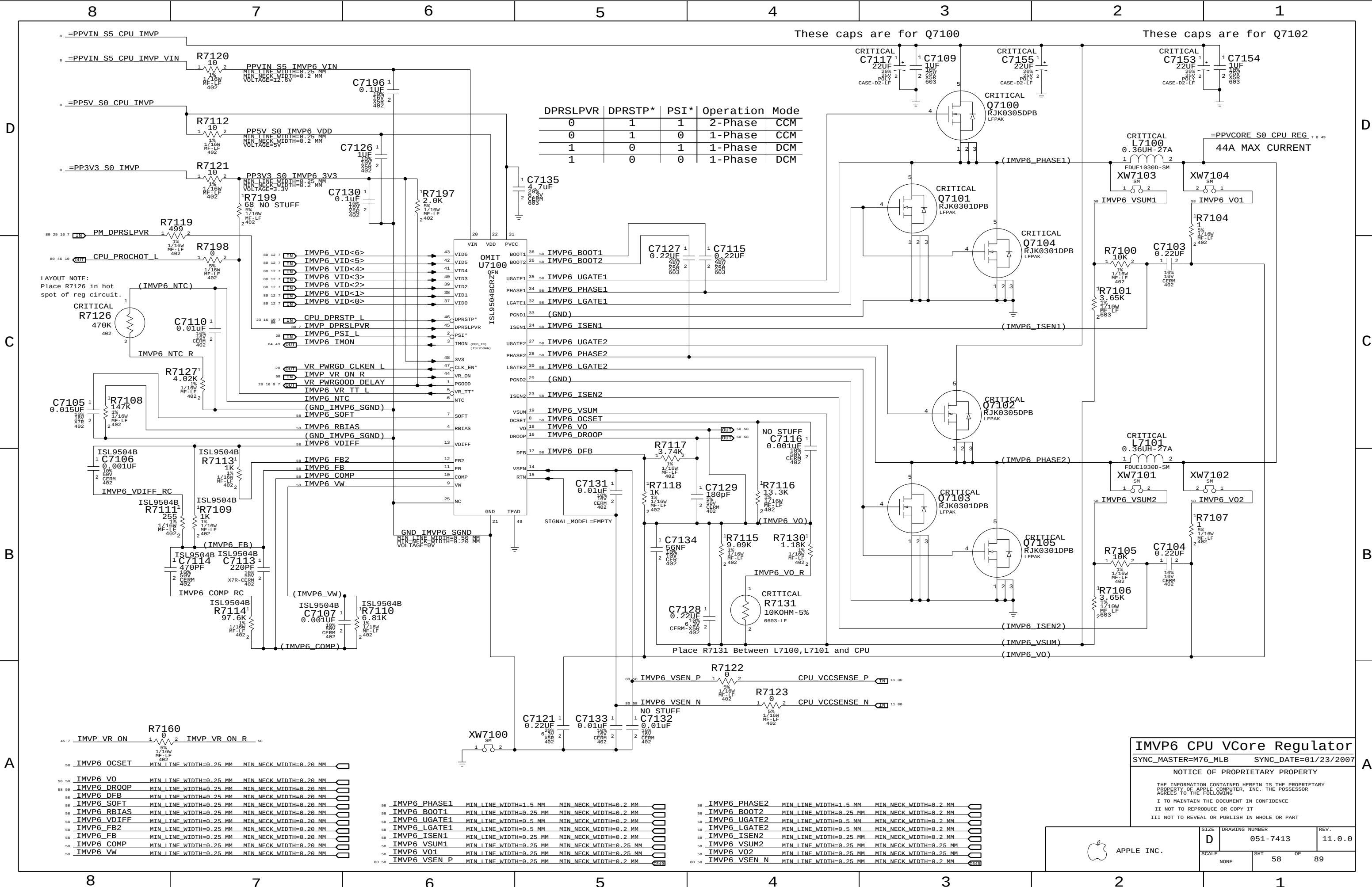
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SCALE	DRAWING NUMBER		REV.
	051-7413	11.0.0	
SHT		OF	
NONE		57	89



APPLE INC.



DPRSLPVR	DPRSTP*	PSI*	Operation Mode
0	1	1	2-Phase CCM
0	1	0	1-Phase CCM
1	0	1	1-Phase DCM
1	0	0	1-Phase DCM

IMVP6 VID<6>	43
IMVP6 VID<5>	42
IMVP6 VID<4>	41
IMVP6 VID<3>	40
IMVP6 VID<2>	39
IMVP6 VID<1>	38
IMVP6 VID<0>	37
CPU DPRSTP L	46
IMVP DPRSLPVR	45
IMVP6 PSI L	2
IMVP6 IMON	3

VR_PWRGD_CLKEN L	48
IMVP VR_ON R	47
VR_PWRGOOD_DELAY	44
IMVP6 VR_TT L	5
IMVP6 NTC	6
(GND_IMVP6_SGND)	7
IMVP6 SOFT	7
IMVP6 RBIAS	4
(GND_IMVP6_SGND)	4
IMVP6 VDIFF	13

IMVP6 FB2	12
IMVP6 FB	11
IMVP6 COMP	10
IMVP6 VW	9
NC	25

IMVP6 FB2	12
IMVP6 FB	11
IMVP6 COMP	10
IMVP6 VW	9
NC	25

IMVP6 FB2	12
IMVP6 FB	11
IMVP6 COMP	10
IMVP6 VW	9
NC	25

IMVP6 FB2	12
IMVP6 FB	11
IMVP6 COMP	10
IMVP6 VW	9
NC	25

IMVP6 FB2	12
IMVP6 FB	11
IMVP6 COMP	10
IMVP6 VW	9
NC	25

IMVP6 BOOT1	36
IMVP6 BOOT2	26
IMVP6 UGATE1	35
IMVP6 PHASE1	34
IMVP6 LGATE1	32
(GND)	33
IMVP6 ISEN1	24
IMVP6 UGATE2	27
IMVP6 PHASE2	28
IMVP6 LGATE2	30
(GND)	29
IMVP6 ISEN2	23
IMVP6 VSUM	19
IMVP6 OCSET	8
IMVP6 VO	16
IMVP6 DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 BOOT1	36
IMVP6 BOOT2	26
IMVP6 UGATE1	35
IMVP6 PHASE1	34
IMVP6 LGATE1	32
(GND)	33
IMVP6 ISEN1	24
IMVP6 UGATE2	27
IMVP6 PHASE2	28
IMVP6 LGATE2	30
(GND)	29
IMVP6 ISEN2	23
IMVP6 VSUM	19
IMVP6 OCSET	8
IMVP6 VO	16
IMVP6 DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 DFB	17
VSUM	19
OCSET	8
VO	16
DROOP	18

IMVP6 CPU VCore Regulator

SYNC_MASTER=M76_MLB SYNC_DATE=01/23/2007

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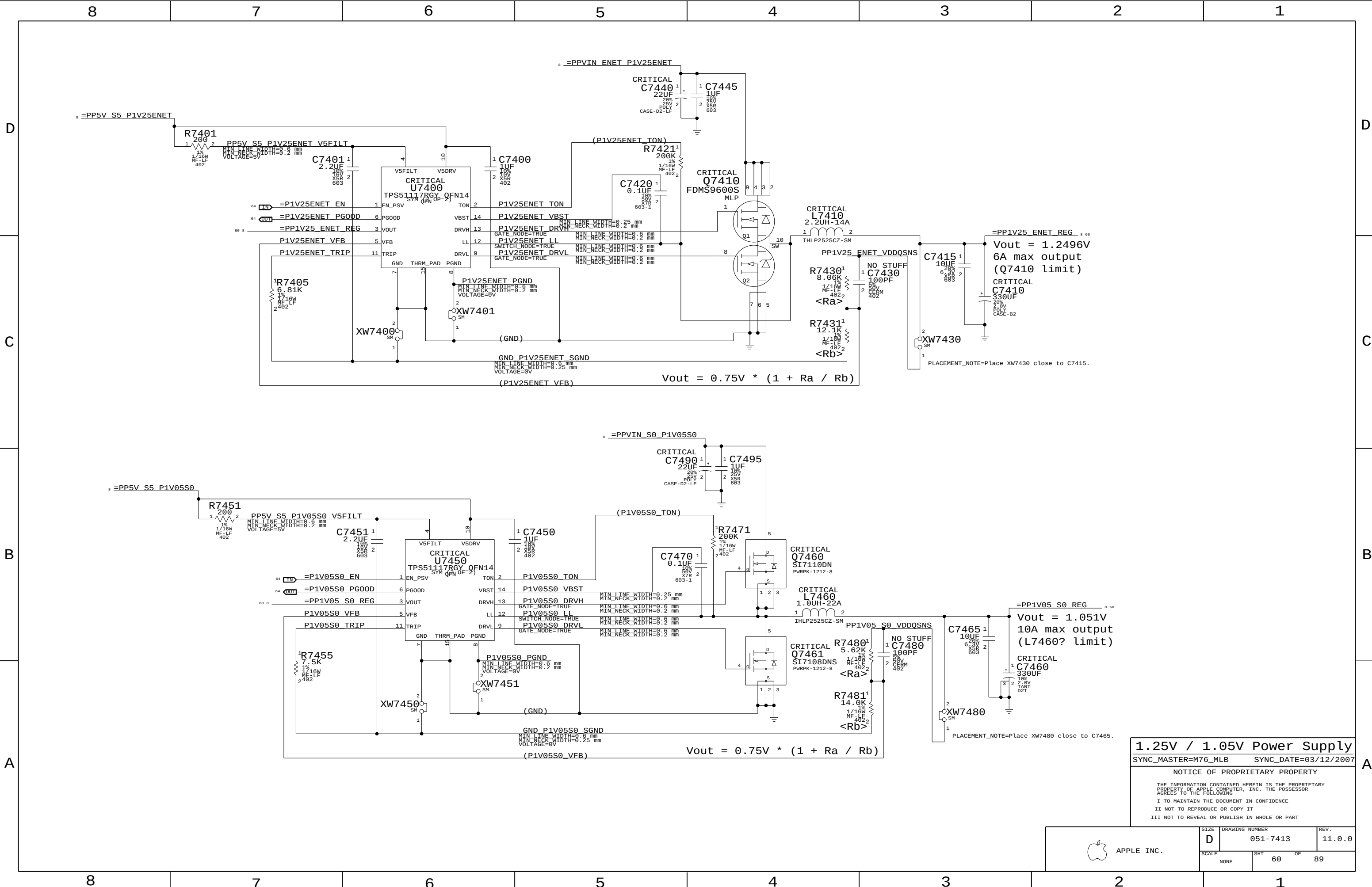
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D 051-7413 11.0.0

NONE 58 OF 89



1.25V / 1.05V Power Supply

SYNC_MASTER=M76_MLB SYNC_DATE=03/12/2007

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SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	60	89

D

C

B

A

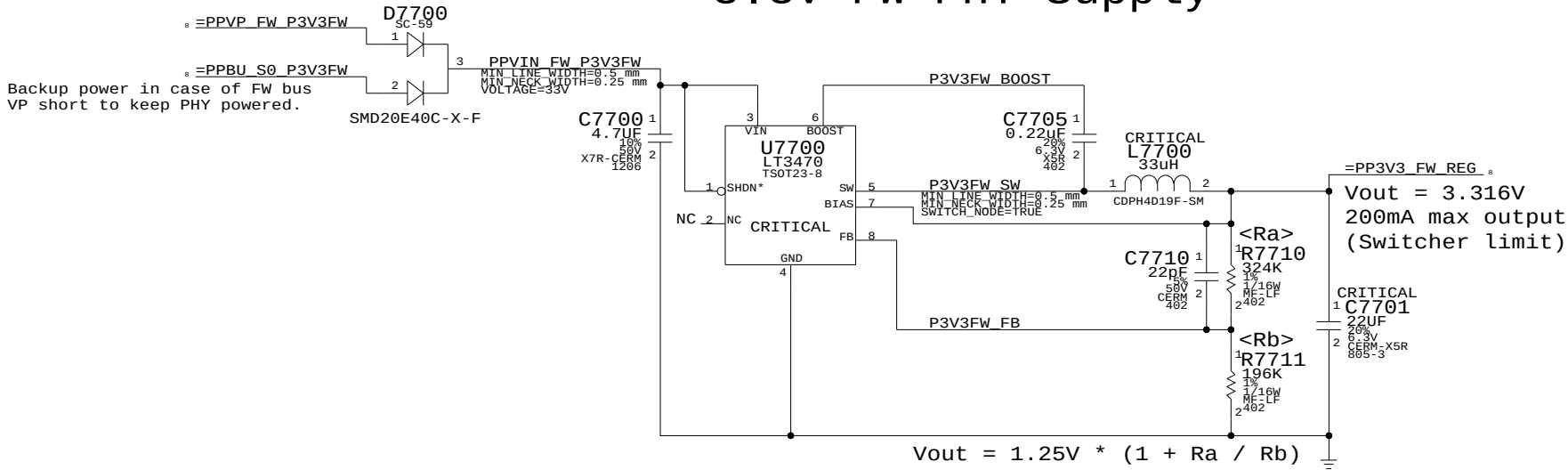
D

C

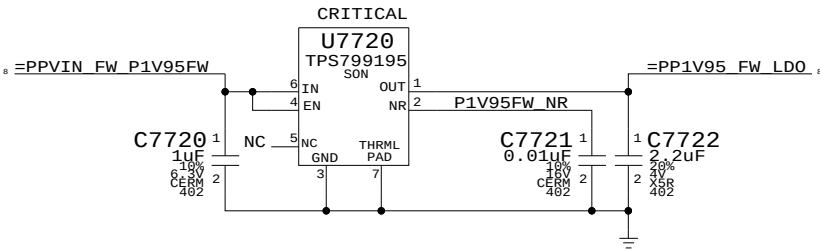
B

A

3.3V FW PHY Supply



1.95V FW PHY Supply



FW PHY Power Supplies

SYNC_MASTER=M76_MLB SYNC_DATE=03/19/2007

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APPLE INC.

SIZE

D

DRAWING NUMBER

051-7413

REV.

11.0.0

SCALE

NONE

SHT

63

OF

89

Power Control Signals

3.425V "G3Hot" Supply

State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3Hot)	0	0	0

Supply needs to guarantee 3.31V delivered to SMC Vref generator

Vout = 3.425
200mA max output
(Switcher limit)

$$V_{out} = 1.25V * (1 + R_a / R_b)$$

Unused PG00D Signals

- =P1V25ENET PG00D == TP P1V25ENET PG00D
- =P1V8_S0GPU PG00D == TP P1V8_S0GPU PG00D

1.5V / 1.05V PWRGD Circuit

Reports when 1.5V S0 and 1.05V S0 are in regulation

NOTE: 0.9V is not checked!
Other S0 Rails PWRGD Circuit

3.425V G3Hot Supply & Power Control

SYNC_MASTER=M88 SYNC_DATE=08/02/2007

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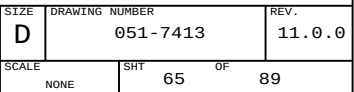
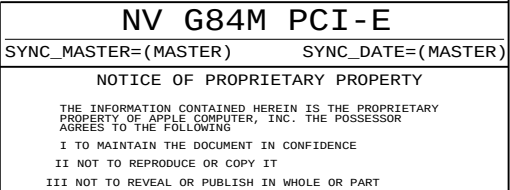


SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	64	89


```
Power aliases required by this page:
- =PP1V2_GPU_PEX_PLLXVDD
- =PP1V2_GPU_PEX_IOVDDQ
- =PP1V2_GPU_PEX_IOVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)
```

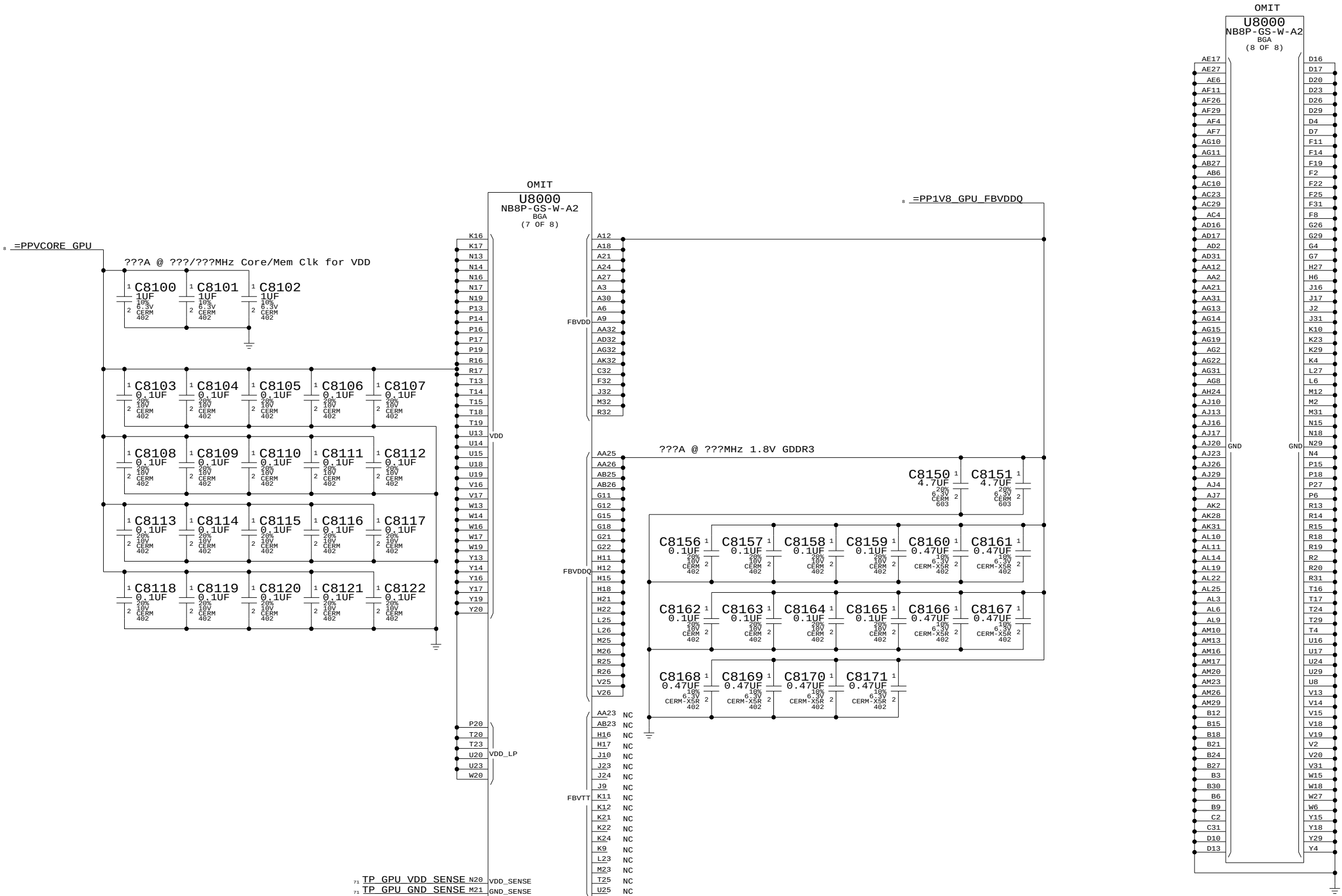


Page Notes

Power aliases required by this page:
- =PPVCORE_GPU
- =PP1V8_GPU_FBVDDQ

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



NV G84M Core/FB Power

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

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D	051-7413	11.0.0
SCALE	SHT	OF
NONE	66	89

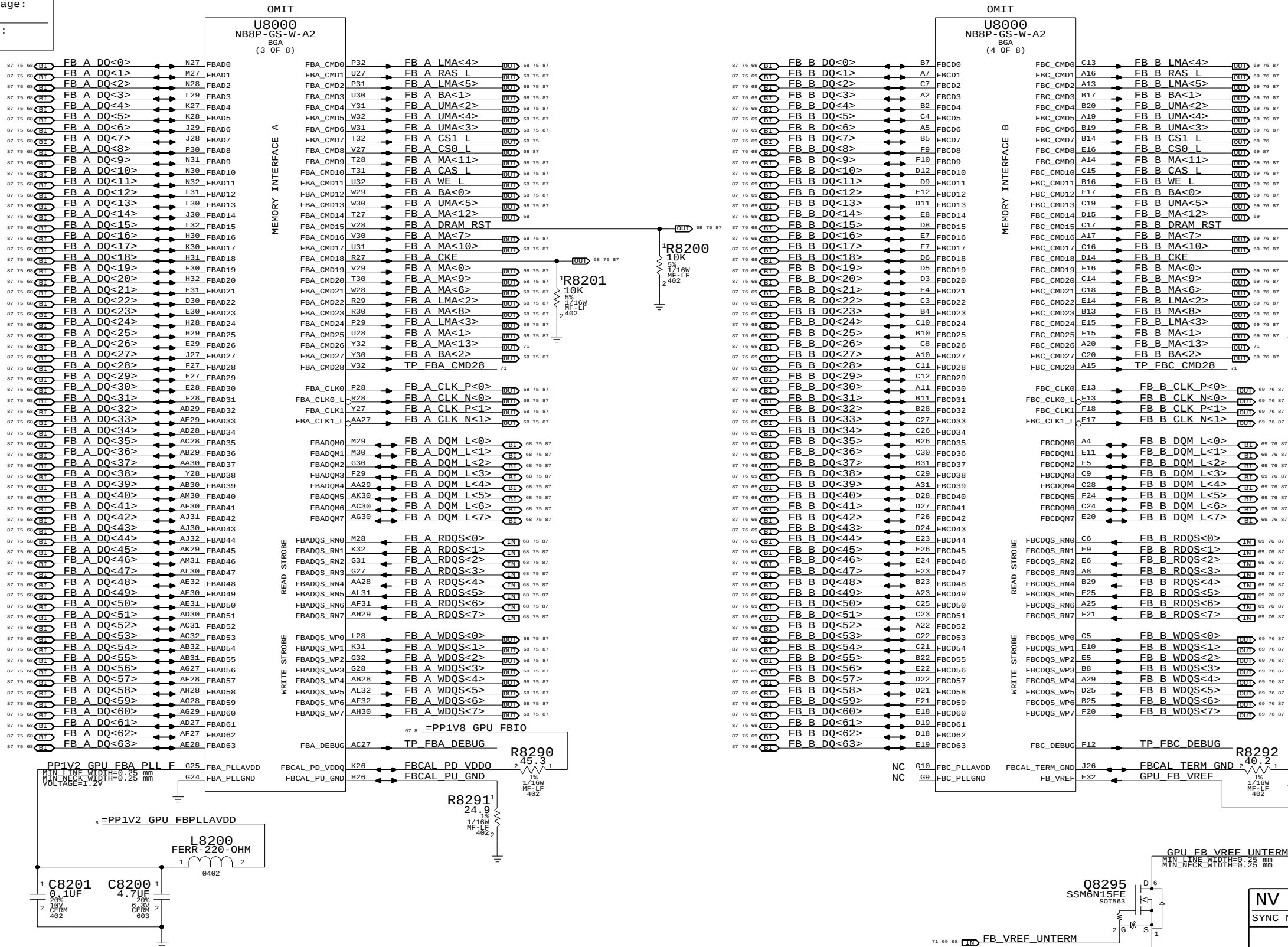
Page Notes

Power aliases required by this page:

- =PP1V2_GPU_FBPLLAVDD
- =PP1V8_GPU_FBI0

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



NV G84M Frame Buffer I/F

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

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SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	67	89



Page Notes

Power aliases required by this page:

- =PP1V8_S0_FB_VDD
- =PP1V8_S0_FB_VDDQ

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

GDDR3 Frame Buffer B (Top)

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

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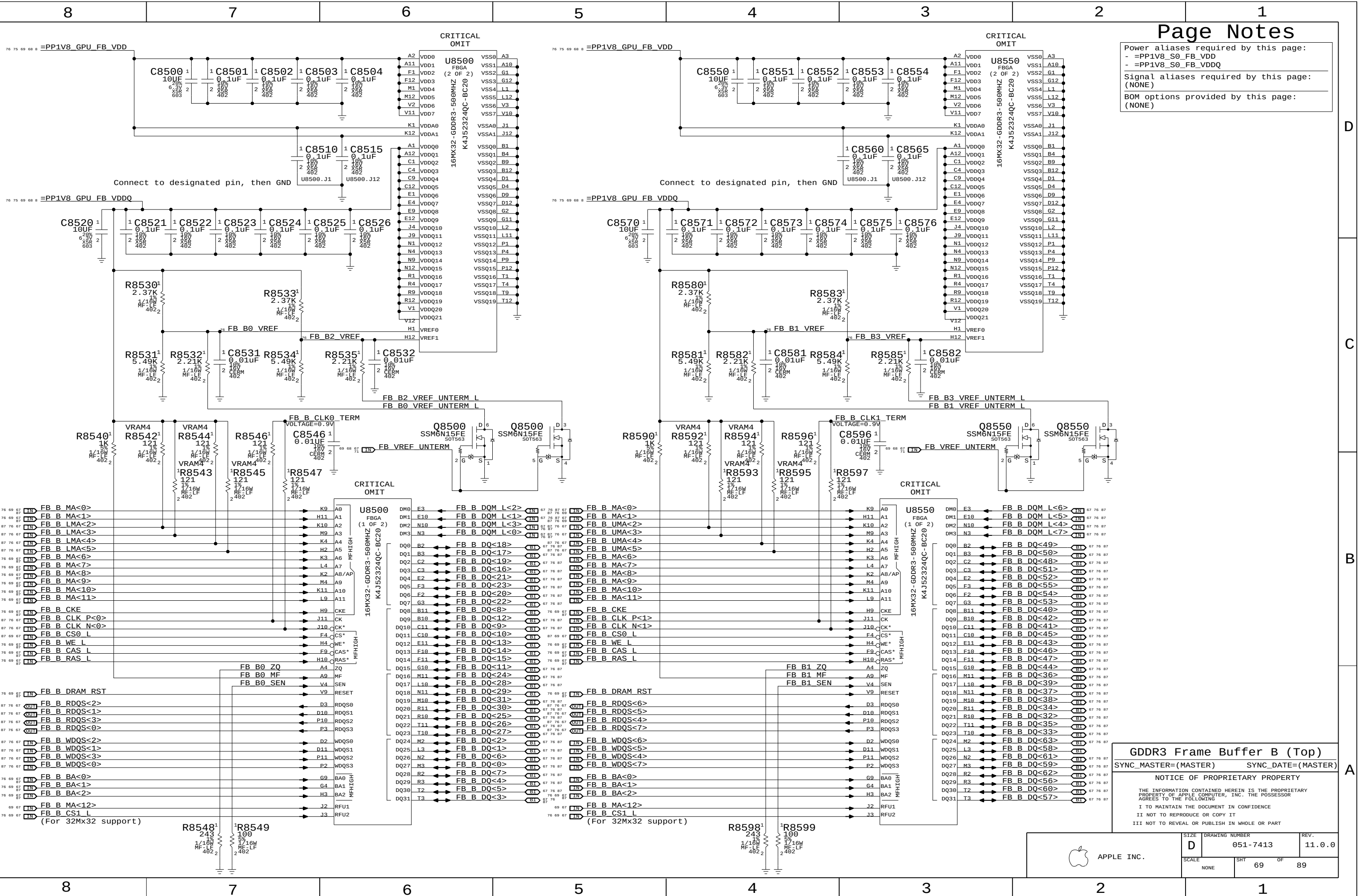
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SIZE D DRAWING NUMBER 051-7413 REV. 11.0.0

SCALE NONE SHT 69 OF 89



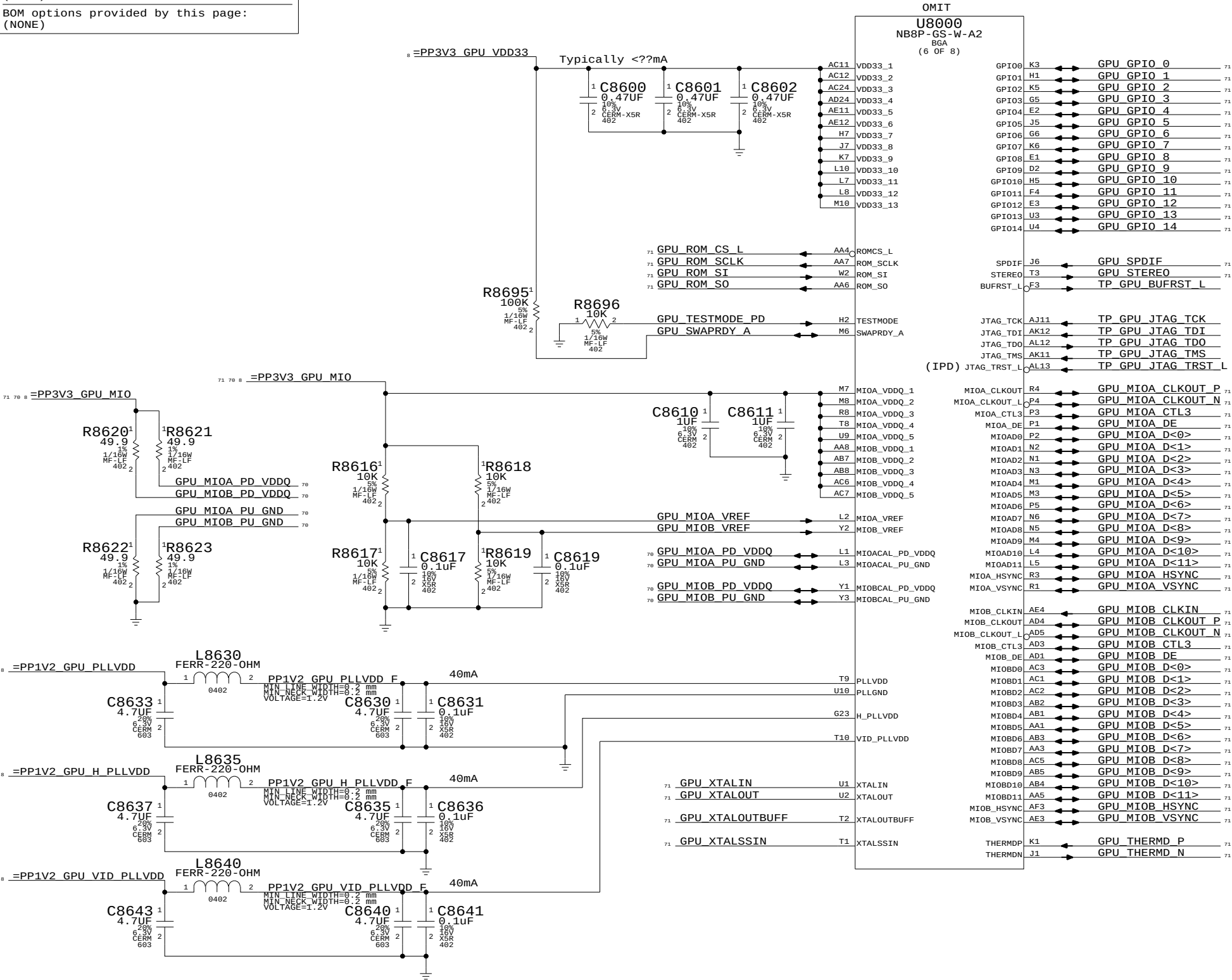
Page Notes

Power aliases required by this page:

- =PP3V3_GPU_VDD33
- =PP3V3_GPI_MIO
- =PP1V2_GPU_PLLVDD
- =PP1V2_GPU_H_PLLVDD
- =PP1V2_GPU_VID_PLLVDD

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



NV G84M GPIO/MIO/Misc

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

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SIZE	DRAWING NUMBER	REV.
D	051-7413	11.0.0
SCALE	SHT	OF
NONE	70	89

Page Notes

Power aliases required by this page:

- =PP1V8_GPU_IFPX
- =PP3V3_GPU_IFPCD_IOVDD
- =PP3V3_GPU_DAC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Sum of peak currents: 240mA

=PP1V8_GPU_IFPX

L8800
FERR-220-OHM
0402

20mA peak per diff pair
160mA peak for all pairs

C8800 4.7UF 20V 603
C8801 0.1UF 20V 402
C8803 0.1UF 20V 402

Place at AF9 Place at AF8

PP1V8_GPU_IFPAB_IOVDD F
MIN LINE WIDTH=0.4 mm
MIN NECK WIDTH=0.2 mm
VOLTAGE=1.8V

L8805
FERR-220-OHM
0402

40mA peak

C8805 4.7UF 20V 603
C8806 0.1UF 20V 402

PP1V8_GPU_IFPAB_PLLVDD F
MIN LINE WIDTH=0.4 mm
MIN NECK WIDTH=0.2 mm
VOLTAGE=1.8V

GPU_IFPAB_VPROBE
GPU_IFPAB_RSET

=PP3V3_GPU_IFPCD_IOVDD

L8810
FERR-220-OHM
0402

20mA peak per diff pair
200mA peak for all pairs

C8810 4.7UF 20V 603
C8811 0.1UF 20V 402
C8813 0.1UF 20V 402

Place at AD6 Place at AE7

PP3V3_GPU_IFPCD_IOVDD F
MIN LINE WIDTH=0.4 mm
MIN NECK WIDTH=0.2 mm
VOLTAGE=3.3V

L8815
FERR-220-OHM
0402

40mA peak

C8815 4.7UF 20V 603
C8816 0.1UF 20V 402

PP1V8_GPU_IFPCD_PLLVDD F
MIN LINE WIDTH=0.3 mm
MIN NECK WIDTH=0.2 mm
VOLTAGE=1.8V

GPU_IFPCD_VPROBE
GPU_IFPCD_RSET

Sum of peak currents: 390mA

=PP3V3_GPU_DAC

L8820
FERR-220-OHM
0402

120mA peak

C8820 4.7UF 20V 603
C8821 0.1UF 20V 402

PP3V3_GPU_DACB_VDD F
MIN LINE WIDTH=0.35 mm
MIN NECK WIDTH=0.25 mm
VOLTAGE=3.3V

GPU_DACB_VREF
GPU_DACB_RSET

L8830
FERR-220-OHM
0402

150mA peak

C8830 4.7UF 20V 603
C8831 0.1UF 20V 402

PP3V3_GPU_DACB_VDD F
MIN LINE WIDTH=0.35 mm
MIN NECK WIDTH=0.25 mm
VOLTAGE=3.3V

GPU_DACB_VREF
GPU_DACB_RSET

NO STUFF
L8840
FERR-220-OHM
0402

120mA peak

C8840 4.7UF 20V 603
C8841 0.1UF 20V 402

PP3V3_GPU_DACC_VDD F
MIN LINE WIDTH=0.35 mm
MIN NECK WIDTH=0.25 mm
VOLTAGE=3.3V

GPU_DACC_VREF
GPU_DACC_RSET

GPU_I2CC_SCL
GPU_I2CC_SDA
GPU_I2CH_SCL
GPU_I2CH_SDA
=GPU_I2CS_SCL
=GPU_I2CS_SDA

I2CS must be pulled up if not used
I2CS addr fixed at 0x9E,0x9F

OMIT

U8000
NB8P-GS-W-A2
BGA
(5 OF 8)

IFPA_TXC_AK9 → LVDS L CLK P
IFPA_TXC_L_AJ9 → LVDS L CLK N

IFPA_TXD0_L_AH6 → LVDS L DATA P<0>
IFPA_TXD0_L_AJ6 → LVDS L DATA N<0>
IFPA_TXD1_L_AH8 → LVDS L DATA P<1>
IFPA_TXD1_L_AJ8 → LVDS L DATA N<1>
IFPA_TXD2_L_AH7 → LVDS L DATA P<2>
IFPA_TXD2_L_AJ7 → LVDS L DATA N<2>
IFPA_TXD3_L_AJ5 → LVDS L DATA P<3>
IFPA_TXD3_L_AH5 → LVDS L DATA N<3>

IFPB_TXC_AK4 → LVDS U CLK P
IFPB_TXC_L_AL4 → LVDS U CLK N

IFPB_TXD4_L_AM6 → LVDS U DATA P<0>
IFPB_TXD4_L_AL6 → LVDS U DATA N<0>
IFPB_TXD5_L_AL7 → LVDS U DATA P<1>
IFPB_TXD5_L_AK6 → LVDS U DATA N<1>
IFPB_TXD6_L_AK5 → LVDS U DATA P<2>
IFPB_TXD6_L_AJ5 → LVDS U DATA N<2>
IFPB_TXD7_L_AK7 → LVDS U DATA P<3>
IFPB_TXD7_L_AL8 → LVDS U DATA N<3>

IFPC_TXC_AM2 → TMDS CLK P
IFPC_TXC_L_AM3 → TMDS CLK N

IFPC_TXD0_L_AE2 → TMDS DATA P<0>
IFPC_TXD0_L_AE1 → TMDS DATA N<0>
IFPC_TXD1_L_AF1 → TMDS DATA P<1>
IFPC_TXD1_L_AF2 → TMDS DATA N<1>
IFPC_TXD2_L_AG1 → TMDS DATA P<2>
IFPC_TXD2_L_AH1 → TMDS DATA N<2>

IFPD_TXC_AG3 → GPU IFPD CLK P
IFPD_TXC_L_AH2 → GPU IFPD CLK N

IFPD_TXD4_L_AK1 → TMDS DATA P<3>
IFPD_TXD4_L_AJ1 → TMDS DATA N<3>
IFPD_TXD5_L_AL2 → TMDS DATA P<4>
IFPD_TXD5_L_AL1 → TMDS DATA N<4>
IFPD_TXD6_L_AJ2 → TMDS DATA P<5>
IFPD_TXD6_L_AJ3 → TMDS DATA N<5>

DACA_RED_AH11 → GPU VGA R
DACA_GREEN_AJ12 → GPU VGA G
DACA_BLUE_AH12 → GPU VGA B

DACA_HSYNC_AF10 → GPU VGA HSYNC
DACA_VSYNC_AK10 → GPU VGA VSYNC

DACB_RED_R6 → GPU TV C
DACB_GREEN_T5 → GPU TV Y
DACB_BLUE_T6 → GPU TV COMP

DACB_CS_Y5 → GPU CSYNC

DACC_RED_AF6 → GPU R2
DACC_GREEN_AG6 → GPU G2
DACC_BLUE_AE5 → GPU B2

DACC_HSYNC_AG7 → GPU H2SYNC
DACC_VSYNC_AG5 → GPU V2SYNC

I2CA_SCL_K2 → GPU I2CA_SCL
I2CA_SDA_J3 → GPU I2CA_SDA

I2CB_SCL_H4 → GPU I2CB_SCL
I2CB_SDA_J4 → GPU I2CB_SDA

Composite/S-Video	VGA	Component
C	R	Pr
Y	G	Y
Comp	B	Pb

NV G84M Video Interfaces

SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

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SCALE	SHT	OF
NONE	72	89

GPU VCore Regulator

D

D

C

C

B

B

A

A

GPU VCore Setpoints

VID3	VID2	VID1	VID0	Voltage	Max Batt	Balanced	Max perf
1	0	0	1	1.05575V	M87,M88	M87	-
0	1	1	0	1.13300V	-	M88	M87
0	0	1	0	1.23600V	-	-	M88

Other VID states may not be valid

M87/M88 Default Vcore Setpoints

BOM GROUP	BOM OPTIONS
GPUVID_1P26V	GPUVID3_0, GPUVID2_0, GPUVID1_1, GPUVID0_0
GPUVID_1P13V	GPUVID3_0, GPUVID2_1, GPUVID1_1, GPUVID0_0
GPUVID_1P05V	GPUVID3_1, GPUVID2_0, GPUVID1_0, GPUVID0_1

GPU (G84M) Core Supply

SYNC_MASTER=M88 SYNC_DATE=08/02/2007

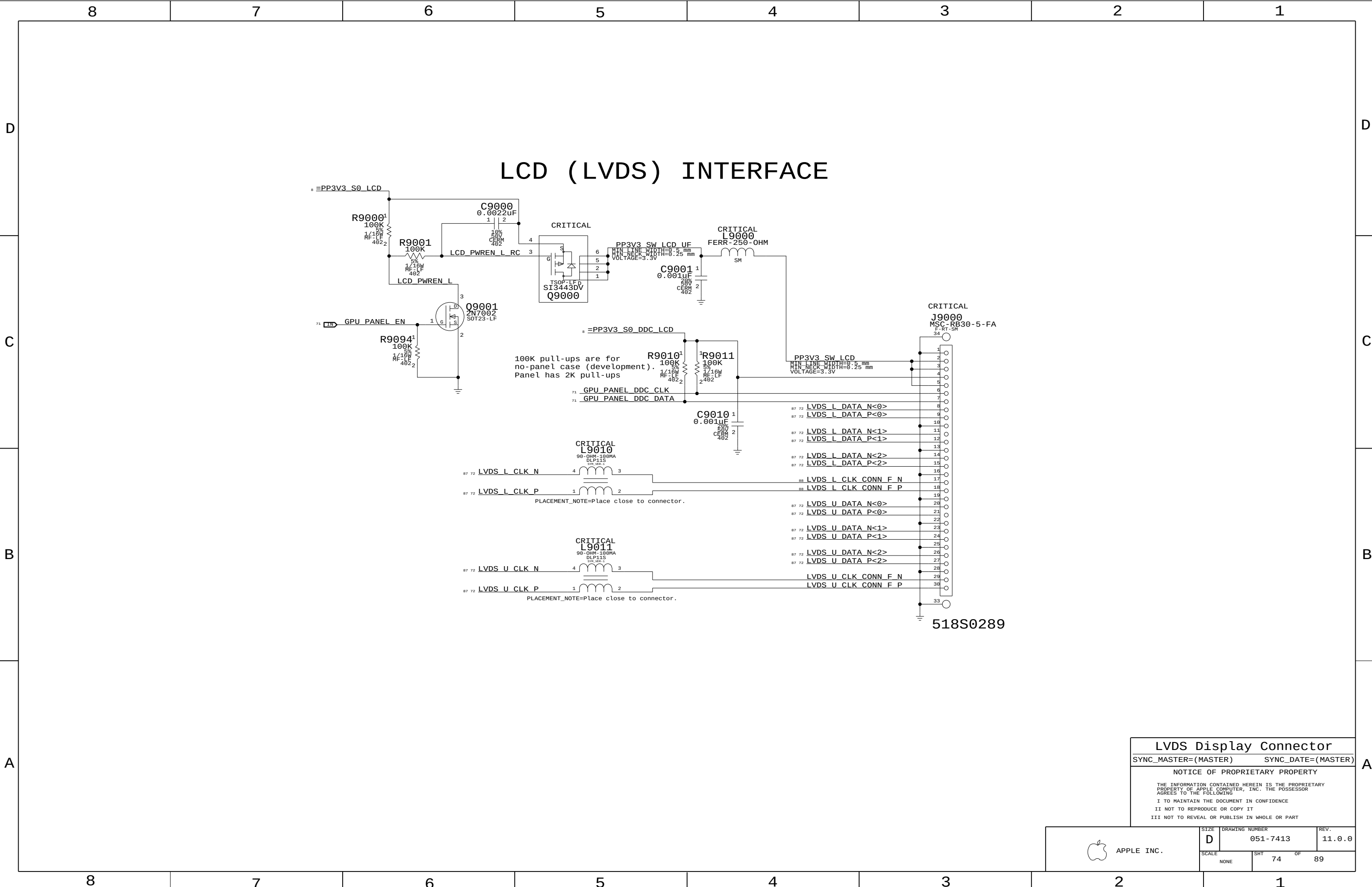
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D	051-7413	11.0.0
SCALE	SHT	OF
NONE	73	89



LVDS Display Connector

SYNC_MASTER=(MASTER)

SYNC_DATE=(MASTER)

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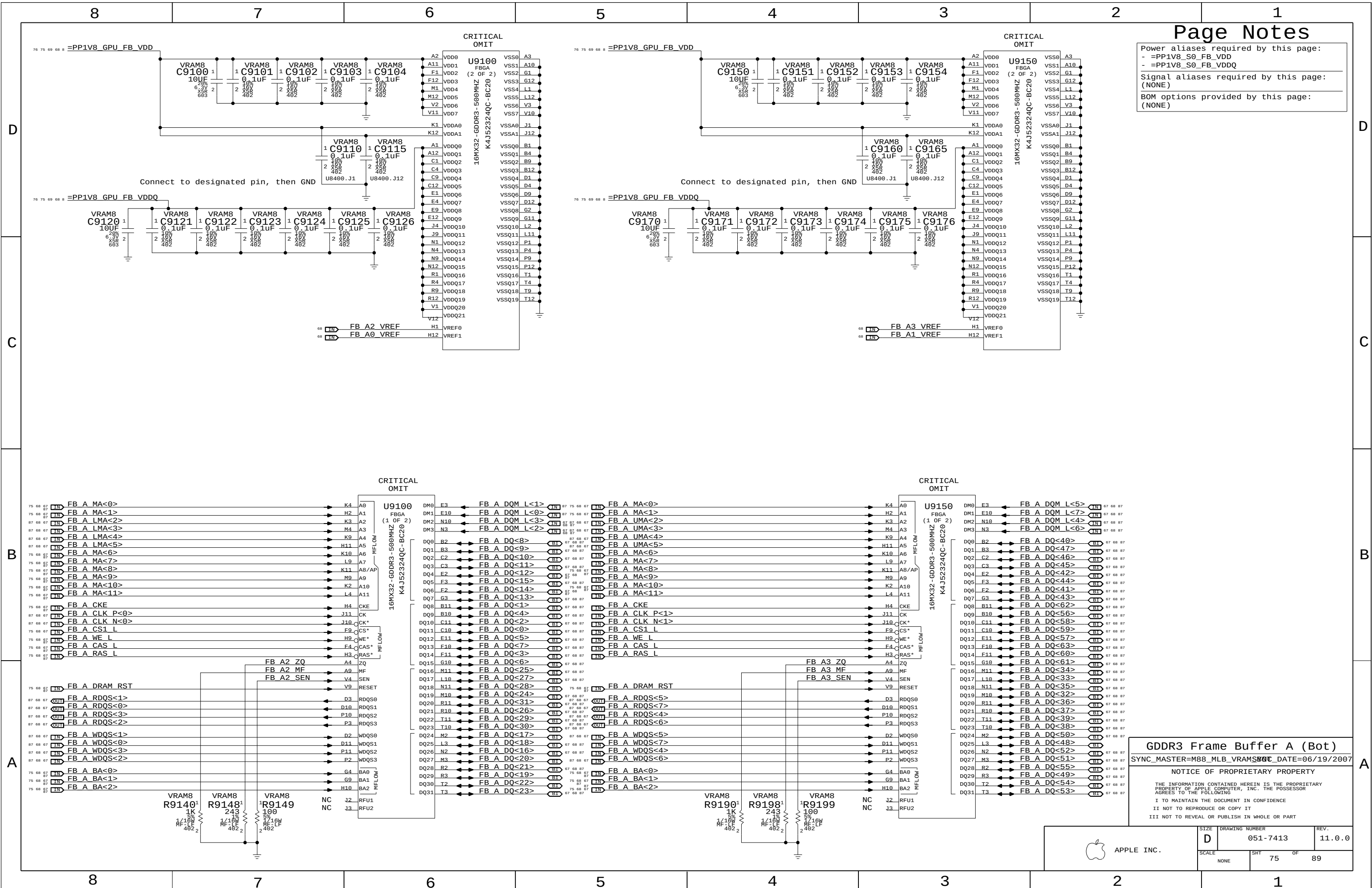
 APPLE INC.	SIZE D	DRAWING NUMBER 051-7413	REV. 11.0.0
	SCALE NONE	SHT 74	OF 89

Page Notes

Power aliases required by this page:
- =PP1V8_S0_FB_VDD
- =PP1V8_S0_FB_VDDQ

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

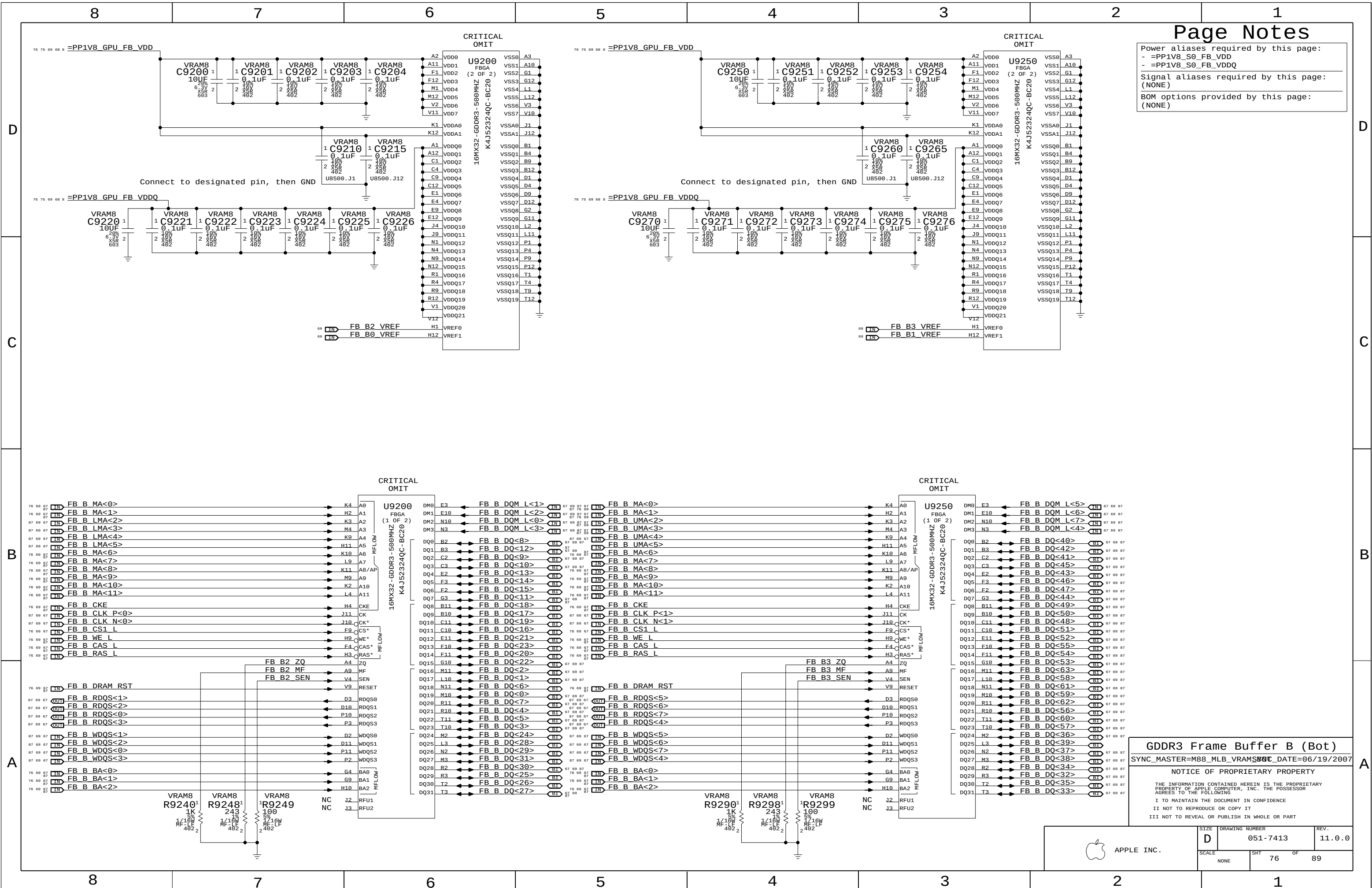


Page Notes

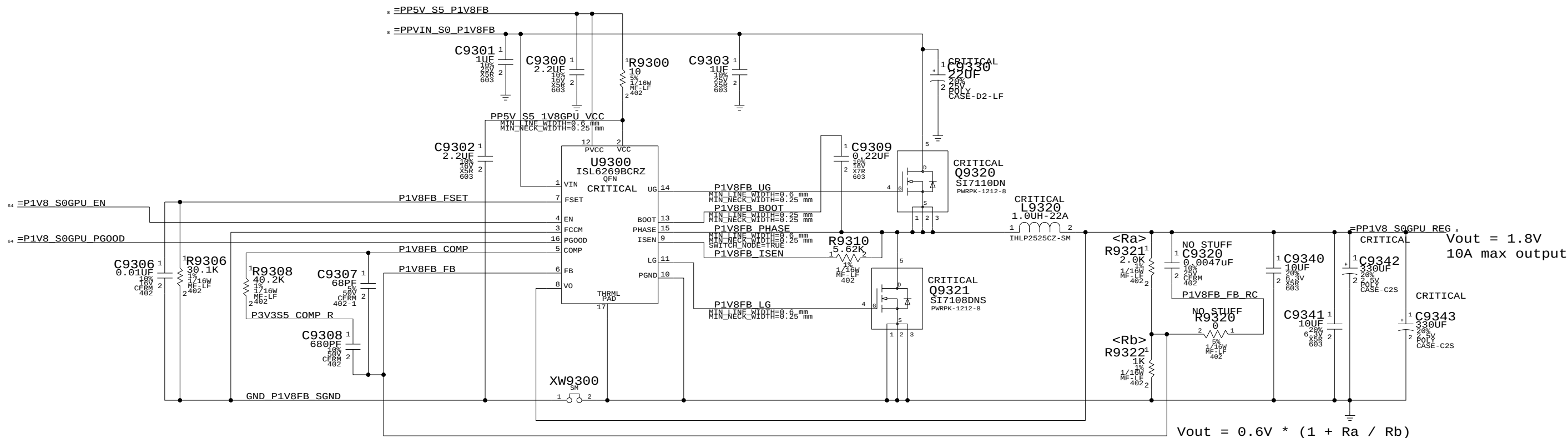
Power aliases required by this page:
- =PP1V8_S0_FB_VDD
- =PP1V8_S0_FB_VDDQ

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

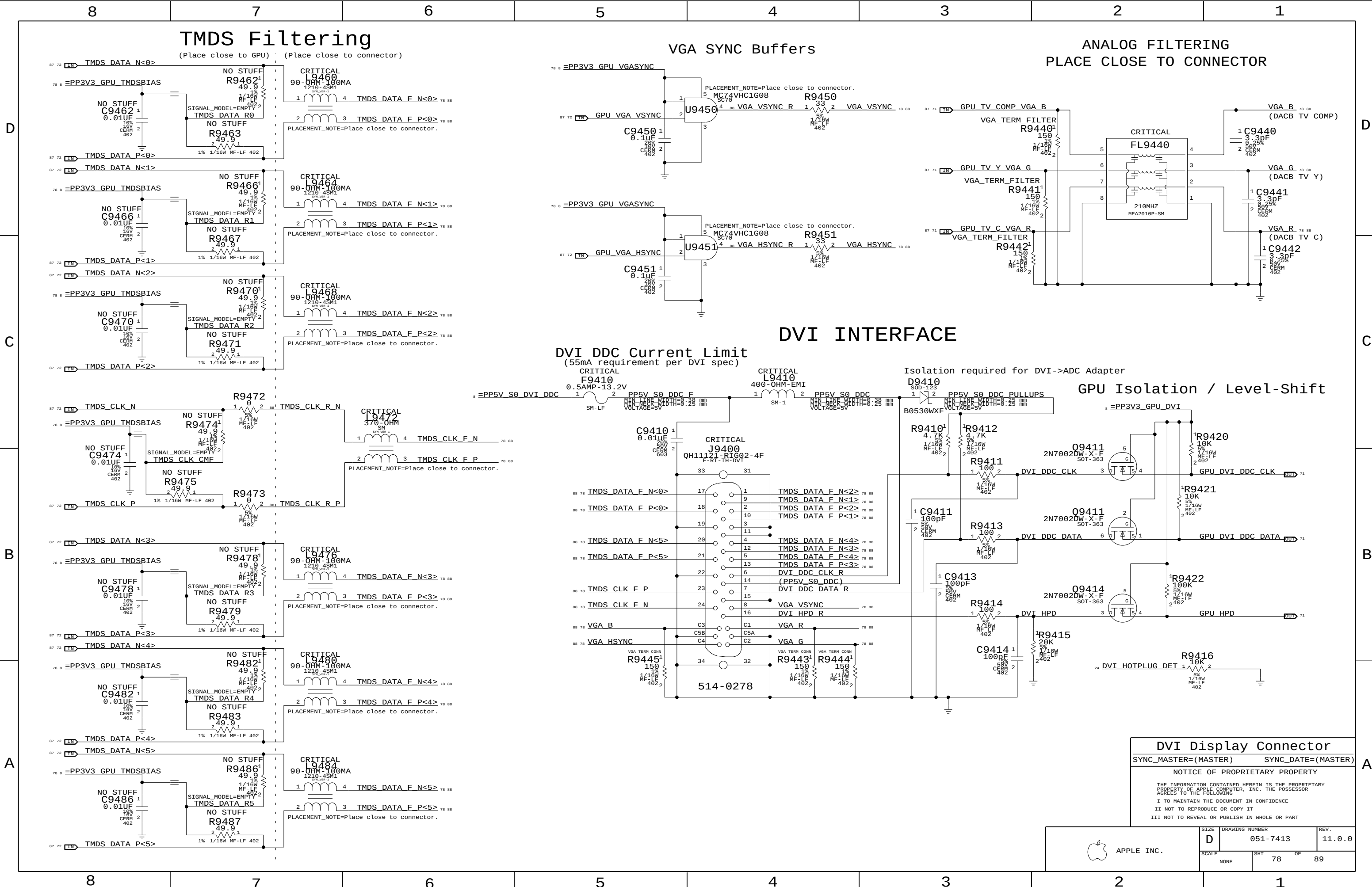


1.8V Frame Buffer Regulator

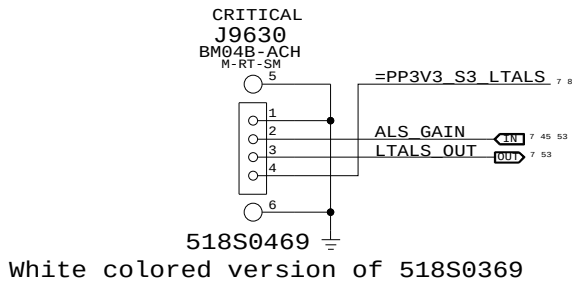


1.8V FB Power Supply
SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)
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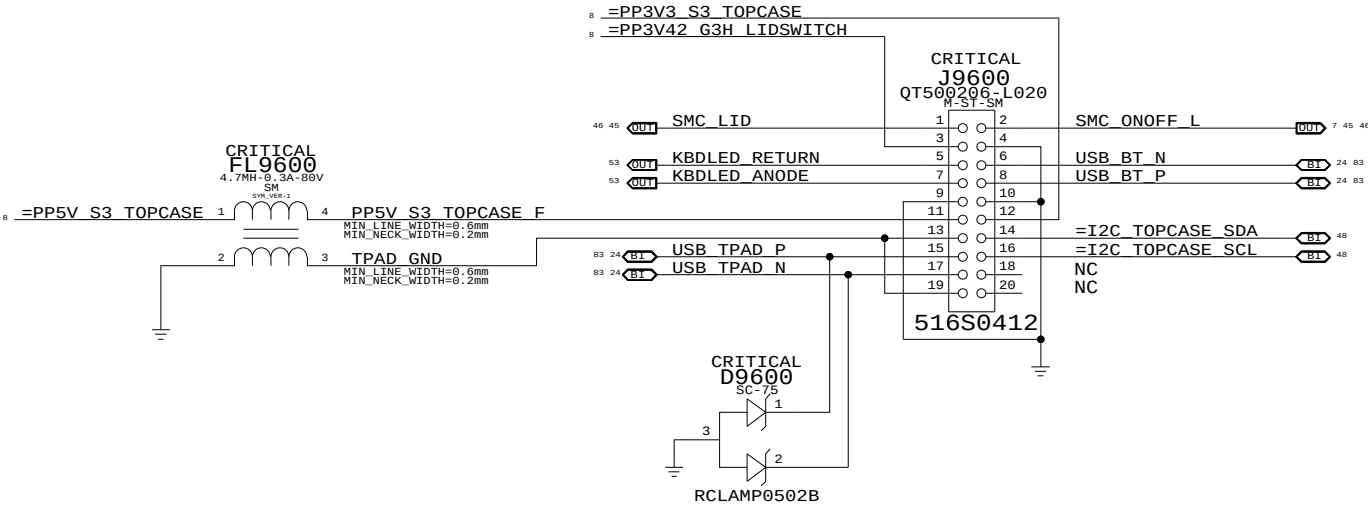
APPLE INC.	SIZE D	DRAWING NUMBER 051-7413	REV. 11.0.0
	SCALE NONE	SHT 77	OF 89



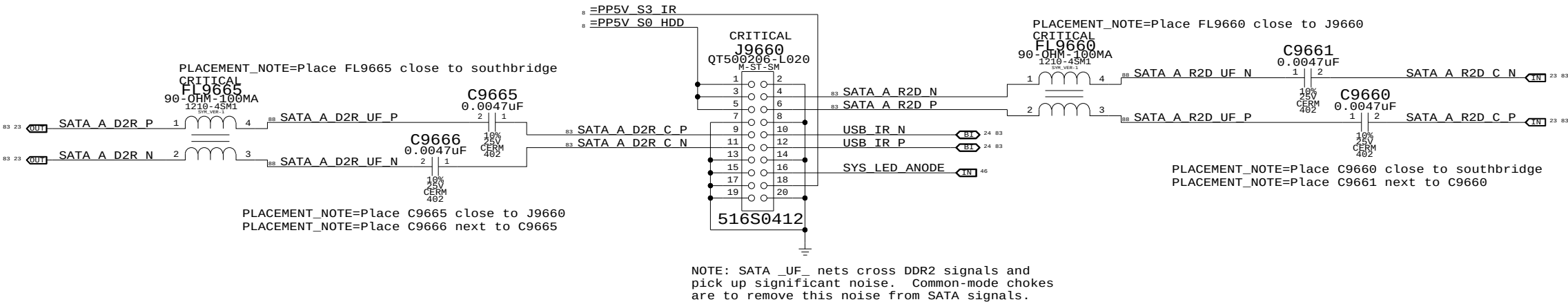
Left ALS Connector



Top-Case Connector



SATA HDD & IR & SIL Flex Connector



Project Specific Connectors

SYNC_MASTER=(M59_SYNC) SYNC_DATE=08/24/2006

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APPLE INC.

SIZE

D

DRAWING NUMBER

051-7413

SCALE

NONE

SHT

79

OF

89

REV.

11.0.0

87654321

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI	*	=2:1_SPACING	?

SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Sections 10.18.1 & 10.19

Controller Link (AMT) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLINK_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLINK_12MIL	*	=STANDARD	12 MILS	5 MILS	300 MILS	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLINK	*	=1.8:1_SPACING	?
CLINK_VREF	*	12 MILS	?

SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Sections 10.27.1.5-7, 10.29 & 10.30

Ethernet (Yukon) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	25 MILS	?

SOURCE: Based on Santa Rosa Platform DG, Rev 1.0 (#21112), Sections 10.27.1.5-7, 10.29 & 10.30

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
PCI_AD	PCI_55S	PCI	PCI_AD<18..0>	24 38
PCI_AD19	PCI_55S	PCI	PCI_AD<19>	24 38
PCI_AD20	PCI_55S	PCI	PCI_AD<20>	24 38
PCI_AD	PCI_55S	PCI	PCI_AD<31..21>	24 38
PCI_AD	PCI_55S	PCI	PCI_PAR	24 38
PCI_C_BE_L	PCI_55S	PCI	PCI_C_BE_L<3..0>	24 38
PCI_CNT1	PCI_55S	PCI	PCI_IRDY_L	24 38
PCI_CNT1	PCI_55S	PCI	PCI_DEVSEL_L	24 38
PCI_CNT1	PCI_55S	PCI	PCI_PERR_L	24 38
PCI_LOCK_L	PCI_55S	PCI	PCI_LOCK_L	24
PCI_CNT1	PCI_55S	PCI	PCI_SERR_L	24 38
PCI_CNT1	PCI_55S	PCI	PCI_STOP_L	24 38
PCI_CNT1	PCI_55S	PCI	PCI_TRDY_L	24 38
PCI_CNT1	PCI_55S	PCI	PCI_FRAME_L	24 38
PCI_FW_REQ_L	PCI_55S	PCI	PCI_FW_REQ_L	24 38
PCI_FW_GNT_L	PCI_55S	PCI	PCI_FW_GNT_L	24 38
PCI_REQ1_L	PCI_55S	PCI	PCI_REQ1_L	24
PCI_GNT1_L	PCI_55S	PCI	PCI_GNT1_L	24
PCI_REQ2_L	PCI_55S	PCI	PCI_REQ2_L	24
PCI_GNT2_L	PCI_55S	PCI	PCI_GNT2_L	24
INT_PIRQA_L	PCI_55S	PCI	INT_PIRQA_L	24
INT_PIRQB_L	PCI_55S	PCI	INT_PIRQB_L	24
INT_PIRQC_L	PCI_55S	PCI	INT_PIRQC_L	24
INT_PIRQD_L	PCI_55S	PCI	INT_PIRQD_L	24 38
INT_PIRQE_L	PCI_55S	PCI	INT_PIRQE_L	9 24
INT_PIRQF_L	PCI_55S	PCI	INT_PIRQF_L	24
PCIE_A_R2D	PCIE_100D	PCIE	PCIE_A_R2D_C_P	
	PCIE_100D	PCIE	PCIE_A_R2D_C_N	
PCIE_A_D2R	PCIE_100D	PCIE	PCIE_A_D2R_P	
	PCIE_100D	PCIE	PCIE_A_D2R_N	
PCIE_B_R2D	PCIE_100D	PCIE	PCIE_B_R2D_C_P	
	PCIE_100D	PCIE	PCIE_B_R2D_C_N	
PCIE_B_D2R	PCIE_100D	PCIE	PCIE_B_D2R_P	
	PCIE_100D	PCIE	PCIE_B_D2R_N	
PCIE_EXCARD_R2D	PCIE_100D	PCIE	PCIE_EXCARD_R2D_C_P	34
	PCIE_100D	PCIE	PCIE_EXCARD_R2D_C_N	34
PCIE_EXCARD_D2R	PCIE_100D	PCIE	PCIE_EXCARD_D2R_P	34
	PCIE_100D	PCIE	PCIE_EXCARD_D2R_N	34
PCIE_FW_R2D	PCIE_100D	PCIE	PCIE_FW_R2D_C_P	
	PCIE_100D	PCIE	PCIE_FW_R2D_C_N	
PCIE_FW_D2R	PCIE_100D	PCIE	PCIE_FW_D2R_P	
	PCIE_100D	PCIE	PCIE_FW_D2R_N	
PCIE_MINI_R2D	PCIE_100D	PCIE	PCIE_MINI_R2D_C_P	24 34
	PCIE_100D	PCIE	PCIE_MINI_R2D_C_N	24 34
PCIE_MINI_D2R	PCIE_100D	PCIE	PCIE_MINI_D2R_P	24 34
	PCIE_100D	PCIE	PCIE_MINI_D2R_N	24 34
GLAN_COMP			GLAN_COMP	23
CLINK_NB	CLINK_55S	CLINK	CLINK_NB_CLK	16 25
CLINK_NB	CLINK_55S	CLINK	CLINK_NB_DATA	16 25
CLINK_NB_RESET_L	CLINK_55S	CLINK	CLINK_NB_RESET_L	16 25
CLINK_WLAN	CLINK_55S	CLINK	CLINK_WLAN_CLK	
CLINK_WLAN	CLINK_55S	CLINK	CLINK_WLAN_DATA	
CLINK_WLAN_RESET_L	CLINK_55S	CLINK	CLINK_WLAN_RESET_L	
NB_CLINK_VREF	CLINK_12MIL	CLINK_VREF	NB_CLINK_VREF	16
SB_CLINK_VREF0	CLINK_12MIL	CLINK_VREF	SB_CLINK_VREF0	25
SB_CLINK_VREF1	CLINK_12MIL	CLINK_VREF	SB_CLINK_VREF1	25
PCIE_ENET_R2D	PCIE_100D	PCIE	PCIE_ENET_R2D_C_P	24 35
	PCIE_100D	PCIE	PCIE_ENET_R2D_C_N	24 35
	PCIE_100D	PCIE	PCIE_ENET_R2D_P	35
	PCIE_100D	PCIE	PCIE_ENET_R2D_N	35
PCIE_ENET_D2R	PCIE_100D	PCIE	PCIE_ENET_D2R_P	24 35
	PCIE_100D	PCIE	PCIE_ENET_D2R_N	24 35
	PCIE_100D	PCIE	PCIE_ENET_D2R_C_P	35
	PCIE_100D	PCIE	PCIE_ENET_D2R_C_N	35
ENET_MDI	ENET_100D	ENET_MDI	ENET_MDI_P<0>	35 37
	ENET_100D	ENET_MDI	ENET_MDI_N<0>	35 37
ENET_MDI	ENET_100D	ENET_MDI	ENET_MDI_P<1>	35 37
	ENET_100D	ENET_MDI	ENET_MDI_N<1>	35 37
ENET_MDI	ENET_100D	ENET_MDI	ENET_MDI_P<2>	35 37
	ENET_100D	ENET_MDI	ENET_MDI_N<2>	35 37
ENET_MDI	ENET_100D	ENET_MDI	ENET_MDI_P<3>	35 37
	ENET_100D	ENET_MDI	ENET_MDI_N<3>	35 37

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SB Constraints (2 of 2)

SYNC_MASTER=T9_NAME SYNC_DATE=01/17/2007

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FireWire Constraints

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SCALE	SHT	OF
NONE	86	89

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8		7		6		5		4		3		2		1		
M75 Board-Specific Spacing & Physical Constraints																
BOARD LAYERS				BOARD AREAS				BOARD UNITS (ALL OF MM)		ALLEGRO VERSION						
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM				NO_TYPE, BGA				MM		15.5.1						
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
DEFAULT		*	Y	=55_OHM_SE		=55_OHM_SE		30 MM		0 MM		0 MM				
STANDARD		*	Y	=DEFAULT		=DEFAULT		12.7 MM		=DEFAULT		=DEFAULT				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
55_OHM_SE		TOP, BOTTOM	Y	0.100 MM		0.100 MM										
55_OHM_SE		ISL2, ISL11	Y	0.250 MM		0.076 MM										
55_OHM_SE		*	Y	0.076 MM		0.076 MM		=STANDARD		=STANDARD		=STANDARD				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
50_OHM_SE		TOP, BOTTOM	Y	0.125 MM		0.125 MM										
50_OHM_SE		*	Y	0.090 MM		0.090 MM		=STANDARD		=STANDARD		=STANDARD				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
46_OHM_SE		TOP, BOTTOM	Y	0.126 MM		0.126 MM										
46_OHM_SE		*	Y	0.100 MM		0.100 MM		=STANDARD		=STANDARD		=STANDARD				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
45_OHM_SE		TOP, BOTTOM	Y	0.150 MM		0.150 MM										
45_OHM_SE		*	Y	0.105 MM		0.105 MM		=STANDARD		=STANDARD		=STANDARD				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
40_OHM_SE		TOP, BOTTOM	Y	0.185 MM		0.185 MM										
40_OHM_SE		*	Y	0.131 MM		0.131 MM		=STANDARD		=STANDARD		=STANDARD				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
27P4_OHM_SE		TOP, BOTTOM	Y	0.335 MM		0.335 MM										
27P4_OHM_SE		*	Y	0.240 MM		0.240 MM		=STANDARD		=STANDARD		=STANDARD				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
70_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
70_OHM_DIFF		ISL3, ISL4	Y	0.149 MM		0.149 MM				0.125 MM		0.125 MM				
70_OHM_DIFF		ISL9, ISL10	Y	0.149 MM		0.149 MM				0.125 MM		0.125 MM				
70_OHM_DIFF		ISL2, ISL11	Y	0.185 MM		0.185 MM				0.125 MM		0.125 MM				
70_OHM_DIFF		TOP, BOTTOM	Y	0.185 MM		0.185 MM				0.125 MM		0.125 MM				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
80_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
80_OHM_DIFF		ISL3, ISL4	Y	0.115 MM		0.115 MM				0.125 MM		0.125 MM				
80_OHM_DIFF		ISL9, ISL10	Y	0.115 MM		0.115 MM				0.125 MM		0.125 MM				
80_OHM_DIFF		ISL2, ISL11	Y	0.140 MM		0.140 MM				0.125 MM		0.125 MM				
80_OHM_DIFF		TOP, BOTTOM	Y	0.140 MM		0.140 MM				0.125 MM		0.125 MM				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
85_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
85_OHM_DIFF		ISL3, ISL4	Y	0.101 MM		0.101 MM				0.125 MM		0.125 MM				
85_OHM_DIFF		ISL9, ISL10	Y	0.101 MM		0.101 MM				0.125 MM		0.125 MM				
85_OHM_DIFF		ISL2, ISL11	Y	0.125 MM		0.125 MM				0.125 MM		0.125 MM				
85_OHM_DIFF		TOP, BOTTOM	Y	0.125 MM		0.125 MM				0.125 MM		0.125 MM				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
90_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
90_OHM_DIFF		ISL3, ISL4	Y	0.102 MM		0.102 MM				0.220 MM		0.220 MM				
90_OHM_DIFF		ISL9, ISL10	Y	0.102 MM		0.102 MM				0.220 MM		0.220 MM				
90_OHM_DIFF		ISL2, ISL11	Y	0.130 MM		0.130 MM				0.220 MM		0.220 MM				
90_OHM_DIFF		TOP, BOTTOM	Y	0.130 MM		0.130 MM				0.220 MM		0.220 MM				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
100_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
100_OHM_DIFF		ISL3, ISL4	Y	0.080 MM		0.080 MM				0.200 MM		0.200 MM				
100_OHM_DIFF		ISL9, ISL10	Y	0.080 MM		0.080 MM				0.200 MM		0.200 MM				
100_OHM_DIFF		ISL2, ISL11	Y	0.099 MM		0.099 MM				0.200 MM		0.200 MM				
100_OHM_DIFF		TOP, BOTTOM	Y	0.099 MM		0.099 MM				0.200 MM		0.200 MM				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
110_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
110_OHM_DIFF		ISL3, ISL4	Y	0.077 MM		0.077 MM				0.330 MM		0.330 MM				
110_OHM_DIFF		ISL9, ISL10	Y	0.077 MM		0.077 MM				0.330 MM		0.330 MM				
110_OHM_DIFF		ISL2, ISL11	Y	0.089 MM		0.089 MM				0.330 MM		0.330 MM				
110_OHM_DIFF		TOP, BOTTOM	Y	0.089 MM		0.089 MM				0.330 MM		0.330 MM				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
100_DIFF_BGA		*	=100_OHM_DIFF	=100_OHM_DIFF		=100_OHM_DIFF		=100_OHM_DIFF		=100_OHM_DIFF		=100_OHM_DIFF				
100_DIFF_BGA		ISL3, ISL4	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM				
100_DIFF_BGA		ISL9, ISL10	Y	0.075 MM		0.075 MM				0.125 MM		0.125 MM				
NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.																
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
100_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
100_OHM_DIFF		ISL3, ISL4	Y	0.102 MM		0.102 MM				0.220 MM		0.220 MM				
100_OHM_DIFF		ISL9, ISL10	Y	0.102 MM		0.102 MM				0.220 MM		0.220 MM				
100_OHM_DIFF		ISL2, ISL11	Y	0.130 MM		0.130 MM				0.220 MM		0.220 MM				
100_OHM_DIFF		TOP, BOTTOM	Y	0.130 MM		0.130 MM				0.220 MM		0.220 MM				
PHYSICAL_RULE_SET		LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH		MINIMUM NECK WIDTH		MAXIMUM NECK LENGTH		DIFFPAIR PRIMARY GAP		DIFFPAIR NECK GAP				
110_OHM_DIFF		*	N	=STANDARD		=STANDARD		=STANDARD		=STANDARD		=STANDARD				
110_OHM_DIFF		ISL3, ISL4	Y	0.077 MM		0.077 MM				0.330 MM		0.330 MM				
110_OHM_DIFF		ISL9, ISL10	Y	0.077 MM		0.077 MM				0.330 MM		0.330 MM				
110_OHM_DIFF		ISL2, ISL11	Y	0.089 MM		0.089 MM				0.330 MM		0.330 MM				
110_OHM_DIFF		TOP, BOTTOM	Y	0.089 MM		0.089 MM				0.330 MM		0.330 MM				
PCB Rule Definitions																
SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)																
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8		7		6		5		4		3		2		1		

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
BGA_P3MM	*	=DEFAULT	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM
MEM_CLK	*	BGA	BGA_P2MM
CLK_FSB	*	BGA	BGA_P2MM
CLK_PCIE	*	BGA	BGA_P2MM
CLK_MED	*	BGA	BGA_P2MM
CLK_SLOW	*	BGA	BGA_P2MM
FSB_DSTB	FSB_DSTB	BGA	BGA_P3MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
1.8:1_SPACING	*	0.18 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

PCB Rule Definitions

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SCALE
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SHT
89

OF
89